

Tuning the machines that tune themselves: a survey of hyperparameter optimization and a plan for building our own tuner

Internal research note, prepared for the workspace projects
(reinforcement learning, Hamiltonian networks, neural-transport samplers,
multi-objective models)

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Abstract

We are planning a shared hyperparameter-tuning system for the training workloads in our workspace: reinforcement learning through RLlib, Hamiltonian neural networks, neural-transport samplers, and multi-objective models. This document surveys the tuning literature through August 2026 at a level meant to teach, with the load-bearing mechanisms derived in place and illustrated on one running example rather than quoted from their papers, audits the open-source tooling library by library against primary sources, and turns both into an implementation plan.

The architectural conclusion is to keep Ray Tune as the execution layer (its distributed trials, check-pointing, and population-based training remain the strongest available, and we already operate Ray) while owning the search layer ourselves, since Tune's algorithm side has been in maintenance while method development moved to Optuna, BoTorch, and the AutoML groups' libraries.

The algorithmic conclusion is a tiered roadmap: asynchronous successive halving plus a carefully configured Bayesian optimizer as the spine; noise-aware multi-objective acquisition with multi-objective early stopping, expert priors, cost-aware search, and the PB2-to-BG-PBT population line for reinforcement learning as the differentiators; freeze-thaw scheduling with a pre-trained learning-curve surrogate as the frontier bet. Each recommendation is tied to the controlled evidence that supports it, and a validation suite is specified before any of it is built.

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Part I

The problem

Chapter 1

Introduction

This document answers two questions we posed ourselves while planning a shared tuning system for the projects in our workspace: whether Ray Tune is the right architecture to build on, given that RLlib already runs our reinforcement-learning work, and which algorithms such a system should implement to serve very different customers, from reinforcement-learning agents through Hamiltonian neural networks and neural-transport samplers to multi-objective models in finance. Both questions turn out to have reasonably confident answers once the literature is laid out, and laying it out carefully is most of the document: our local paper collection thins out around 2023, and several things a 2023 reader would recommend are no longer what a 2026 reader should build.

The short answers, defended over the following chapters; the method names that follow are only names for now, each taught from zero in its chapter, so read this section as a map to return to rather than an argument to absorb. Ray Tune: yes, as the execution layer, and deliberately not as the source of search algorithms. Tune’s distributed trial execution, checkpoint machinery, and population-based training support remain the strongest in open source and compose naturally with RLlib, but its own algorithm layer has been in maintenance for roughly two years while the method frontier moved into other projects (Chapters 10 and 12).

The algorithms: a spine of asynchronous successive halving plus a well-configured Bayesian optimizer covers the routine work at low cost; the differentiators worth building are noise-aware multi-objective acquisition with a multi-objective early-stopping scheduler, expert priors, cost-aware acquisition, and a population line for reinforcement learning. The method names that follow are only names for now, each taught from zero in its chapter: qLogNEHVI with MO-ASHA (Chapters 7), priors in the PriorBand style (Chapter 8), and the population line as PB2 extended to mixed spaces and then BG-PBT, the method we were specifically asked to assess (Chapter 6). Freeze-thaw scheduling with a pre-trained learning-curve surrogate is the one frontier bet we recommend (Chapter 5), and Chapter 15 puts the whole list in build order with the evidence attached to each item.

The document is written to teach as well as to conclude, because the people extending this system next year will need the reasoning, not just the verdicts, and a verdict whose derivation you cannot reconstruct is one you cannot safely modify. Three habits follow, and knowing them up front makes the document easier to read.

First, one running example carries the whole text: tuning PPO on a Brax locomotion task, in the fifteen-dimensional search space introduced in Section 2.1, with a Hamiltonian network as the second thread wherever several objectives compete. When a formula arrives, its symbols are read against that example before anything is done with them.

Second, every load-bearing equation is derived or unpacked in place rather than quoted: the Gaussian-

process posterior, expected improvement and the floating-point failure that motivates its modern replacement, the tree-structured Parzen estimator’s ratio rule, the budget arithmetic of successive halving, the geometry that breaks weighted-sum scalarization.

Third, the figures are schematic drawings of mechanisms, not plots of measured data, and their captions say what each one is supposed to show; when a number matters, it is computed in the text where the reader can check it.

The document also carries a decision layer, kept visually separate from the teaching so that neither gets in the other’s way. Each method chapter ends in a boxed *Decisions from this chapter* marked by a thick left rule: plain-language verdicts, include or exclude, with the cost to build, the evidence behind the call, and the comparison against the alternatives, written to be judged without opening any cited paper.

The box is earned rather than asserted: each method chapter closes by first teaching the experiments its verdict rests on, their designs, arenas, and sponsors, so the evidence can be judged rather than taken on citation, and then arguing the verdict from several independent angles, ending with the criterion that would overturn it in our own validation suite.

Table 15.1 in Chapter 15 collects every verdict on one page. A glossary at the end defines every term of art in plain language. And because several of our readers come from economics and statistics rather than machine learning, the text flags the bridges where they exist: the Gaussian process is kriging, expected improvement is an option value, the Pareto front (the set of configurations undominated in objective space) is the welfare-theoretic one.

Chapter 2 pins down the problem, fixes the running example, and maps the four ways our workloads bend the textbook formulation. Chapters 3 through 8 build up the method families in increasing order of structure: searching without a model, modeling the objective, exploiting partial evaluations, tuning while training, handling several objectives, and importing knowledge from priors, past studies, and scale. Each chapter opens with the failure of the previous chapter’s best answer, which is not a storytelling flourish: it is the actual dependency structure of the field, and it is why the chapters are in this order.

Chapter 9 turns back to our four workload families and says what to do for each. Chapter 10 audits the open-source field library by library, with maintenance status checked against primary sources on August 22, 2026, and re-verified on August 23. The closing part turns the verdicts into a concrete product proposal: Chapter 11 states what we ship and walks a first study through it, Chapter 12 gives the architectural design, Chapter 13 pins the interfaces and data schemas, Chapter 14 specifies the test suite, and Chapter 15 the build plan and the bill.

A reader who wants only decisions can read Chapter 2, the decision boxes at the ends of Chapters 3 through 9, and then Chapters 10, 12, and 15 with Table 15.1; a reader implementing a searcher should read the method chapters in order, since each one leans on the machinery of its predecessors.

A note on sourcing. Claims about specific methods are tied to the papers that introduced or tested them, and the load-bearing ones (BG-PBT, the RL tuning study of Eimer, Lindauer, and Raileanu, and the current defaults of the major libraries) were checked against the primary texts and pages rather than summaries; a companion file records, for every source, what was inspected and at what depth. The derivations added for teaching (Gaussian conditioning, the expected-improvement closed form and its underflow arithmetic, the successive-halving budget count, the hypervolume and scalarization geometry) are elementary and stand on their own; they explain the cited results rather than extend them. Where a comparison rests on one paper’s experiments we say so, and where the honest answer is that two methods are indistinguishable on current evidence, we say that too, in preference to manufacturing a ranking.

Chapter 2

The problem we are actually trying to solve

Every project in our workspace trains something: a reinforcement-learning (RL) agent through RLlib (Ray’s RL library), a Hamiltonian neural network on trajectory data (a network whose architecture builds in energy conservation), a normalizing flow that preconditions a sampler (a neural map that reshapes a hard posterior into an easier one), a forecasting model with several competing losses. Each of these is taught properly where it matters (Chapter 9 devotes a section to each); for now they are simply the customers. Each of these training procedures exposes ten to twenty knobs, and the knobs come in four kinds. The kinds are worth meeting one at a time, because every method in this document has to handle all four, and more than one otherwise-good method stumbles on exactly one of them.

A *continuous* knob takes any value in an interval. The learning rate is the canonical case: 3×10^{-4} is a legal setting, so is 2.9×10^{-4} , and so is everything between them. Entropy coefficients and integrator step sizes are the same kind. Continuous knobs are the friendliest kind, because “nearby” means what it seems to mean: settings close together usually behave similarly, and every model-based method in this document leans on that.

An *ordinal* knob is an integer with a natural order. Batch size runs 128, 256, 512; network width and the number of integrator steps inside a sampler are the same kind. The order carries real information, since 256 genuinely sits between 128 and 512 and performance usually moves through it on the way from one to the other, but the grid is coarse, and treating the values as just another continuum papers over the coarseness. How much that ordering is worth respecting returns as a concrete design choice in Chapter 4.

A *categorical* knob is a choice among labels with no “between” at all. There is no activation function halfway from tanh to ReLU, and no gradual path between using spectral normalization (a training stabilizer we will meet again in the running example) and not using it. Methods built on continuity must do something special here, and what they do (overlap kernels, per-category bandits) occupies real space in Chapters 4 and 6.

A *conditional* knob exists only when another knob takes a particular value. Learning-rate annealing is the standing instance: if annealing is switched off, the final learning-rate multiplier is not merely irrelevant, it is undefined, and a tuner that keeps proposing values for it is spending budget on configurations that do not differ. (Off-policy algorithms supply the same structure elsewhere in our workloads: the temperature of a soft target update is undefined when hard updates are selected.) Conditional structure is why realistic search spaces are trees rather than boxes, and it decides real verdicts later; it is much of the case for one of Chapter 4’s searchers.

Choosing these values well is routinely worth more than the next clever idea, and choosing them badly can make a sound method look broken. Eimer et al. (2023) swept eleven hyperparameters of PPO, the same reinforcement-learning algorithm our running example will tune below, across standard

environments and found that almost every one of them had a large effect on final return in at least one environment, including knobs that most papers never tune, such as the clipping range. The same study found that the worst value of a swept hyperparameter landed within one standard deviation of the best value in only 7 of 126 sweeps. Tuning is not optional in our line of work; the only question is how much compute and attention it consumes.

2.1 A running example, and the loop every method lives in

Abstract definitions slide off the mind unless they have something to stick to, so we fix a concrete case now and return to it in every section. Our running example is the one our own cluster runs most often: tuning PPO, the workhorse reinforcement-learning algorithm of our RLlib stack, as it learns a locomotion task in Brax, a suite of fast physics-simulated environments.

Since PPO’s knobs will follow us through the whole document, one plain-language pass through the algorithm is worth its space; nothing deeper than this pass is ever assumed. The agent holds two neural networks. The *policy* maps what the agent observes to a distribution over actions; the *value network* predicts how much reward is still to come from a given situation. Training alternates two activities: the agent acts in the simulated environment for a while, collecting observations, actions, and rewards, and then the optimizer takes gradient steps that make the actions which worked out better than the value network expected (the *advantage*) more probable, with a clip that keeps the updated policy close to the policy that collected the data. The name PPO, proximal policy optimization, is a description: optimize the policy, but stay proximal. A trained agent’s score, its *return*, is the total reward it accumulates over an episode of play.

The full search space, taken from the experiments behind BG-PBT, the population-based method Chapter 6 examines in detail, has fifteen dimensions: nine training hyperparameters (learning rate, discount, entropy coefficient, unroll length, reward scaling, batch size, update epochs, the generalized-advantage-estimation parameter GAE λ , clipping ϵ) and six architecture parameters (width, depth, and a spectral-normalization flag, separately for the policy and value networks) (Wan et al., 2022).

2.1.1 Reading the fifteen knobs

Fifteen numbers is a lot to hold, so read them in groups, by what each group controls. The aim is not to memorize the list but to see that each entry is a lever a practitioner can explain, not a mystery constant.

Three knobs govern the size of an update. The learning rate scales how far every gradient step moves the network, and it is the most notorious hyperparameter in deep learning for a reason: too large diverges, too small never arrives, and the usable range spans decades. PPO’s clipping ϵ bounds how far one update may move the *policy* away from the policy that collected the current data, a trust leash rather than a step size. Reward scaling sets the units of the training signal itself, and every gradient inherits those units.

Two knobs decide what the future is worth. The discount γ weights a reward t steps ahead by γ^t , so it is exactly a time-preference parameter, and the conventional $\gamma = 0.99$ gives an effective horizon of about $1/(1 - \gamma) = 100$ steps. GAE’s λ (generalized advantage estimation) sets how far into the observed future the advantage estimate looks, trading the bias of short lookaheads against the variance of long ones.

One knob buys experimentation. The entropy coefficient pays the policy a bonus for staying random, so that it keeps trying actions it would otherwise have abandoned. Too little and the agent commits early to a mediocre habit; too much and it never commits at all.

Three knobs meter the data per update: the unroll length (how many environment steps are collected before an update), the batch size (how many of those steps each gradient estimate averages over), and

the number of update epochs (how many passes the optimizer makes over the same collected data before discarding it).

The remaining six are the architecture: width and depth for the policy network and again for the value network, capacity in four integers, plus a spectral-normalization flag for each, a stabilizer and our standing example of a categorical knob.

None of these levers acts alone. The size of a stable gradient step depends on how much data estimated it, so batch size and learning rate tend to move together; a longer unroll changes what γ and λ can see; extra capacity changes how much restraining the entropy bonus and the clip must do. This coupling is why turning one knob at a time misleads, and why methods that model the joint surface (Chapter 4) earn their complexity. Whenever a figure needs a space we can draw, we will slice out the two knobs everyone recognizes: the learning rate, searched on a log scale from 10^{-5} to 10^{-2} , and the entropy coefficient.

2.1.2 Expensive, noisy, and without gradients

One evaluation of one configuration means training an agent for up to 150 million environment steps, hours of wall-clock on several accelerators. Everything else in this document is downstream of that number: a hundred-trial study is hundreds of GPU-hours, a line item someone has to defend, and Section 2.3 prices it out.

The resulting score is noisy. Each training run reports the mean evaluated return over a handful of episodes, and two agents differing only in random seed can land far apart. When the formal statement below takes an expectation over seeds, this spread is what the expectation is pricing.

And the score arrives without gradients in the knobs. Backpropagation supplies derivatives of the loss with respect to the network's weights, not with respect to the batch size that fed it or the discount that shaped its targets; between λ and the final return stands an entire non-differentiable training procedure. Nothing like gradient descent is available in the space we are actually searching.

Expensive, noisy, and gradient-free: those three properties are the whole reason this document exists.

A second, smaller case will carry the multi-objective thread: a Hamiltonian neural network from our physics projects, where one training run is minutes rather than hours, and where nobody can say in advance how many units of long-rollout energy drift one unit of short-horizon prediction error is worth. We will meet it properly in Chapters 7 and 9.

2.1.3 The loop, and the words for its parts

Every method in this document, from grid search to the most elaborate population scheme, is an instance of one loop, and it is worth drawing once so that later chapters can point at its parts (Figure 2.1). A *tuner* holds two responsibilities that we will keep separate throughout, because the literature develops them separately and our own design (Chapter 12) implements them behind separate interfaces. The *searcher* decides which configuration to try next. The *scheduler* decides how long each trial runs, which trials to stop early, and which paused trials to resume. The executor trains the model and streams back intermediate scores; everything observed lands in a store that the tuner (and, later, other studies) can consult. Grid search is this loop with a trivial searcher and no scheduler; Bayesian optimization upgrades the searcher (Chapter 4); successive halving upgrades the scheduler (Chapter 5); population-based training entangles the two on purpose (Chapter 6).

Three words from the figure recur on nearly every page, so we fix them now. A *trial* is a single training run with a fixed configuration (and, we will insist later, a recorded seed). The *incumbent* is the best configuration found so far. The *fidelity* of a partial evaluation is its resource level: epochs trained, environment steps consumed, chain length run. In the running example, a full-fidelity trial is 150 million

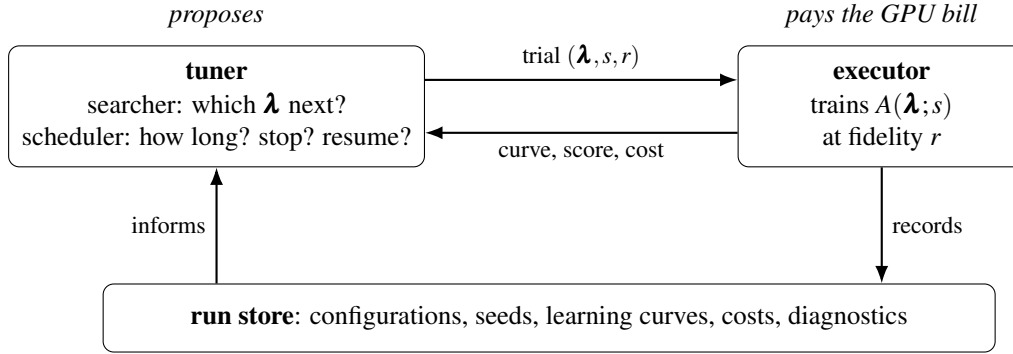


Figure 2.1: The loop every tuning method instantiates. The searcher and the scheduler are separate decisions: one picks configurations, the other allocates training budget across them. Everything observed flows into a store, which is what later lets a new study start smarter than its predecessor (Chapter 8).

steps; a one-tenth-fidelity trial of the same configuration costs one tenth of the GPU bill and tells us something, but not everything, about the full run.

To see the loop turn once, walk a young study through it. The searcher, knowing nothing yet, proposes a first configuration: learning rate 3×10^{-4} , width 256, annealing off. The scheduler grants it a tenth of full fidelity, 15 million steps. The executor trains, streaming back a learning curve, and the store records the configuration, the seed, the curve, and the bill. At the next decision point the scheduler looks at that curve and either buys the trial more steps or stops payment, and the searcher, now holding one completed record rather than none, proposes a second configuration a little less blindly. Every chapter that follows upgrades one arrow of Figure 2.1 and leaves the rest of the loop alone.

2.2 The formal problem, and where it bends

The textbook formulation is simple, and we will need it on the page because every method chapter refines some part of it. A training algorithm A maps a configuration $\lambda \in \Lambda$ and a random seed s to a trained model, and a cost function c scores that model on held-out data, so we want

$$\lambda^* \in \arg \min_{\lambda \in \Lambda} \mathbb{E}_s [c(A(\lambda; s))], \quad (2.1)$$

where the expectation runs over seeds (and, when relevant, over draws of the training data or of environment instances). Read the pieces against the running example: Λ is the fifteen-dimensional product of intervals (learning rate), integer ranges (width, batch size), and finite sets (the spectral-normalization flag), sometimes with conditional branches; A is the machine-learning training loop as a whole, here PPO and everything around it; c is the negative of the evaluated return; the expectation over s is the part beginners drop and RL practitioners regret dropping. Each evaluation of the objective means training a model, so evaluations are expensive, noisy, and yield no gradients with respect to λ . This is black-box optimization under noise, and Chapter 4 describes the machinery built for exactly this setting.

The expectation in (2.1) deserves one concrete reading before it starts doing work. Suppose, schematically, that configuration A trains to evaluated returns of 900 and 300 on two seeds, while configuration B trains to 640 and 660. A leaderboard that remembers best single runs crowns A, whose 900 is the best number anyone has seen. The expectation averages instead, 600 against 650, advantage B, and that is the right verdict for anyone who must ship one configuration and then live with whatever seed the production

run draws. Optimizing the average of a noisy objective and optimizing its luckiest observed value are different problems, and (2.1) commits us, deliberately, to the first.

Formulation (2.1) bends in four places for our workloads, and each bend picks out a branch of the literature. The four bends are worth reading slowly: they are the map of Chapters 3 through 8, and each one will reappear as the opening problem of its chapter.

2.2.1 Bend one: the seed is part of the problem

The expectation over seeds is not a nuisance term; in reinforcement learning it is often the dominant source of variation. Eimer et al. (2023) observed single seeds crashing while sibling seeds of the same configuration finished well, and found that a configuration selected on one seed can sit far below average on others. Concretely: if we tune the running example on seed 1 and the winning configuration happened to draw a lucky initialization, we have optimized for the luck, not the configuration, and the “win” evaporates on seeds 2 through 5. Their remedy, which we adopt in Chapter 15, is to treat the seed budget as part of the tuning protocol: evaluate candidates on several training seeds, and report final results on test seeds that the tuner never saw. The structure is the same train/test split that protects supervised learning from overfitting, applied one level up, to the tuner itself. The same discipline appears in the sampler world, where a single chain can look excellent by luck.

2.2.2 Bend two: the best configuration drifts during training

The best configuration is not constant in time. Learning-rate schedules exist because the loss surface a network sees at the end of training is not the one it saw at the start, and in reinforcement learning the data distribution itself shifts as the policy improves: the running example’s agent collects its own training data, so better play produces different data, which changes what the next gradient step should be. Eimer et al. (2023) formalize the distinction as algorithm configuration, which seeks one fixed λ^* , versus dynamic algorithm configuration, which seeks a policy mapping training state to configuration; landscape studies that refit the response surface at several points during RL training find the good region genuinely drifting (Mohan et al., 2023). Population-based training and its descendants (Chapter 6) live in the dynamic setting: they output a schedule of hyperparameters rather than a point, which is precisely why they appeal for RL and why they need more parallel compute than pointwise methods.

2.2.3 Bend three: several costs at once

We often care about several costs at once. A Hamiltonian network trades prediction error against violation of energy conservation; a trading model trades expected return against drawdown; any model on shared hardware trades quality against training or inference cost. When the exchange rate between objectives is unknown in advance, the honest target is the Pareto front, the set of configurations that cannot be improved in one objective without paying in another; fixing weights up front would answer by assumption the question the study exists to ask. Chapter 7 develops this properly, including the geometry of why the obvious fix (just add the objectives with weights) quietly discards solutions.

2.2.4 Bend four: cheap evidence is biased evidence

A partial evaluation is cheap and informative. A learning curve at epoch ten says a great deal about epoch one hundred; a short MCMC run says something (though, as Chapter 9 discusses, not everything) about a long one. Methods that exploit partial evaluations, called multi-fidelity methods, are the single largest lever for cutting tuning cost, routinely saving an order of magnitude of compute over full-length

evaluations (Chapter 5). The catch, previewed here because it shapes everything in that section: cheap evidence is biased evidence, and the bias has a sign. Configurations that invest early (large learning rates) look better at low fidelity than configurations that invest late (small learning rates, heavy regularization), so a scheduler that trusts early scores too much systematically kills the slow starters. Keeping that bias in view is most of the art.

2.3 What “fast and cheap” should mean

The phrase “low computation cost” hides three different quantities, and a tuner can be good at one while being poor at another. Before comparing any two methods we need to know which bill we are trying to shrink.

The first is the compute consumed by the trials themselves: the sum of GPU-hours over every training run the tuner launches, including the runs it kills early. This is where essentially all the money goes, and it is the quantity multi-fidelity methods attack. In the running example, a 100-trial study at full fidelity is hundreds of GPU-hours; every other cost in this subsection is rounding error next to it. A useful way to compare tuners is the total trial compute needed to reach a target quality with, say, probability one half, estimated over repeated tuning runs.

The second is wall-clock time to a good configuration, which depends on how well the tuner exploits parallel workers. A method that needs its evaluations one after another (classical Bayesian optimization in its sequential form) can lose to a dumber parallel method simply by leaving GPUs idle; asynchronous variants exist precisely to close that gap. The arithmetic separating the two bills is worth one line: a sixty-four-trial study at full fidelity burns the same GPU-hours on one worker as on eight, but the calendar time differs by a factor of eight, and a tuner that can only propose sequentially cannot use the other seven workers at all. [Wan et al. \(2022\)](#) report that a sequential Bayesian optimization baseline needed roughly six times more resources than their population method to reach comparable quality on Brax tasks, and more still in wall-clock terms, because the population trains in parallel while sequential optimization cannot. The distinction matters to us operationally: cluster GPU-hours and a researcher’s calendar week are different budgets, held by different people.

The third is the tuner’s own overhead: fitting whatever internal model the tuner carries, scoring candidates with it, moving checkpoints. This is usually negligible next to training. [Eimer et al. \(2023\)](#) measured minutes of optimizer overhead against days of trial compute in their RL study, with only their most elaborate population method (BG-PBT) reaching about two hours of overhead at their largest budget of 64 runs. Overhead only starts to matter when trials themselves are short, in the regime of small Hamiltonian networks or quick supervised probes, where a heavyweight distributed stack can dominate the cost of the work. This is worth remembering when we weigh Ray against lighter execution backends in Chapter 12.

2.4 What this document does not cover

Three neighboring topics are out of scope, with pointers for the curious. Gradient-based hyperparameter optimization differentiates the validation loss through the training procedure itself, either exactly on small problems or by implicit differentiation at scale; [Lorraine et al. \(2020\)](#) tune millions of regularization hyperparameters this way. It is powerful when the hyperparameters enter the training loss smoothly, but it requires intrusive access to the training loop and does not apply to discrete choices, so we treat it as a specialist tool rather than product scope.

Learned optimizers and meta-gradient methods that adapt hyperparameters inside a single run, such as the self-tuning actor-critic of [Zahavy et al. \(2020\)](#), sit inside the training algorithm rather than outside it; they compete with outer-loop tuning and we note in Chapter 9 where internal adaptation should simply be switched on and left alone.

Full neural architecture search with weight sharing is a large field of its own; Chapter 9 takes from it only the pieces that have survived independent evaluation and that fit a tuning product, and explains why we do not recommend building a differentiable-NAS capability.

Part II

Methods that choose configurations

Chapter 3

Search without a model: grid, random, quasi-random, evolutionary

The simplest tuners evaluate configurations chosen without reference to any model of the objective. They are worth understanding not because we would ship them as the product’s centerpiece but because they are the baselines every fancier method must beat, they parallelize perfectly, and one of them (random search) is quietly load-bearing inside several of the modern methods in later chapters.

3.1 Why random beats grid: an argument in one picture

Grid search evaluates the Cartesian product of a few values per axis. Its appeal is legibility: the results form a table a human can read. Its flaw is arithmetic. With a budget of n trials spread over d axes, each axis receives only $n^{1/d}$ distinct values. In the running example’s full fifteen-dimensional space, a generous budget of $n = 100$ trials gives each axis $100^{1/15} \approx 1.36$ distinct values: the “grid” cannot even afford two settings per knob. Even on the two-dimensional slice we can draw, $n = 9$ trials buys a 3×3 grid, three distinct learning rates in total.

Why is that so bad? [Bergstra and Bengio \(2012\)](#) made the now-standard observation that machine-learning objectives typically have low *effective* dimensionality: a couple of hyperparameters dominate, but which ones dominate varies by problem. Suppose, as is typical for PPO, the learning rate matters enormously and the entropy coefficient barely at all. Then what determines a trial’s quality is essentially its learning-rate coordinate alone, and the right question is not “how well do the trials cover the plane” but “how well do their *projections* cover the learning-rate axis.” Figure 3.1 shows the answer. The 3×3 grid projects onto three distinct learning rates, each tested three redundant times; nine random points project onto nine distinct learning rates. The grid wastes two thirds of its budget re-measuring values of the axis that matters while random search, which draws each coordinate afresh for every trial, places n distinct values on every axis simultaneously, without needing to know in advance which axis is the important one.

The same point can be made with elementary probability, and the number is worth carrying around. Suppose “good” configurations occupy some fraction p of the search space, say the top 5 percent. Each uniform random trial lands in the good region with probability p independently, so the chance that n trials all miss it is $(1 - p)^n$, and

$$\Pr[\text{at least one good trial}] = 1 - (1 - p)^n. \quad (3.1)$$

With $p = 0.05$, sixty trials give $1 - 0.95^{60} \approx 0.95$: sixty random trials find a top-5-percent configuration with 95 percent probability, in any number of dimensions.

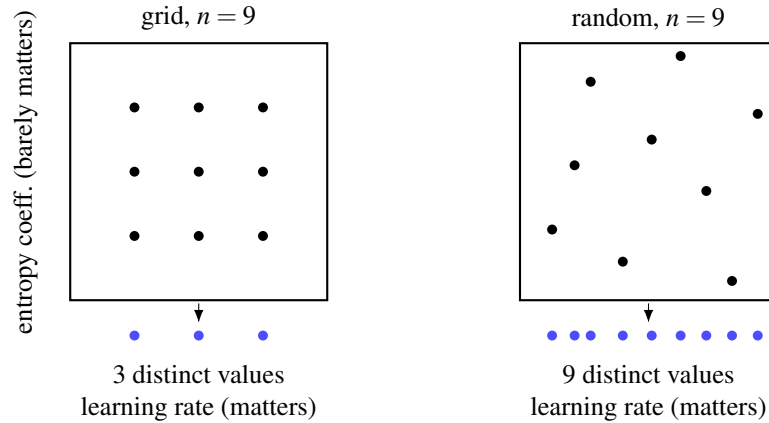


Figure 3.1: The projection argument for random search, after [Bergstra and Bengio \(2012\)](#), drawn schematically. When one axis dominates the objective, only the coverage of that axis’s projection (blue) matters. The same nine trials give the important axis three distinct values under a grid and nine under random sampling, and random search does not need to know which axis is important to collect this advantage.

The dimension-free character of (3.1) is the honest content of the phrase “random search scales”: what it cannot do is exploit structure, so when the good region is much smaller than 5 percent, or when we want the best of the good region rather than a member of it, the model-based methods of Chapter 4 earn their complexity.

The empirical side of [Bergstra and Bengio \(2012\)](#) showed random search matching or beating grid search on neural-network tuning at a fraction of the cost, and the result has aged well: fifteen years later, tuning studies in reinforcement learning still reach for random search as the honest cheap baseline ([Eimer et al., 2023](#)), and it remains the recommended floor in methodological guides ([Bischi et al., 2023](#)).

3.2 Two free upgrades

Two refinements cost nothing and should be our defaults wherever plain random sampling appears. First, sample in a transformed space: learning rates and other scale parameters should be drawn log-uniformly, since the difference between 10^{-3} and 10^{-4} matters as much as the difference between 10^{-4} and 10^{-5} . A uniform draw from $[10^{-5}, 10^{-2}]$ puts 90 percent of its samples above 10^{-3} , crowding into the top decade of the range; a log-uniform draw spends a third of its budget in each of the three decades. The transform matters more than any sophistication layered on top of it, and the same logic will reappear as input warping inside Gaussian-process surrogates (Chapter 4).

Second, replace independent draws with a low-discrepancy sequence (Sobol points), which spreads samples more evenly than chance and removes the unlucky clusters and gaps of independent sampling: independent uniform draws are perfectly happy to put two of nine samples almost on top of each other, which purchases the same information twice. The gain is modest but free, and quasi-random initialization is already the default first phase inside several Bayesian optimization packages.

3.3 Respecting the baseline

Random search has one property that deserves respect when we evaluate our own product: it is unbiased and easy to reason about. [Wan et al. \(2022\)](#) found random search over their full 15-dimensional joint space to be, in their words, a surprisingly strong baseline on several Brax environments, on par with or better than the original population-based training on four of seven tasks; that is our running example’s own search space, and the lesson transfers directly.

When a proposed method fails to beat well-configured random search at matched compute, the method, not the baseline, is on trial. That said, model-based methods do beat it when configured sensibly: the analysis of the 2020 black-box optimization challenge, run on real tuning problems, found Bayesian optimization entries clearly ahead of random search across the board ([Turner et al., 2021](#)).

Both facts fit in one sentence: random search is hard to beat when its budget is generous relative to the difficulty of the space, and clearly beatable when trials are scarce enough that each one has to be chosen with care. Scarce, expensive trials are our situation, which is why Chapter 4 exists.

3.4 Adapting the sampler without modeling the objective

Evolutionary and direct-search methods occupy a middle ground: no fitted model of the objective, but the sampling distribution adapts. CMA-ES (covariance matrix adaptation evolution strategy) maintains a multivariate Gaussian over a continuous search space and, at each generation, samples a population from it, ranks the samples by objective value, and updates the Gaussian toward the successful ones: the mean moves to a weighted average of the best samples, and the covariance is updated so that directions in which good steps were recently taken become directions the sampler explores more boldly ([Hansen, 2016](#)).

Only the *ranking* of samples enters the update, never the raw objective values, which buys invariance to any monotone rescaling of the objective, a genuinely useful robustness when losses are heavy-tailed.

The price is appetite: covariance adaptation needs generations of populations to learn the geometry, so CMA-ES is a strong choice when trials are cheap enough to afford hundreds to thousands of evaluations, when the space is purely continuous, and when noise is moderate ([Hansen, 2016](#)). In the running example’s regime of hours-long trials it is mis-matched; in the Hamiltonian-network regime of minutes-long trials it is a live option.

Differential evolution, the other family member we will need, proposes new points by recombining coordinate differences of existing population members: a mutant is formed by adding the scaled difference of two population members to a third, then crossed with a parent coordinate by coordinate. Its role for us is as the engine inside DEHB (Chapter 5), currently one of the most reliable budget-constrained tuners in reinforcement learning ([Eimer et al., 2023](#)). The practical reading: evolutionary machinery earns its place at low fidelities and inside hybrid methods, not as a standalone tuner for expensive trials, where its appetite for evaluations collides with our budgets.

3.5 The baseline’s case, argued from four angles

The chapter’s verdict, transforms and Sobol sampling on by default, random search enthroned as the mandatory baseline, grid retired to a debugging view, the evolutionary methods wrapped or embedded rather than built, is modest, which is why it is worth noticing that four separate arguments underwrite it.

From arithmetic: the two computations of this chapter are one line each and dimension-free. The exponent $n^{1/d}$ says a grid starves every axis long before budgets we can afford; the bound (3.1) says sixty random trials find the top twentieth with 95 percent confidence in any dimension. Between them they fix

both the floor’s strength and its limit, since nothing in (3.1) improves when the good region shrinks or when we want its best member rather than any member.

From evidence: the floor held up in the two most adversarial arenas this document has. In the BG-PBT authors’ own fifteen-dimensional space, random search matched or beat the original population methods on four of seven tasks (Section 6.4); in the outsiders’ challenge on real tuning problems, the model-based entries beat it across the board (Section 4.5). The two results are one fact seen from its two sides: generous budgets make the floor hard to beat, scarce trials make deliberation pay, and our studies live on the scarce side.

From operations: random search has no state, no hyperparameters worth arguing about, no failure modes, and perfect parallelism, which makes it the one component whose behavior never needs explaining; the log and Sobol upgrades are free and are inherited by every later method, since quasi-random points are also the GP searcher’s opening phase; and grid’s single virtue, a table a human can read, survives intact as the two-knob visual sweep.

From decision theory: the baseline’s real product is epistemic. Every verdict in the chapters ahead has the form “this method beats the honest cheap alternative at matched budget,” and that sentence is only checkable if the honest cheap alternative is permanently installed in the validation suite. Carrying it costs nothing; dropping it would convert every later claim from a measurement into an opinion.

The falsification here runs in both directions. The chapter’s own positive claims are the cheapest in the document to test, the same searcher with and without the log transform on a pinned task settles the transform’s value in an afternoon, and the chapter’s main export is the standing criterion the rest of the document is judged by: any searcher that cannot beat the upgraded floor at matched budget, with seeds and uncertainty reported, is not shipped.

Decisions from this chapter. For each method: what it is in one sentence, and whether it goes in the package.

- **Log-scale and Sobol sampling** (include, days of work). Drawing scale-type knobs on a log scale, and drawing the opening batch from a scrambled Sobol sequence rather than independently. An engineering default, not a theorem: the log transform applies to knobs that span decades (learning rates, coefficients, step sizes) and not to ordinals or categories, and Sobol’s even coverage is a claim about initialization batches, where it is also the GP searcher’s opening phase anyway. Two caveats worth carrying: scrambling matters, since an unscrambled sequence’s low-order points are visibly structured, and at budgets of a handful of trials the difference from independent sampling is below the noise we are measuring against.
- **Random search** (include, trivial). Not as a product feature but as the mandatory baseline in our validation suite: any searcher that cannot beat it at matched budget is not shipped. The arithmetic of (3.1) is why it is harder to beat than its reputation suggests.
- **Grid search** (exclude as a search method). Keep only as a debugging utility for two-knob visual sweeps; Figure 3.1 is the entire case against it as a tuner.
- **CMA-ES** (include by wrapping an existing implementation, low priority). Only competitive when a study can afford hundreds of cheap, continuous trials, which for us means the Hamiltonian-network regime and nothing else. Building it ourselves would buy nothing.
- **Differential evolution** (no standalone entry). It enters the package only inside DEHB in Chapter 5.

3.6 What would overturn these verdicts

The floor baseline is structural, not empirical: if the GP searcher cannot beat log-scale Sobol at matched budget on the pinned tasks of Chapter 14, it is demoted and this chapter rewritten. CMA-ES’s inclusion is conditional on the wrap remaining thin; if the dependency becomes unmaintained or the adapter requires

ongoing surgery, the verdict changes to exclude and the register records it. Grid search's debug-view limit holds unless a colleague requests it as a search method with a use case the decision boxes did not consider, at which point the box is reopened.

Chapter 4

Model-based search: Bayesian optimization and its relatives

Random search treats trial fifty exactly like trial one: it has learned nothing. When each trial costs GPU-hours, that indifference is expensive, and it pays to think hard between trials. Bayesian optimization (BO) spends that thought in two steps: fit a probabilistic model of the objective from the trials seen so far (the *surrogate*), then choose the next configuration by maximizing an *acquisition function* that scores candidate points by how useful evaluating them would be. The surrogate is cheap to evaluate, so the inner maximization can afford thousands of candidate evaluations to decide where to spend one real trial. Modern references are the book by [Garnett \(2023\)](#) and the survey by [Shahriari et al. \(2016\)](#); here we develop the machinery far enough that the design choices in our product are not acts of faith, starting from the question the surrogate must answer: after eight trials of the running example, what do we believe about the loss at a learning rate we have never tried, and how sure are we?

4.1 Gaussian process surrogates

The default surrogate is a Gaussian process (GP), and its entire content is one modeling assumption stated positively: we model the objective f as a random function such that any finite collection of values $(f(\boldsymbol{\lambda}_1), \dots, f(\boldsymbol{\lambda}_n))$ is jointly Gaussian, with prior mean zero and covariance given by a kernel, $\text{Cov}[f(\boldsymbol{\lambda}), f(\boldsymbol{\lambda}')] = k(\boldsymbol{\lambda}, \boldsymbol{\lambda}')$. The kernel is where all the prior knowledge lives: it says how similar the objective's values at two configurations should be as a function of how far apart the configurations are. The common default, and ours, is the Matérn-5/2 kernel with a lengthscale per dimension,

$$k(\boldsymbol{\lambda}, \boldsymbol{\lambda}') = \sigma_f^2 \left(1 + \sqrt{5} \rho + \frac{5}{3} \rho^2\right) e^{-\sqrt{5} \rho}, \quad \rho^2 = \sum_{j=1}^d \frac{(\lambda_j - \lambda'_j)^2}{\ell_j^2}, \quad (4.1)$$

which produces functions that are twice differentiable but not infinitely smooth, a better match for loss surfaces than the very smooth squared exponential ([Rasmussen and Williams, 2006](#)). Each lengthscale ℓ_j has a direct reading: it is the distance along axis j over which the objective's values decorrelate. A long lengthscale on the entropy-coefficient axis says that moving the entropy coefficient barely changes the loss.

It is worth pausing on why the model is nonparametric, because the obvious alternative is one every applied statistician has fit. Response surface methodology would posit a global parametric form, a quadratic in the knobs, and estimate its coefficients by least squares. That imposes one curvature on the

entire space: the model must believe the loss bends the same way near 10^{-5} as near 10^{-2} , and eight trials estimating a fifteen-dimensional quadratic are hopeless anyway (the quadratic alone has over a hundred coefficients). The kernel assumption is weaker and better matched to what we actually know: not the shape of the surface, only that configurations close together perform similarly, with “close” learned per axis from the data. The GP is what Bayesian regression becomes when the parametric form is dropped and only that smoothness belief is kept; the price is that predictions lean on nearby data rather than on a global formula, which is exactly the right epistemic posture for a surface we know nothing else about.

We assume observations are the true objective plus Gaussian noise, $y_i = f(\boldsymbol{\lambda}_i) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, which is our stand-in for seed variation; it is a modeling choice, adopted because it keeps every formula below in closed form, and Section 4.1.1 returns to what to do when the noise is visibly not Gaussian. Under these assumptions the prediction problem solves itself by pure linear algebra. The observed vector $\mathbf{y} = (y_1, \dots, y_n)$ and the unknown value $f(\boldsymbol{\lambda})$ at a new point are jointly Gaussian,

$$\begin{pmatrix} \mathbf{y} \\ f(\boldsymbol{\lambda}) \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \mathbf{K} + \sigma^2 \mathbf{I} & \mathbf{k}(\boldsymbol{\lambda}) \\ \mathbf{k}(\boldsymbol{\lambda})^\top & k(\boldsymbol{\lambda}, \boldsymbol{\lambda}) \end{pmatrix}\right), \quad \mathbf{K}_{ij} = k(\boldsymbol{\lambda}_i, \boldsymbol{\lambda}_j), \quad \mathbf{k}(\boldsymbol{\lambda})_i = k(\boldsymbol{\lambda}_i, \boldsymbol{\lambda}), \quad (4.2)$$

and the standard formula for conditioning one block of a Gaussian on another (condition the second block on the first) gives the posterior at $\boldsymbol{\lambda}$ as another Gaussian,

$$\mu_n(\boldsymbol{\lambda}) = \mathbf{k}(\boldsymbol{\lambda})^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}, \quad s_n^2(\boldsymbol{\lambda}) = k(\boldsymbol{\lambda}, \boldsymbol{\lambda}) - \mathbf{k}(\boldsymbol{\lambda})^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}(\boldsymbol{\lambda}) \quad (4.3)$$

(Rasmussen and Williams, 2006). Readers who know kriging from spatial statistics or Bayesian linear regression from econometrics are on home ground here: (4.3) is exactly that conditioning, with the kernel playing the role of the spatial covariance function.

The two formulas carry the whole story, and they are worth reading term by term. The posterior mean is a weighted combination of the observed values, with weights $\mathbf{k}(\boldsymbol{\lambda})^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1}$ set by similarity under the kernel: observations near $\boldsymbol{\lambda}$ (as measured in lengthscale units) dominate the prediction.

The posterior variance starts from the prior variance $k(\boldsymbol{\lambda}, \boldsymbol{\lambda})$ and *subtracts* a term that grows as data approaches $\boldsymbol{\lambda}$; at fixed kernel hyperparameters nothing about the observed *values* appears in it, only the observed *locations*, so the model knows where it is ignorant before it knows what the answer is. The qualifier is not decoration: once the lengthscales and the noise variance are estimated from the data by (4.4), the observed values do influence the variance, through the fitted hyperparameters rather than through the conditioning formula.

The noise term $\sigma^2 \mathbf{I}$ deserves its own sentence: with $\sigma^2 = 0$ the posterior mean passes exactly through every observation, treating each score as gospel, while $\sigma^2 > 0$ turns interpolation into smoothing, letting the mean pass *near* noisy scores instead of through them. Readers who know ridge regression will recognize the mechanism, since $(\mathbf{K} + \sigma^2 \mathbf{I})^{-1}$ applies precisely ridge’s shrinkage, with the noise variance as the regularization constant; estimating σ^2 from the data by (4.4) below is what lets the model decide for itself how much of the spread in RL returns is signal and how much is seed luck.

Figure 4.1 shows the shape this produces on the running example’s learning-rate axis: the band of uncertainty is pinched near the trials we have paid for and swells between and beyond them, which is exactly the information an acquisition function needs.

The kernel hyperparameters (the lengthscales ℓ_j , signal variance σ_f^2 , and noise σ^2) are learned by maximizing the marginal likelihood of the data,

$$\log p(\mathbf{y} | \boldsymbol{\ell}, \sigma_f, \sigma) = -\frac{1}{2} \mathbf{y}^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y} - \frac{1}{2} \log \det(\mathbf{K} + \sigma^2 \mathbf{I}) - \frac{n}{2} \log 2\pi, \quad (4.4)$$

whose two data-dependent terms pull in opposite directions: the first rewards kernels under which the observed values are unsurprising (fit), the second penalizes kernels that could explain anything

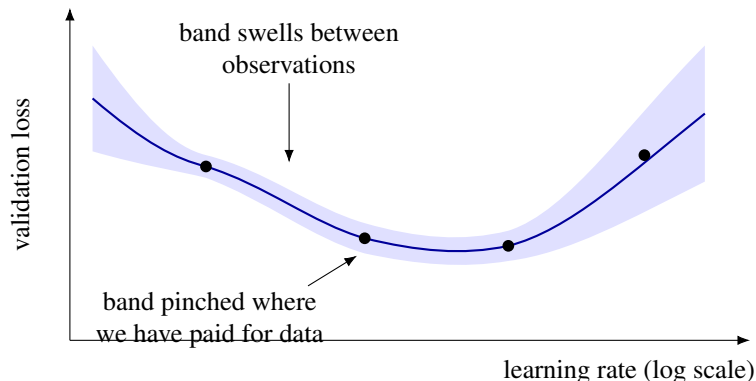


Figure 4.1: A Gaussian-process posterior on the running example’s learning-rate axis after four trials (schematic illustration of the mechanism, not measured data). The mean (4.3) passes near the observations; the shaded one-standard-deviation band shows $\mu_n \pm s_n$, which is pinched near observed points and grows away from them. The figure is drawn in the low-noise regime for legibility: with $\sigma^2 > 0$ the mean does not interpolate exactly and the variance at an observed location stays strictly positive, and only in the noiseless limit $\sigma^2 \rightarrow 0$ does the band close completely. Given the kernel hyperparameters, the posterior variance depends only on *where* we have evaluated, not on the values we saw, so the model maps its own ignorance; once those hyperparameters are themselves fitted by marginal likelihood (below), the observed values re-enter the variance indirectly through them.

(complexity), so maximizing the sum performs a built-in Occam trade-off (Rasmussen and Williams, 2006).

This is how a GP quietly performs the sensitivity analysis that grid search cannot: if the data can be explained with a very long lengthscale on some axis, the likelihood prefers it, and that axis is effectively switched off. The fitted numbers are worth reading, not just using: if the running example’s model learns a lengthscale of half a decade on the log learning-rate axis, then trials a decade apart are nearly independent evidence and the axis needs dense coverage; a fitted lengthscale of five decades says the axis is effectively flat and the search can spend its trials elsewhere.

The $n \times n$ solve in (4.3) and (4.4) costs $O(n^3)$ time, irrelevant at our trial counts (tens to low hundreds) but a real constraint for methods that model learning curves point by point, a fact that returns in Chapter 5.

4.1.1 What actually goes wrong in practice

Two practical amendments matter more than any exotic kernel choice, and both are the same move we made for random search: respect the data’s geometry. First, inputs should be warped: search axes like learning rate live on a log scale, and beyond fixed transforms it helps to learn a monotone warping of each axis (HEBO uses a Kumaraswamy distribution function per axis) so that the objective looks stationary to the kernel; a kernel with one lengthscale per axis assumes the objective wiggles at the same rate everywhere along that axis, and warping is how we make that assumption true rather than hoping it holds.

Second, outputs should be transformed toward Gaussianity by a power transform, because raw losses are often heavy-tailed and heteroscedastic: one diverged trial with a loss of 10^6 will otherwise dominate the fit of (4.4) and flatten the model’s view of the interesting region.

HEBO’s authors open their paper with hypothesis tests on 108 real tuning tasks showing that heteroscedasticity and non-stationarity are the rule, not the exception, and HEBO won the NeurIPS 2020 black-box optimization challenge with exactly these amendments plus one more hedge: instead of trusting

any single rule for choosing the next trial, HEBO scores candidates by three standard such rules at once (the acquisition functions of the next section) and picks candidates that no rival beats on all three (Cowen-Rivers et al., 2022; Turner et al., 2021). The lesson for implementers is encouraging: the wins come from respecting the data’s geometry, not from acquisition novelty.

4.2 Acquisition functions

The surrogate summarizes belief; the acquisition function turns belief into a decision, and the decision has a tension in it. Sampling where the mean μ_n is best exploits what we know; sampling where the variance s_n^2 is largest explores what we do not. Every acquisition function is a different exchange rate between the two.

4.2.1 Expected improvement, derived and then broken

Expected improvement (EI), the traditional default, scores a candidate by the expected amount by which it beats the best value seen so far. Write f^* for the incumbent value (in minimization). If we evaluate at λ and observe $f(\lambda)$, the improvement is $\max(f^* - f(\lambda), 0)$: the amount by which we beat the incumbent, or zero if we fail to. An economist will recognize the shape immediately: this is an option payoff with strike f^* , and EI is its expected value under the posterior, so Bayesian optimization spends its trials the way one prices options, paying for upside while the downside is capped at the cost of the trial.

Under the posterior (4.3), $f(\lambda) \sim \mathcal{N}(\mu_n, s_n^2)$, and the expectation of the improvement has a closed form. Substituting $z = (f - \mu_n)/s_n$ and writing $u = (f^* - \mu_n)/s_n$ for the incumbent’s position in posterior standard units,

$$\alpha_{\text{EI}}(\lambda) = \int_{-\infty}^{f^*} (f^* - f) \frac{1}{s_n} \varphi\left(\frac{f - \mu_n}{s_n}\right) df = s_n \int_{-\infty}^u (u - z) \varphi(z) dz = s_n [u \Phi(u) + \varphi(u)], \quad (4.5)$$

using $\int_{-\infty}^u \varphi(z) dz = \Phi(u)$ and $\int_{-\infty}^u z \varphi(z) dz = -\varphi(u)$, the latter because $\varphi'(z) = -z\varphi(z)$. Rearranged into its usual form,

$$\alpha_{\text{EI}}(\lambda) = (f^* - \mu_n(\lambda)) \Phi(u) + s_n(\lambda) \varphi(u), \quad u = \frac{f^* - \mu_n(\lambda)}{s_n(\lambda)}, \quad (4.6)$$

with φ and Φ the standard normal density and distribution function (Jones et al., 1998). The two terms are the two halves of the tension made explicit: the first rewards a promising mean (how much better than the incumbent we expect to be, times the probability of being better at all), the second rewards uncertainty, and the balance between them is automatic, with no exploration knob to tune.

A small worked case shows the balance operating, and it is worth doing once by hand. Suppose the incumbent loss in the running example is $f^* = 0.60$. Candidate A sits in unexplored territory: the posterior says $\mu_n = 0.62$, $s_n = 0.05$, so $u = -0.4$, and (4.6) gives $\alpha_{\text{EI}} = (-0.02)(0.345) + (0.05)(0.368) \approx 0.012$. Candidate B sits next to the incumbent: $\mu_n = 0.59$, $s_n = 0.005$, so $u = 2$, and $\alpha_{\text{EI}} = (0.01)(0.977) + (0.005)(0.054) \approx 0.010$.

Candidate A has the *worse* expected loss, yet EI ranks it slightly ahead, because its wide posterior leaves real probability of a large improvement, while candidate B can improve on the incumbent by at most a hair. That preference for informative gambles over safe pennies is the entire personality of EI, and Figure 4.2 draws where it comes from: the acquisition is the mass of the posterior lying below the incumbent line, weighted by how far below it lies.

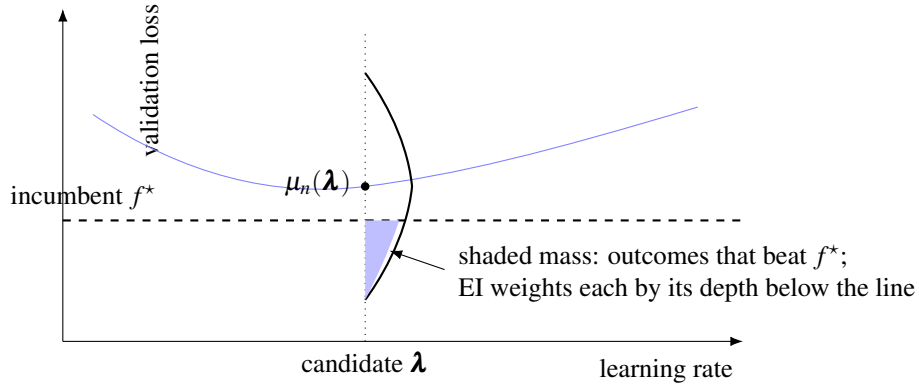


Figure 4.2: What expected improvement measures (schematic). At the candidate, the posterior over the unknown loss is the sideways bell curve. Only the shaded tail below the incumbent line counts as improvement, and (4.6) is exactly this tail’s mass weighted by its depth. When the whole bell sits far above the line, the shaded region, and with it EI and its gradient, vanishes; that observation is the doorway to the numerical failure discussed next.

Now the flaw, which is not conceptual but floating-point, and which matters because we will inherit it the day we ship a naive implementation. Once a decent incumbent exists, most of the search space has μ_n far above f^* , so u is a large negative number. For large negative u the standard tail expansion of Φ gives $\alpha_{\text{EI}} \approx s_n \varphi(u)/u^2$, and $\varphi(u)$ decays like $e^{-u^2/2}$: at $u = -6$ the acquisition is already of order $10^{-10} s_n$, and near $u \approx -38$ the density $\varphi(u)$ leaves the range of double-precision numbers entirely, so (4.6) evaluates to exactly zero, its gradient with it. A value of $u = -6$ is not exotic; it just means the posterior mean is six posterior standard deviations worse than the incumbent, which describes most of the domain once the search has found anything good. In that regime multi-restart gradient ascent on the acquisition degenerates into random search among its own starting points.

Ament et al. (2023) analyze this failure, show that the affected region grows with observations, dimensions, and constraints, and repair it by optimizing numerically careful log transforms of the whole EI family (analytic, batch, constrained, and hypervolume variants): $\log \alpha_{\text{EI}}$ is computed directly from stable asymptotic expansions rather than by taking the logarithm of an underflowed number, and smooth approximations replace the hard maximum inside the Monte Carlo versions, the sampling-based batch variants that arrive later in this section. The logarithm does not change where the maxima are; it changes whether an optimizer can find them, because gradients of order 10^{-300} become gradients of order one.

The repaired acquisitions match or beat more elaborate alternatives across standard benchmarks, which suggests much of EI’s mixed reputation was numerics all along. Our implementation should treat the LogEI variants as the default and never optimize raw EI.

4.2.2 The rest of the family, and when we need it

A different family spends its budget on information rather than improvement: predictive entropy search values a candidate by how much its observation would shrink uncertainty about the optimum’s location (Hernández-Lobato et al., 2014), max-value entropy search replaces the location (a d -dimensional object) with the optimum’s value (a scalar), making the computation far cheaper (Wang and Jegelka, 2017), and joint entropy search conditions on fantasized optimal input-output pairs to capture both at once (Hvarfner et al., 2022a).

These acquisitions are elegant and occasionally worth their cost on very expensive problems, but

given the LogEI result above we treat them as optional extras rather than defaults; [Frazier \(2018\)](#) is the standard tour of the whole acquisition zoo, including the knowledge gradient that Chapter 7 meets again in multi-objective form.

Two families beyond EI matter for us directly. Upper confidence bound (UCB) style rules score a candidate by an optimistic quantile of the posterior,

$$\alpha_{\text{UCB}}(\boldsymbol{\lambda}) = \mu_n(\boldsymbol{\lambda}) - \beta^{1/2} s_n(\boldsymbol{\lambda}) \quad (4.7)$$

for minimization: pretend every point is as good as its posterior allows at confidence level β , then pick the best pretend value. Larger β explores harder. The rule comes with regret guarantees under GP-bandit assumptions, with β growing slowly over time so that no region is starved forever ([Srinivas et al., 2010](#)), and it extends naturally to the time-varying setting that population-based methods inhabit; PB2, a population-based tuner taught in Chapter 6, uses exactly such a rule, so the formula is worth remembering.

Noise-aware variants of EI stop pretending the incumbent value f^* is known, and the reason they must is a bias economists teach under its own name. With noisy observations the “best value seen” is the *minimum of noisy draws*, and the minimum of noisy draws is optimistically biased: the trial that reported the best return in the running example is disproportionately likely to be a decent configuration that also drew a lucky seed. This is the winner’s curse, and plugging the lucky number into (4.6) as f^* transmits the curse to every later decision: the reported incumbent is better than the true incumbent, so genuine improvements look smaller than they are, the acquisition under-rewards them, and the search stalls defending a score that no configuration, including the incumbent itself, can reproduce.

The repair is to stop conditioning on the noisy draws and integrate over what the true values might be. Let $f(\boldsymbol{\lambda}_{1:n})$ be the (unknown) true objective values at the evaluated points; the noisy-EI acquisition is

$$\alpha_{\text{NEI}}(\boldsymbol{\lambda}) = \mathbb{E} \left[\max \left(\min_{i \leq n} f(\boldsymbol{\lambda}_i) - f(\boldsymbol{\lambda}), 0 \right) \right], \quad (4.8)$$

where the expectation runs over the *joint* posterior of $f(\boldsymbol{\lambda})$ and $f(\boldsymbol{\lambda}_{1:n})$: the incumbent value inside the expectation is the true best, not the lucky observed best, and the average over the posterior prices the luck out.

The integral has no closed form and is estimated by Monte Carlo, jointly sampling the true values at old and candidate points; this is the formalization of the approach used in production at Meta and now standard in BoTorch ([Letham et al., 2019](#); [Balandat et al., 2020](#)), and its log-stabilized version inherits the LogEI repair ([Ament et al., 2023](#)). Reinforcement learning returns and effective-sample-size measurements are noisy enough that a tuner without the noise-aware acquisition can waste budget refining estimates of points the GP already knows about rather than exploring genuinely new candidates, so we ship the noise-aware variant by default. A user whose observations are nearly deterministic may switch it off, but RL and sampling workloads keep it on.

Batch and asynchronous proposals fall out of the same Monte Carlo formulation, and they answer the wall-clock concern of Section 2.3. To propose q points jointly, one optimizes the expected improvement of the *best* of the q , which automatically spreads the batch out (two candidates next to each other mostly duplicate each other’s improvement, so the joint objective prefers diversity without any explicit diversity penalty).

To propose while k trials are still running, one conditions on their unknown outcomes by sampling them from the posterior (so-called fantasies) and averages the resulting decisions ([Balandat et al., 2020](#)). This is the machinery that lets model-based search keep a pool of workers busy rather than idling between sequential suggestions, and it is a hard requirement for us, not a luxury.

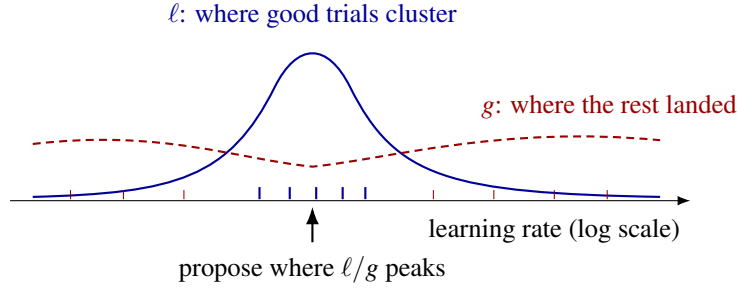


Figure 4.3: TPE’s view of the running example’s learning-rate axis (schematic). Trials are split at a score quantile; the configurations of the good fraction (solid ticks) get density ℓ , the rest (dashed ticks) get g , and the next candidate maximizes ℓ/g , which by (4.9) is the same as maximizing expected improvement under TPE’s model. The construction never represents interactions between axes; that is the price of its speed and its ease with categorical and conditional parameters.

One mechanical detail completes the picture, because “maximize the acquisition” is itself an optimization problem and its failure mode is the one LogEI fixed. In continuous spaces the standard recipe scores a few thousand quasi-random candidates (Sobol points again), keeps the best handful, and polishes each by gradient ascent, which is possible because μ_n and s_n in (4.3) are differentiable in λ and the Monte Carlo acquisitions are built to be differentiated (Balandat et al., 2020). In mixed spaces the gradient steps apply to the continuous coordinates and local moves to the discrete ones, the interleaving that Casmopolitan formalizes (Section 4.4). The polish step is exactly where a flat, underflowed acquisition strands the optimizer at its starting points, which is why the numerics of the previous subsection are an engineering requirement rather than a refinement.

4.3 Density-ratio and tree-based alternatives

The tree-structured Parzen estimator (TPE) inverts the modeling question, and the inversion is worth deriving because it explains both TPE’s strengths and its blind spot. Instead of modeling $p(y | \lambda)$ as the GP does, TPE splits the observed trials at the γ -quantile of their scores y^* (say the best 25 percent versus the rest) and fits kernel density estimates to the two groups of *configurations*: $\ell(\lambda)$ for configurations that scored well, $g(\lambda)$ for the rest (Bergstra et al., 2011). New candidates are proposed where the ratio $\ell(\lambda)/g(\lambda)$ is largest: where good trials cluster and bad trials are rare (Figure 4.3).

The connection to EI is not a heuristic. Under TPE’s model, $p(\lambda | y)$ equals $\ell(\lambda)$ when $y < y^*$ and $g(\lambda)$ otherwise, so by Bayes’ rule the marginal is $p(\lambda) = \gamma\ell(\lambda) + (1 - \gamma)g(\lambda)$, and the expected improvement below the threshold becomes

$$\alpha_{\text{EI}}(\lambda) = \int_{-\infty}^{y^*} (y^* - y) \frac{p(\lambda | y) p(y)}{p(\lambda)} dy = \frac{\ell(\lambda) \int_{-\infty}^{y^*} (y^* - y) p(y) dy}{\gamma\ell(\lambda) + (1 - \gamma)g(\lambda)} \propto \left(\gamma + (1 - \gamma) \frac{g(\lambda)}{\ell(\lambda)} \right)^{-1}, \quad (4.9)$$

since the integral in the numerator does not depend on λ . The right-hand side is a monotone increasing function of $\ell(\lambda)/g(\lambda)$: maximizing the density ratio *is* maximizing EI under this model (Bergstra et al., 2011).

What the model gives up is visible in the same derivation: the densities are typically factorized per coordinate, so the ratio cannot represent “this learning rate is good only when the batch size is large,” an interaction a GP kernel captures without effort.

Three properties make TPE the workhorse of Optuna, the open-source tuning library where sampler development is most active (its record is audited in Chapter 10). It handles conditional (tree-structured) spaces natively, because densities are fit per branch of the configuration tree; it tolerates categorical and integer parameters without kernel gymnastics, since a kernel density over a finite set is just a weighted histogram; and its cost grows gently with both dimension and trial count, with none of the $O(n^3)$ algebra of (4.3). Its weakness is the flip side of the factorized densities above: blindness to interactions, and in low dimensions with smooth objectives a GP squeezes more out of each trial. The recent tutorial by Watanabe (2023) documents the many small design choices (bandwidths, splitting quantile, prior weight) that separate a good TPE from a poor one; since Optuna implements these carefully, wrapping Optuna’s sampler is more sensible than reimplementing TPE ourselves.

SMAC replaces the GP with a random forest surrogate, which natively digests mixed and conditional spaces and yields uncertainty from the spread across trees (Lindauer et al., 2022). Forest surrogates are less sample-efficient than GPs on smooth low-dimensional problems but degrade gracefully on the awkward spaces where GPs struggle, and SMAC3 pairs the surrogate with multi-fidelity racing (Chapter 5). We cite it here both as an algorithmic reference point and because it is the tuner that Wan et al. (2022) used as their sequential BO baseline.

4.4 High-dimensional and mixed spaces

Vanilla GP BO has a reputation for failing beyond ten or twenty dimensions, which would be disqualifying for the running example’s fifteen. The reputation is only half deserved, and the recent literature sorts out which half; the sorting matters to us because it changes what we build first.

The mechanism behind the failures is worth stating before the fixes. In high dimension, with standard lengthscale priors, the fitted GP tends toward one of two degenerate postures: lengthscales so short that every observation is an island (posterior variance is high everywhere, and the acquisition chases pure exploration into the corners), or a fit that contorts itself around noise. Each published fix restricts the model in a different way, and each restriction is a bet about the objective.

TuRBO bets that good local models beat bad global ones. It abandons the global search problem and maintains local trust regions: boxes around good points, inside each of which an independent BO runs with its own GP (Figure 4.4). A region’s side length grows after runs of successes (the local model is trustworthy, let it see further) and shrinks after runs of failures (the local model is misleading, tighten the box); a region that shrinks below a floor is abandoned and restarted elsewhere, and a bandit allocates trials across regions (Eriksson et al., 2019). This concentrates samples where the model has earned trust and scales to hundreds of dimensions; it is the backbone that BG-PBT adapts for population training (Chapter 6), which is the specific reason this document cares about its details.

SAASBO (sparse axis-aligned subspace BO) bets that only a few axes matter. It keeps the global view but places sparsity-inducing half-Cauchy priors on the kernel’s inverse squared lengthscales, so that every axis starts effectively switched off (infinite lengthscale) and only axes with evidence of mattering wake up; inference is fully Bayesian over these priors, which buys striking sample efficiency in hundreds of nominal dimensions and costs enough per iteration to confine the method to small evaluation budgets (Eriksson and Jankowiak, 2021). If the effective-dimension picture behind Figure 3.1 is right for a problem, SAASBO is that picture taken seriously as a prior. A related line grows a random embedding of the space as evidence accumulates rather than committing to one up front, with a mixed-space successor for combinatorial problems (Papenmeier et al., 2022, 2023).

Then the twist that reorders the whole subsection. Hvarfner et al. (2024) showed that much of the classical failure was a prior mistake rather than a fundamental one: simply scaling the GP’s lengthscale

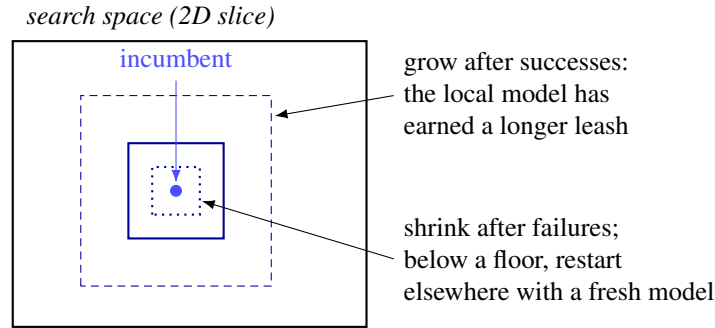


Figure 4.4: TuRBO’s trust region around the incumbent (schematic). All modeling and acquisition happen inside the current box; success and failure counters resize it, and collapse triggers a restart. The bet is that a GP asked only local questions stays reliable in dimensions where a global GP degenerates, and the same box-resizing logic reappears inside BG-PBT in Chapter 6.

prior with the dimension, so that the model’s complexity stays controlled as axes are added, lets plain BO with standard acquisitions outperform the specialized high-dimensional methods on common benchmarks, with no structural assumption about the objective. The intuition connects straight back to (4.4): the complexity penalty a kernel pays scales with how much the function can wiggle across the whole space, which grows with dimension unless lengthscales grow too, so the fix is a prior that expects longer lengthscales in higher dimension. The practical ordering for us follows: fix the priors first (cheap, one line of configuration), reach for trust regions when the space is genuinely large or the budget is generous, and reserve SAASBO for sample-starved, high-stakes searches.

Mixed spaces need kernels that understand their coordinates. Continuous axes take standard kernels; categorical axes take overlap or Hamming-style kernels (two configurations are similar to the extent they agree, with no notion of “between”); ordinal axes (integers with order, like batch size or network width) should be treated as coarse continuous axes rather than unordered categories, because width 256 really is between width 128 and width 512, and throwing that ordering away discards exactly the structure a kernel can use.

Casmopolitan combines these with trust regions and interleaved acquisition optimization, alternating local search over the discrete coordinates with gradient steps over the continuous ones (Wan et al., 2021), and the BG-PBT paper reports that explicitly respecting ordinal structure gives measurable gains in the RL setting (Wan et al., 2022).

An alternative with BoTorch support is probabilistic reparameterization, which relaxes discrete choices into distributions over them and optimizes the relaxation with gradients (Daulton et al., 2022b). Our product needs exactly one good mixed-space GP searcher; Casmopolitan-style kernels inside a trust region, reusing BoTorch primitives, is the path of least resistance and doubles as the core of a future BG-PBT implementation.

4.5 The evidence, and how it was produced

Everything so far argues that model-based search *should* work. A skeptical reader is entitled to ask what would count as evidence that it does, and to ask it before accepting any verdict, so this section describes the experiments themselves, not just their conclusions.

The evidence problem in this field is selection bias, and it is worth naming before weighing any result. A paper introducing a method chooses its own benchmarks, its own baselines, and its own tuning of those

baselines, and only successful comparisons reach print; a reader cannot tell how much of a reported gain is method and how much is choice of arena. The strongest evidence therefore comes from settings the method’s authors did not control. Two such settings anchor this chapter’s verdicts.

The first is the NeurIPS 2020 black-box optimization challenge, a public competition in which entrant tuners were run on real machine-learning hyperparameter tuning problems at matched evaluation budgets, with the organizers rather than the entrants choosing the tasks. The organizers’ published analysis found the Bayesian optimization entries clearly ahead of random search across the board (Turner et al., 2021): an adversarial arena, fixed budgets, and the baseline of Chapter 3 still cleanly beaten. The winning entry, HEBO, is instructive for *why* it won. Its authors’ paper opens not with a new acquisition function but with hypothesis tests on 108 real tuning tasks showing that heteroscedastic, non-stationary objectives are the rule rather than the exception, and the winning design is essentially the standard GP machinery of this chapter plus the input and output transformations that make such objectives look well behaved to it (Cowen-Rivers et al., 2022). The reading we take: the competition validates the framework, and the margin came from respecting the data’s geometry, which is an engineering discipline we can copy, not a secret.

The second setting is the controlled RL tuning study of Eimer et al. (2023). Three algorithms (PPO, SAC, DQN) across eleven environments; tuners compared at matched budgets of ten to sixty-four full training runs, which is the regime our routine studies actually inhabit; candidates evaluated on several tuning seeds with final results reported on disjoint test seeds the tuner never saw; and the tuners’ own overhead measured. The finding that matters here: sweeping hyperparameters across eleven environments, the fitted response surfaces looked smooth and well behaved, with the difficulty of RL tuning coming overwhelmingly from seed noise rather than from a pathological objective. Smooth surfaces are exactly what the kernel assumption (4.1) posits, and severe observation noise is exactly what the noisy-EI machinery (4.8) prices in, so the one controlled study of our hardest workload lands on both of this chapter’s design choices. (The study’s full design and its budget-regime findings occupy Chapter 6, where the population line’s evidence lives.)

Two honest boundaries on this evidence. It does not rank GP against TPE universally: which surrogate wins depends on the space’s shape and the budget, which is why the package carries both and why the comparison that matters will be run on our own pinned tasks. And the LogEI results, showing the repaired classic acquisitions matching or beating the more elaborate ones across standard benchmarks (Ament et al., 2023), are the authors’ own experiments; we treat them as strong because the claim is negative and numerical (a bug explained away a literature’s worth of differences) and because the mechanism, the underflow arithmetic of Section 4.2, can be checked from first principles on one page, which we did.

4.6 The recommendation, argued from four angles

The chapter’s verdict is that a carefully configured GP searcher with log-stabilized, noise-aware acquisitions is the core of the package, with TPE wrapped alongside for conditional spaces. Four independent lines of argument arrive at that verdict, and a reader should demand that they agree, because each could fail separately.

From decision theory: when a trial costs GPU-hours, the value of deliberation between trials is high, and (4.6) is literally the option value of the next experiment under current beliefs. Any method that ignores posterior uncertainty is discarding the one quantity that says where deliberation pays.

From statistics: the machinery is not exotic. The surrogate is kriging, the noise handling is ridge shrinkage with an estimated regularizer, the winner’s-curse correction is the same bias argument auction theory makes. A reader who trusts those tools separately is most of the way to trusting their composition;

nothing in the chapter asks for a new inferential principle.

From evidence: an author-independent competition on real tuning tasks put the framework clearly ahead of the honest baseline, the winner’s margin traced to checkable preprocessing rather than proprietary insight, and the one controlled study of our own hardest workload found exactly the surface smoothness and noise structure this design assumes.

From operations: every component the chapter recommends exists, maintained, in BoTorch and Optuna (Chapter 10), so the build cost is composition rather than research; batch and asynchronous proposals keep a cluster busy; and the known numerical failure mode has a published, adopted repair.

And the check on all four: the verdict is falsifiable in our own validation suite. The GP searcher must beat Sobol sampling at matched budgets on the pinned tasks of Chapter 15, with the comparison repeated across seeds and reported with uncertainty. If it cannot, the verdict, not the suite, gives way; that is the standard we would demand of anyone else’s recommendation, so the document applies it to its own.

Decisions from this chapter. The choice among surrogates comes down to four manager-checkable questions: how few trials before the model helps, what kinds of knobs it digests, how it copes with noise, and what it costs us to ship.

- **GP searcher with qLogEI on BoTorch** (include; the core build, weeks). Extracts the most from the fewest trials on mostly continuous spaces of up to a few dozen dimensions, handles noise honestly via the noisy-EI machinery, proposes batches so workers never idle. This is the searcher for expensive trials, which is to say for the studies where the tuner’s quality is actually worth money. The challenge evidence is on its side: model-based entries beat random search across the board, and the winner was a GP method (Turner et al., 2021; Cowen-Rivers et al., 2022).
- **TPE by wrapping Optuna** (include; days). Useful from a handful of trials, natively handles conditional and categorical spaces that GPs find awkward, and costs us almost nothing because we wrap the best-maintained implementation rather than writing one. Its known blind spot, interactions between knobs, is the price (equation 4.9); for smooth low-dimensional problems the GP squeezes more from each trial.
- **Raw EI, anywhere** (exclude). The underflow analysis is not a nicety: past a modest incumbent quality the naive formula returns exactly zero over most of the space and the optimizer stalls (Ament et al., 2023). Log variants only.
- **Entropy-search acquisitions** (exclude). Elegant, costly to implement well, and the controlled comparisons show the repaired EI family matching or beating them (Ament et al., 2023). No capability we need survives their exclusion.
- **Dimension-scaled lengthscale priors** (include; one configuration line). The cheapest high-dimensional fix on record (Hvarfner et al., 2024). Always on.
- **Trust-region searcher with mixed-space kernels** (include; weeks). Justified twice over: it is the credible high-dimensional searcher, and it is a prerequisite we would build anyway for BG-PBT in Chapter 6.
- **SAASBO** (defer). Striking sample efficiency when trials are precious and axes are few-but-unknown, at a per-iteration compute cost that only suits small, high-stakes studies. Revisit when such a study appears.
- **SMAC** (do not adopt as a dependency). We use it as a correctness oracle in validation; its random-forest surrogate niche (huge conditional spaces) is not where our workloads live.

4.7 What would overturn these verdicts

The GP searcher’s tier-0 status is tied to V04: if it cannot beat the Sobol floor on our pinned tasks, the complexity is not justified and it is demoted to an option. TPE’s wrapping strategy holds as long as Optuna’s maintenance remains current (audited in Chapter 10); if Optuna’s TPE implementation stagnates

or breaks backward compatibility repeatedly, the verdict changes to a native build. The trust-region searcher's shared role (high-dimensional search and population explore) is load-bearing; if the population line is demoted by V08, the trust-region design is re-argued on single-objective merits alone. Raw EI and entropy-search exclusions are reopened if a published study on RL or sampler workloads shows either beating qLogEI at matched budgets, at which point the evidence section is updated and the box reconsidered.

Part III

Methods that manage the budget

Chapter 5

Spending less per trial: multi-fidelity methods

Everything in Chapter 4 treats a trial as atomic: pick a configuration, pay full price, observe the final score. But training produces a stream of intermediate evidence, and most bad configurations announce themselves early. In the running example, a learning rate of 10^{-2} that sends PPO’s policy loss to infinity does so within the first million environment steps; paying for the remaining 149 million to confirm the diagnosis is charity. Methods that act on partial evaluations are called multi-fidelity methods, and they are the largest single source of savings available to us.

The catch, which every method in this chapter manages in its own way, is that low-fidelity performance is only a proxy, and the proxy is biased in a known direction: rankings at epoch five need not match rankings at epoch fifty, and the disagreement is not symmetric noise. Small learning rates and heavy regularization look bad early and win late; aggressive settings sprint and stall. Figure 5.1 draws the failure we must design against: at the first checkpoint the eventual winner is in last place, so a scheduler that culls hard on early evidence systematically kills slow starters. Whether early rankings can be trusted is an empirical property of each workload, worth a pilot measurement (a rank correlation between fidelities on a handful of configurations) before committing to an aggressive schedule.

5.1 Successive halving: the arithmetic of courage

Successive halving spends a budget on many cheap looks and a few expensive ones (Jamieson and Talwalkar, 2016). Fix a reduction factor η (usually 3). Give every starter a small budget r ; keep the best $1/\eta$ fraction and give each survivor η times more budget; repeat. The rounds of this ladder are called *rungs*, and the race ends when one rung remains. Rung sizes shrink by η while per-trial budgets grow by η , so every rung costs about the same, and there are about $\log_\eta n$ rungs of them.

The numbers deserve one concrete pass, because the savings are the whole argument. Take the running example with $n = 81$ starting configurations, $\eta = 3$, and an initial budget of $r = 1$ unit, where 81 units is a full training run (so one unit is just under two million environment steps). The rungs then look like this: 81 trials at 1 unit each (cost 81), the best 27 continue to 3 units (cost 81), the best 9 continue to 9 units (cost 81), the best 3 continue to 27 units (cost 81), and the single survivor runs to the full 81 units (cost 81). Total: 405 units, about five full-length runs, to triage 81 configurations. Evaluating all 81 at full length would have cost $81 \times 81 = 6561$ units, a factor of sixteen more; even the standard sixteen-full-runs budget of the RL study in Chapter 9 could not have screened a fifth as many candidates at full fidelity. Figure 5.2 draws the resulting shape: the budget is a rectangle of constant area per rung, spent mostly on

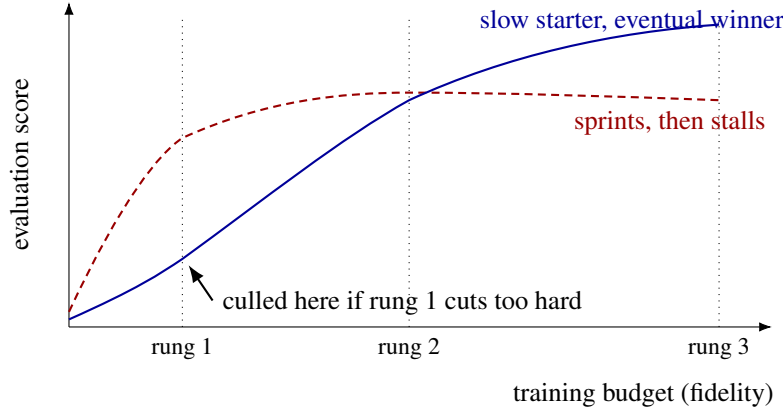


Figure 5.1: The multi-fidelity hazard (schematic). At rung 1 (the first of the budget checkpoints that this chapter’s ladders call rungs) the aggressive configuration leads and the eventual winner trails; their curves cross between rungs 1 and 2. Every method in this chapter is a different answer to the question “how hard may we cut on early evidence, given that curves cross?”, and the honest answer starts with measuring how often they cross on the workload at hand.

configurations that survived at least one cut.

The rub is the choice of the starting budget r : too small and slow starters die unfairly (the hazard of Figure 5.1 at its sharpest), too large and the savings evaporate. Hyperband declines to choose. It runs several successive-halving brackets, from aggressive (many configurations, tiny initial budget) to conservative (few configurations, nearly full initial budget), giving each bracket the same total budget, and provably loses only a logarithmic factor to the best bracket in hindsight (Li et al., 2018). Continuing the numbers above ($\eta = 3$, full budget 81 units), the five brackets start 81 trials at 1 unit, 34 at 3, 15 at 9, 8 at 27, and 5 at the full 81, each bracket then halving as before; each costs about 405 units, so the whole portfolio costs about 2025 units, twenty-five full-run equivalents to hedge every plausible answer to “how early is too early to judge.” On a workload where early rankings are honest, the aggressive brackets deliver; on one where curves cross late, the conservative brackets are the insurance policy that was bought in advance.

5.2 The asynchronous turn

Both methods as stated are synchronous: a rung must fully finish before promotions happen, so a single straggler idles the fleet. With hundreds of workers this is not a corner case; it is the steady state.

ASHA, the asynchronous successive halving algorithm, removes the barrier (Li et al., 2020). Whenever a worker frees up, look at the topmost rung that has a promotable trial: promote any configuration that is in the top $1/\eta$ of the results recorded so far at its rung; otherwise start a fresh configuration at the bottom rung. Concretely: with $\eta = 3$, the moment the fourth result lands on a rung, its best member is in the top third of four and can be promoted immediately, without waiting for trials five through eighty-one.

Early promotions are sometimes wrong (judged against a rung that was not full yet), and ASHA accepts this deliberately: the authors’ experiments show the aggressiveness costs little in final quality while keeping hundreds of workers busy, and ASHA has become the standard early-stopping scheduler in distributed tuners, including Ray Tune’s implementation. The design lesson generalizes and will return in Chapter 6: at scale, a slightly worse decision now beats a slightly better decision after a barrier.

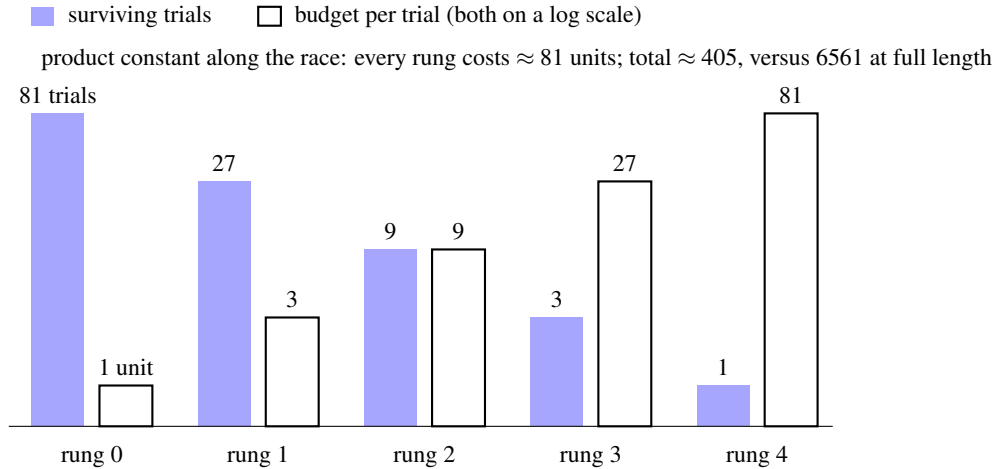


Figure 5.2: One successive-halving bracket on the running example ($n = 81$, $\eta = 3$). At each rung the surviving trials shrink by the factor the per-trial budget grows by, so the two staircases mirror each other and every rung costs the same 81 units: the whole screen of 81 configurations costs about five full-length runs. The scheme’s entire risk is concentrated in the cuts between rungs, where Figure 5.1’s crossing curves get judged early.

Two refinements are worth knowing. PASHA grows the maximum fidelity progressively and stops growing it once the ranking of survivors stabilizes, saving the top-rung budget when the decision is already clear (Bohdal et al., 2023). The median stopping rule, older and model-free, simply kills any trial whose running average falls below the median of completed trials at the same step (Golovin et al., 2017); it composes with any searcher and is the cheapest insurance policy we can offer users.

5.3 Putting the model back: BOHB, DEHB, and racing

Bandit schedulers decide when to stop but sample new configurations blindly: the bottom rung of every bracket above was filled by random search. After Chapter 4, the obvious repair is to let a model choose what enters the race. BOHB, Bayesian optimization plus Hyperband as its name says, runs Hyperband’s brackets but draws new configurations from a TPE fit at the highest fidelity that has enough observations, so the racing machinery keeps its budget arithmetic while the entrants improve over time (Falkner et al., 2018). DEHB replaces the sampler with differential evolution (Chapter 3): a subpopulation evolves at each rung, and information flows across fidelities through the evolutionary operations themselves, survivors of cheap rungs seeding the recombination that proposes candidates for expensive ones, rather than through a fitted density model (Awad et al., 2021).

DEHB deserves particular attention from us: in the controlled comparison of Eimer et al. (2023) across PPO and two other standard RL algorithms (SAC and DQN) on eleven environments, DEHB was the most consistent tuner at budgets of ten to sixty-four full training runs, matching or beating hand tuning, random search, and the population methods, all at negligible optimizer overhead. That is precisely the budget regime our routine RL studies live in, and Chapter 9 builds the recommendation on it. SMAC3 reaches a similar design point from the BO side, pairing its random forest surrogate with successive-halving style racing of incumbents (Lindauer et al., 2022).

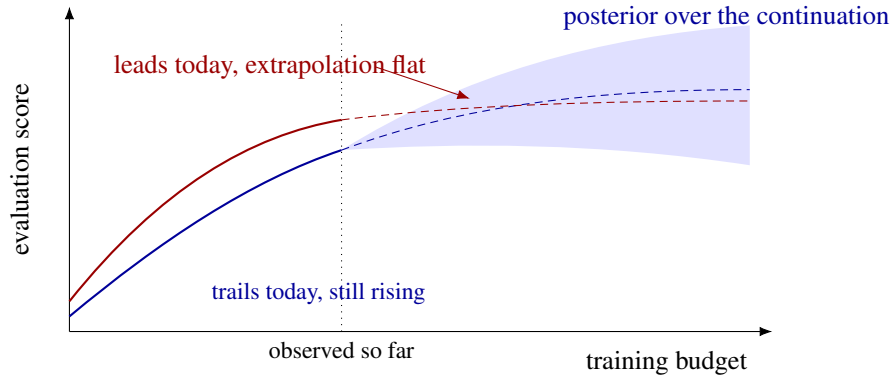


Figure 5.3: The freeze-thaw decision (schematic). Both trials are paused at the dotted line. The red trial leads today but its extrapolation is flat; the blue trial trails with a rising curve and a wide posterior over where it ends up. A learning-curve method allocates the next budget slice by comparing these posteriors, information that rung-based promotion never sees.

5.4 Modeling the learning curve itself

Rung-based methods compare trials only at matching fidelities and throw away the shape of the curve between rungs; in Figure 5.1, the information that the blue curve was still rising steeply at rung 1 while the red curve had flattened is exactly what the rung comparison discards. Learning-curve methods keep it. They model the whole trajectory and ask a sharper question: given everything observed so far, which paused trial (or which new one) should receive the next slice of budget? Figure 5.3 shows the decision problem: from a partial curve, extrapolate a distribution over its continuation, and weigh thawing the promising-but-uncertain run against extending the solid-but-plateauing one.

The idea goes back to freeze-thaw Bayesian optimization (Swersky et al., 2014), which kept a basket of paused runs (freeze a run to pause it, thaw it to resume) and used a GP with an exponential-decay kernel over training curves to decide which to thaw. Every modern instance keeps that decision structure and changes one component, the curve model, and the choice matters more than it sounds, because the curve model is refit inside the decision loop: whatever it costs is paid again at every allocation decision.

DyHPO makes the curve model a deep-kernel GP over the triple of configuration, budget, and partial curve, and greedily advances whichever candidate has the best expected improvement one step ahead (Wistuba et al., 2022). The price follows from Chapter 4: GP fitting is cubic in the number of observations, and a curve observed at every epoch multiplies observations quickly, so the surrogate’s bill grows with exactly the data that makes it useful.

DPL swaps the GP for an ensemble of power-law extrapolations, motivated by the empirical regularity that neural training curves follow approximate power laws; the spread of the ensemble supplies the uncertainty that the acquisition needs (Kadra et al., 2023). The model is lighter than a deep-kernel GP, but the ensemble is still refit inside the loop.

ifBO (in-context freeze-thaw BO) removes the in-loop fitting entirely. Its curve model is a prior-fitted network: a transformer (the neural architecture behind modern language models) pre-trained once on synthetic learning curves, which then performs Bayesian curve extrapolation in-context, in a single forward pass, one to two orders of magnitude faster than fitting DyHPO’s deep GP or DPL’s ensembles inside the loop. The acquisition paired with it randomizes its lookahead horizon and improvement threshold (Rakotoarison et al., 2024). Chapter 8 explains what a prior-fitted network is and why “pre-trained once” is the phrase that changes the economics.

The evidence and the price, briefly, before the evidence section weighs both. On the multi-fidelity benchmarks used in those papers, the learning-curve methods dominate rung-based baselines when curves are informative. The price is operational: the tuner must be able to pause, checkpoint, and resume trials at fine granularity, which is exactly the machinery that Ray Tune trials and RLlib checkpoints already expose.

The honest summary for our roadmap: ASHA plus a good searcher is the sturdy default that captures most of the savings; DEHB is the evidence-backed choice at small RL budgets; learning-curve methods are the frontier worth one focused implementation (Chapter 15) once the pause and resume plumbing is solid, with ifBO the most attractive because the surrogate arrives pre-trained rather than being refit inside the loop.

5.5 The evidence, and how much of it is arithmetic

Those verdicts lean on four different grades of support, and the grades are worth keeping separate because each properly carries a different weight of decision: arithmetic, which requires no trust in anyone; theory, which is genuine but conditional; the method authors’ own benchmarks; and a single independent study on live training runs. Section 4.5 argued that evidence should be weighed by how it was produced. Applied here, that principle sorts the chapter’s claims cleanly.

The arithmetic came first, and it is the part that cannot be argued with. The 405-versus-6561 comparison under Figure 5.2 is not an experimental finding; given the rung geometry, the factor of sixteen follows by multiplication, and no benchmark can add to it or take it away. What multiplication cannot supply is a promise that the cheap rungs *select* well, and the theory is candid about the same limit. The successive-halving guarantee is of a deliberately modest kind, bounding identification error under assumptions on how fast each training curve settles toward its final value, and Hyperband’s theorem promises performance within a logarithmic factor of the best bracket in hindsight, not good performance outright (Jamieson and Talwalkar, 2016; Li et al., 2018). Neither theorem knows whether the curves of our workloads settle quickly enough for early rankings to be trusted. That is a property of the workload, not of the algorithm, which is why the rank-correlation pilot suggested at the chapter’s opening is the one empirical measurement the whole apparatus genuinely needs.

ASHA’s evidence is of another kind again, and the paper it comes from announces what kind: this is a systems paper, written around the production tuning service its authors were building, and its central claims are operational (Li et al., 2020).

The distributed experiments compare ASHA against synchronous successive halving, BOHB, and PBT (the population method Chapter 6 teaches) with twenty-five workers on two image-classification tuning tasks (CIFAR-10), five runs of each method, and the finding is the one the mechanism predicts: on the task where training times vary across configurations, the synchronous methods idle at their barriers and ASHA pulls cleanly ahead, while on the task where they do not, the schedulers roughly agree, ASHA reaching a good configuration about 1.5 times sooner and BOHB ending slightly ahead on final quality. On two architecture search spaces with sixteen workers, ASHA reached a below-ten-percent test error roughly twice as fast as synchronous halving and BOHB and ended slightly ahead on average. The demonstration that justifies the word “massive” ran five hundred workers for weeks on a recurrent language-model space.

All of this is the authors’ own arena, and the selection-bias caution of Section 4.5 applies; two things temper it here. The claim under test is throughput rather than statistical cleverness, so a skeptic can audit the mechanism (promote now against the top third observed so far, never wait) instead of trusting a tuned baseline. And the operational half of the claim has been replicated the way operational claims are: ASHA is the standard early-stopping scheduler in the distributed tuners we audited (Chapter 10), adopted by

teams with no stake in the paper.

DEHB’s support arrives in two installments, and the order matters. The authors’ own paper is a breadth exercise: toy functions, surrogate and tabular benchmarks, and thirteen tabular architecture-search benchmarks, against random search, TPE, GP-based BO, SMAC, Hyperband, and BOHB, with headline speedups of up to a factor of a thousand over random search and thirty-two over BOHB (Awad et al., 2021; Falkner et al., 2018).

Two features of that design are worth noticing before the numbers settle in. Nearly all of the benchmarks are tabular or surrogate, meaning the training runs were paid for once, offline, and the tuners replay or interpolate them, with optimizer overhead excluded from the cost axis by construction (a choice that, if anything, understates DEHB’s cheapness relative to the model-based methods). And the paper reports its own boundary case plainly: on NAS-Bench-101, where cheap-fidelity and full-fidelity results correlate weakly, DEHB behaves like random search early on, which is Figure 5.1’s hazard surfacing in the method’s own results.

The second installment is the independent one, and it is what this chapter’s verdict actually stands on: the budget-matched, seed-split RL study of Eimer et al. (2023), run on live training at ten to sixty-four full runs, whose design is laid out in Section 6.4, found DEHB the most consistent tuner in exactly the regime our routine studies inhabit. The authors’ benchmarks nominate the method; the independent live study is what promotes it.

The learning-curve methods sit at the far end of the scale. Their support is the authors’ own, on multi-fidelity benchmark suites built for the purpose, with no independent budget-matched test on live RL training at our budgets yet (Wistuba et al., 2022; Kadra et al., 2023; Rakotoarison et al., 2024). That is not a criticism; it is where every young line of methods starts, and it is precisely the evidentiary grade at which a sensible product buys an option rather than installs a default. The chapter’s decisions read directly off this scale. The always-on default rests on arithmetic, a measurable hazard, and an operational record; the small-budget RL pick rests on one independent controlled study; the frontier bet rests on promising author-run comparisons, and is gated accordingly.

5.6 The default, argued from four angles

The chapter’s verdict is that early stopping is not optional equipment: ASHA with the median rule as the always-on default, DEHB where a study is a few dozen full runs of RL, the learning-curve line as a gated bet. As in Chapter 4, four independent lines of argument arrive at the same place, and a reader should demand that they agree.

From arithmetic: the savings factor is an identity, not an estimate. Sixteenfold on the worked bracket, an order of magnitude in general, and the only empirical quantity the identity leans on, whether early rankings predict late ones, is measurable on the workload at hand for the price of a pilot. A default whose upside is multiplication and whose risk is one measurable number stands on firmer ground than anything else in this document.

From evidence: the sources agree at the boundary. The theory conditions its guarantees on curves settling; DEHB’s own benchmarks degrade toward random search exactly where fidelities decorrelate; the one independent live-RL study finds the rung family most consistent at exactly the budgets where the arithmetic bites hardest. Three kinds of source, none sharing authors, point at the same governing variable, the trustworthiness of cheap evaluations, and that is the variable our pilot measures and our defaults respect.

From operations: ASHA and the median rule ask one thing of training code, that it report a metric as it goes, and nothing at all of the searcher, so they compose with every method of Chapters 3 and 4

rather than competing with them. The asynchronous design is what lets a schedule survive contact with a shared cluster, where stragglers are the steady state and every barrier converts one slow trial into fleet-wide idleness. The learning-curve methods’ operational price, pause and resume at fine granularity, is real, which is why their entry is gated on plumbing the roadmap builds anyway for the population methods of Chapter 6.

From decision theory: the two ways to be wrong cost differently. Stopping too eagerly risks killing a slow starter, a bounded loss that the rank-correlation pilot prices in advance and a conservative bracket insures against; stopping too late, or not at all, pays the full factor of sixteen with certainty. When one error is bounded, diagnosable, and insurable and the other is certain and large, the default must accept the first risk, which is exactly what an aggressive-by-default, insurable-by-option scheduler does.

And the criterion that could overturn the verdict, stated in advance: the validation suite of Chapter 15 races ASHA plus a searcher against the same searcher at full fidelity on the pinned tasks at matched total budgets, repeated across seeds; it logs between-rung rank correlations on every study it touches, so the crossing-curves hazard becomes a measured quantity of our workloads rather than a literature anecdote; and DEHB is re-raced against BO plus ASHA at sixteen to sixty-four runs on the pinned RL task. If early stopping costs final quality on our workloads, the default flips to conservative brackets, and the suite, not this chapter, has the last word.

Decisions from this chapter. Multi-fidelity is where the money is saved, so every verdict here is really a claim about compute bills.

- **ASHA plus the median stopping rule** (include; days; tier 0). The workhorse pair. Between them they capture the bulk of the order-of-magnitude savings this chapter promises, compose with any searcher, and ask almost nothing of the training code beyond reporting a metric as it goes. If the package shipped only one scheduler, it would be this one.
- **Hyperband as a distinct feature** (exclude). Its insurance-portfolio idea survives inside our defaults (run a conservative bracket when curve crossing is suspected), but as a product surface ASHA superseded it and no user-facing knob is worth the confusion.
- **BOHB** (exclude). The idea, model-guided entrants into the race, lives on in our BO-plus-ASHA composition; the reference implementation is formally unmaintained (Chapter 10), and adopting it would import a dead dependency for no capability we lack.
- **DEHB, implemented against our interfaces** (include; weeks; tier 3). The one scheduler-searcher hybrid with controlled RL evidence behind it at exactly our routine budgets: most consistent tuner at ten to sixty-four full runs across PPO, SAC, and DQN (Eimer et al., 2023). The upstream package is frozen, so this is a reimplement with the original as reference, not a wrap.
- **PASHA** (include as an ASHA option flag; small). Stops paying for top-rung training once the ranking has stabilized; cheap to add because it is a stopping condition, not a new scheduler.
- **Learning-curve scheduling, specifically ifBO** (include; tier 3 bet; weeks, gated on pause/resume plumbing). The only entry that uses the shape of the curve between rungs (Figure 5.3), and the variant whose surrogate is pre-trained once rather than refit inside our loop, which is what makes its overhead tolerable (Rakotoarison et al., 2024). DyHPO and DPL taught the field the idea; we adopt the version that ships.

5.7 What would overturn these verdicts

ASHA’s tier-0 status depends on V06: if early stopping cannot match full-fidelity final quality at a fraction of compute on our tasks, or if rung-final rank correlations on real studies fall consistently below the crossing-curves threshold, the scheduler is demoted and budgets revert to full runs. DEHB’s small-budget

RL verdict is tied to V07; if it loses to BO+ASHA at matched sixteen-run budget, it is excluded and the register updated. The ifBO curve model's gating (V15) is the designed check: if it does not match ASHA on public suites before integration, the build is canceled and the tier-2 slot reallocated. BOHB's explicit exclusion is reopened if the Optuna wrap becomes negligible-cost and a colleague requests it with a use case the box did not consider.

Chapter 6

Tuning while training: the population-based family

Everything so far searches for one fixed configuration, and the second bend of Section 2.2 has been waiting patiently: in the running example, the best learning rate at step zero is not the best learning rate at step 100 million, because the agent’s own improving policy keeps changing the data it trains on. Population-based training (PBT) is the family that takes this seriously. It searches for a *schedule*, and it does so inside a single training run, which is why it occupies a special place in reinforcement learning, where the optimal learning rate, entropy bonus, or batch size at the start of training is rarely optimal at the end.

6.1 PBT: exploit, explore, inherit

PBT trains a population of B agents in parallel, each with its own hyperparameters and its own network weights (Jaderberg et al., 2017). Every t_{ready} steps, two operations run (Figure 6.1). In the *exploit* step, each agent in the bottom fraction of the population (by current evaluation score) abandons its own progress: it copies both the network weights and the hyperparameters of an agent sampled from the top fraction, a rule known as truncation selection. In the *explore* step, the copied hyperparameters are perturbed, in the original paper by random multiplicative jitter or resampling from the prior. Training then continues from the copied weights. Because weights are inherited rather than restarted, the population performs a single collective training run whose hyperparameters adapt on the fly, and the lineage of the final best agent traces out a hyperparameter schedule that no pointwise tuner could have expressed: in Figure 6.1, the winning agent’s history runs through two exploit events and three different learning rates, and that whole trajectory, not its endpoint, is the method’s output.

The figure repays one slow read-through, event by event, because every later argument about this family refers back to it. Four agents train in parallel, each with its own learning rate. At the first ready event, the population is ranked by current evaluated return; the bottom agent (lane two) is culled, receives a byte-for-byte copy of the top agent’s network weights and hyperparameters, has those hyperparameters perturbed, and resumes training from the copied weights rather than from scratch.

Between the first and second ready events, four *different* hyperparameter settings are being evaluated while four networks make training progress; no compute is spent on rehearsal runs that will be thrown away, which is the family’s core economy.

At the second ready event the same surgery repeats on the new laggard. The winning agent’s bold lineage, read left to right, is a sequence of (weights inherited, hyperparameters revised) segments: a learning-rate *schedule* discovered by selection, with no schedule ever specified. That is the object no

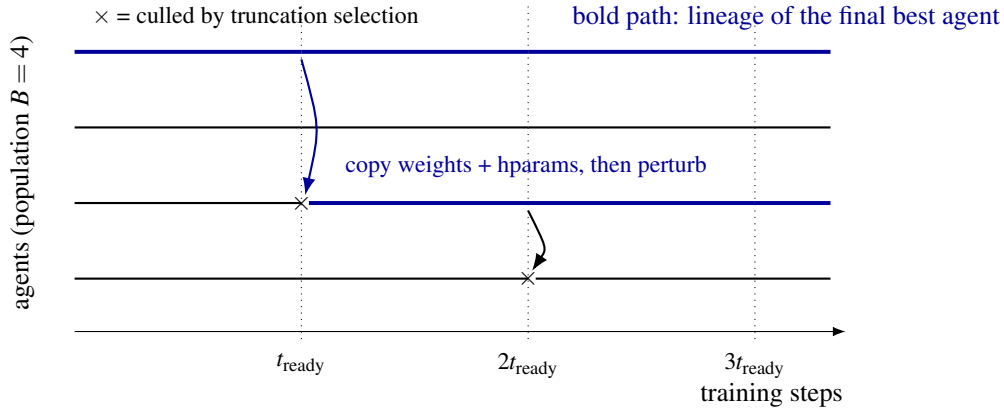


Figure 6.1: One PBT run (schematic, $B = 4$). At each ready event the bottom agents are culled and restart from copies of top agents’ weights and hyperparameters, perturbed before training resumes. The bold lineage is the method’s real product: a hyperparameter *schedule* that switched values twice mid-run, expressed by no pointwise method in this document.

method in the previous chapters can produce, and it is also, as the next paragraphs show, the source of the family’s characteristic failures.

Two ablations in the original paper pin down where the value lives, and they are the reason we drew the lineage rather than the final point. Rerunning training from scratch with the final PBT-found hyperparameters underperformed PBT itself, so the schedule rather than the endpoint carries the gain; and populations of around twenty gave consistent improvements with diminishing returns beyond. The cost structure is equally distinctive: one PBT run costs B training runs of wall-clock length one, fully parallel, rather than a long sequence of trials. When a project has B machines and one calendar week, that shape is not a detail; it is the argument.

Two failure modes matter as much as the mechanism. First, PBT is greedy: truncation selection optimizes the *current* score, and Figure 5.1 already showed what current scores do to configurations that invest now to pay off later. A small learning rate or a heavy regularizer is exactly the blue curve of that figure, and truncation selection is the cull arrow. FIRE-PBT (faster improvement rate PBT) counters this by scoring sub-populations on their rate of improvement rather than their level, and the size of the greediness cost is worth quoting: on ImageNet, the learning-rate schedule found by greedy PBT reached 71.7 percent top-1 accuracy where FIRE-PBT reached 76.5, essentially matching the hand-tuned schedule at 76.6 (Dalibard and Jaderberg, 2021).

Second, with the modest populations we can afford (the experiments below use $B = 8$), copying weights across the population collapses diversity in weight space: after a few exploit events, every agent descends from the same early winner, and the search degenerates into local perturbation around one basin. Wan et al. (2022) observed both effects directly: on four of their seven Brax tasks, plain random search matched or beat both PBT and PB2 at the same compute, which they attribute to exactly this premature-convergence behavior.

6.2 PB2: replacing random perturbation with a bandit

The explore step is where PBT wastes its evidence: jitter treats the population’s accumulated experience as worthless. PB2 repairs it with the machinery of Chapter 4 (Parker-Holder et al., 2020). PB2 records

every observed (hyperparameters, training-progress) increment across the population and fits a Gaussian process whose kernel includes a time coordinate, so that old evidence decays: an observation from early training informs, but does not dictate, beliefs about what works now.

When an agent needs new hyperparameters after an exploit event, PB2 chooses them by maximizing a UCB acquisition, the rule of (4.7) applied to this time-varying surrogate: for the increment model $\hat{f}_t$ with posterior mean μ_t and spread s_t , pick

$$\boldsymbol{\lambda}_{\text{new}} \in \arg \max_{\boldsymbol{\lambda} \in \Lambda} \mu_t(\boldsymbol{\lambda}) + \beta_t^{1/2} s_t(\boldsymbol{\lambda}), \quad (6.1)$$

maximization now because the increment is a reward. Members of the batch whose outcomes are still pending are handled by updating the posterior variance as if their results were already in (the fantasy device of Section 4.2), so the whole population can be proposed at once.

The quantity modeled is deliberately the *increment* of performance over one interval rather than its level, which is what makes the bandit framing clean: total performance is the sum of increments, so maximizing final performance equals minimizing cumulative regret over increments, and the authors prove a sublinear-regret bound for the resulting time-varying GP bandit. Sublinear regret has a plain reading: the average quality gap between the bandit’s choices and the best choices available at each moment goes to zero as training proceeds.

A follow-up, PB2-Mix, adds categorical hyperparameters through a hierarchical explore step: a time-varying adversarial bandit picks the categories, then the GP picks the continuous coordinates conditioned on that choice, with the regret guarantee extended to the mixed space (Parker-Holder et al., 2021).

The empirical promise is sample efficiency at small population sizes, but two independent observations temper it: the BG-PBT ablations show a plain GP surrogate limits how much of the space PB2 can cover (Wan et al., 2022), and in the budget-constrained study of Eimer et al. (2023), PB2’s tuned configurations changed at most once over a run (so the schedule advantage, the entire point of Figure 6.1, went unused) and its incumbents transferred poorly to fresh seeds.

6.3 BG-PBT: trust-region BO plus generational architecture search

BG-PBT, the method we were specifically asked to evaluate, upgrades both halves of PBT and adds architecture search (Wan et al., 2022). We read the paper in full; what follows is its actual content rather than its abstract. Readers will recognize every ingredient: the search space is the running example’s, the searcher is Section 4.4’s trust-region machinery, and the schedule-versus-point distinction is Figure 6.1’s.

The search space is the full RL configuration. For their PPO-on-Brax experiments this means nine training hyperparameters (learning rate, discount, entropy coefficient, unroll length, reward scaling, batch size, update epochs, GAE λ , clipping ϵ) plus six architecture parameters (width, depth, and a spectral-normalization flag for policy and value networks separately), the 15-dimensional mixed space of continuous, ordinal, and categorical coordinates that Section 2.1 adopted as our running example.

The explore step becomes a high-dimensional mixed-space BO agent built on Casmopolitan (Wan et al., 2021): kernels tailored to each coordinate type (their extension treats ordinals as ordered rather than categorical, with measurable gains in their ablation), acquisition optimization that interleaves local search over discrete coordinates with gradient steps over continuous ones, and trust regions around the current best configuration exactly as in Figure 4.4, expanding on successes, shrinking on failures, and triggering a restart with a fresh surrogate when they collapse, guided by a UCB rule on an auxiliary global GP. Restarts matter more here than in the static setting of Section 4.4: in a time-varying problem they are what stops the surrogate from anchoring on stale evidence.

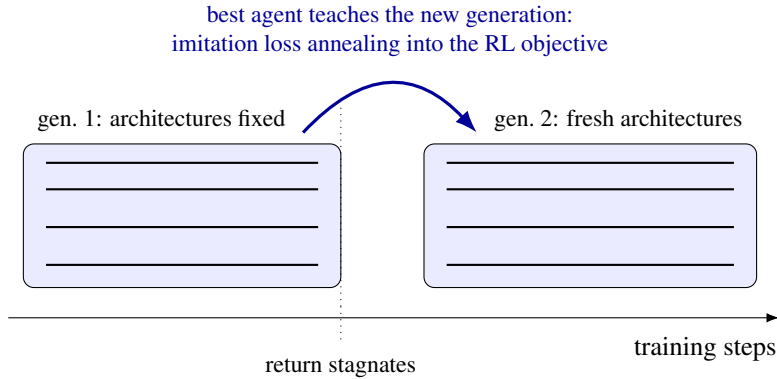


Figure 6.2: BG-PBT’s generational structure (schematic). Hyperparameters adapt continuously inside a generation via the trust-region BO explore step; architectures change only at generation boundaries, where new networks are warm-started by distilling the outgoing best agent rather than training from scratch. Distillation is the piece that makes mid-run architecture change affordable.

The authors extend the regret guarantee of PB2 to this trust-region, mixed-space, time-varying setting (their Theorem 3.1), with the usual caveat that the guarantee covers the version without architecture transfer.

Architecture search enters through *generations*, and the device that makes it affordable is distillation (Figure 6.2). Within a generation, architectures are fixed per agent; truncation selection copies weights *and architecture*, so poor architectures are squeezed out of the population, and the BO agent conditions its hyperparameter suggestions on the architecture coordinates as context. When the population’s evaluated return stagnates past a patience threshold, a new generation begins: a fresh set of architectures is proposed (by random search plus successive halving over the architecture subspace, later informed by a BO model fit to the best score each architecture achieved), and the new agents are trained initially by on-policy distillation from the best agent of the outgoing generation, with a joint loss that anneals from imitation (value regression plus a KL term to the teacher policy) into the plain RL objective. Distillation is what makes changing architecture mid-run affordable: the new network does not start from zero, it starts from an imitation of the best agent so far.

The reported results, on seven Brax environments with population size 8 and 150M environment steps: BG-PBT reaches the best mean return on five of seven tasks against tuned PPO, random search, PBT, and PB2, with the remaining two split; against a sequential SMAC3 baseline given fifty full runs (about six times the compute), BG-PBT still wins on four of seven. The discovered schedules are interpretable and match folklore: learning rates fall over training and batch sizes grow, without any schedule having been specified. The paper’s own limitations section notes that on Humanoid the ablation without architecture search did better, that generational transfer voids the theoretical guarantee, and that the distillation hyperparameters are themselves fixed by hand. That last item deserves a smile: the state of the art in hyperparameter optimization contains, at its center, a handful of hand-tuned hyperparameters. The regress has to stop somewhere, and knowing where it stops is part of understanding the method.

6.4 The two experiments this chapter’s verdict rests on

The verdict this chapter builds toward, build BG-PBT but confine the whole population family to studies with real parallelism, rests on two experiments. Section 4.5 argued that evidence should be weighed by

its design, not its abstract, so both designs are laid out here in enough detail to be judged, and then the argument that reconciles them is made explicitly, because at first reading they appear to disagree.

The first experiment is the BG-PBT paper’s own (Wan et al., 2022). Seven Brax locomotion environments; PPO; a population of eight agents; 150 million environment steps per run; the fifteen-dimensional mixed search space of our running example, nine training hyperparameters plus six architecture parameters. The comparisons: PPO with tuned fixed hyperparameters, random search over the same full space, original PBT, PB2, and, importantly, a sequential SMAC3 baseline given fifty complete training runs, roughly six times the compute of one population. The scoring is mean evaluated return, and the headline results were quoted in the previous section: best on five of seven tasks against the population-budget baselines, still ahead on four of seven against the six-fold-compute sequential baseline.

Three details of the design speak well of the authors. Random search was run over the *full* joint space rather than a crippled subset, which is how the paper itself surfaced the uncomfortable fact that random search matched or beat PBT and PB2 on four of seven tasks. The ablations separate the contributions (ordinal-aware kernels help; on Humanoid the version *without* architecture search did better). And the limitations section concedes the points a referee would raise: the theory does not cover generational transfer, and the distillation hyperparameters are set by hand. A paper that reports where its own method loses earns more trust than one that does not.

The second experiment is the controlled RL tuning study of Eimer et al. (2023), from a different group with no shared authors. Three algorithms (PPO, SAC, DQN) across eleven environments, Brax tasks among them; tuners compared at matched budgets of ten to sixty-four *full training runs*, which is the regime our routine studies actually inhabit; candidates evaluated on several tuning seeds with final results reported on disjoint test seeds the tuner never saw; and the tuners’ own overhead measured (minutes for most methods, about two hours for BG-PBT at the largest budget, against days of trial compute).

The design’s discipline is the point: matched budgets stop a hungrier method from winning by eating more, and the seed split stops a tuner from being rewarded for overfitting luck. In this regime, DEHB was the most consistent tuner, random search was respectable, and the population methods trailed: BG-PBT sat behind DEHB and even random search at sixteen and sixty-four runs, and PB2’s tuned configurations changed at most once per run, so the schedule advantage that justifies the family’s existence went unused. The authors note, fairly, that BG-PBT would likely benefit from larger budgets.

Now the reconciliation, because a reader holding both results needs it. The two experiments occupy different budget regimes. One population run in the Wan design consumes eight parallel full-length trainings, and its evolutionary machinery gets hundreds of exploit events over 150 million steps; the Eimer design’s entire budget is ten to sixty-four full runs total, which starves a population method of exactly the parallelism and step count its mechanism needs.

Where the regimes overlap, the studies agree: both find random search over the full space embarrassingly strong at modest budgets, and both find plain PBT and PB2 fragile with small populations. So the apparent contradiction is not one paper being wrong; it is one boundary seen from both sides, and the boundary is the finding.

Below roughly eight concurrent full-length workers or a few dozen full-run equivalents, rung-based methods with a model (DEHB, or BO plus ASHA) are the reliable choice; above it, with a schedule-shaped question, BG-PBT is the strongest design published for that regime, on the evidence of Brax locomotion at population sizes of eight and generous step budgets, and its official implementation is available to build from. The package should enforce that boundary in software rather than trust each user’s optimism, and our validation suite re-tests it on our own pinned tasks, since a boundary established on Brax deserves a check before it governs a default.

6.5 The family since 2022

The line has stayed active, and the newer entries share one diagnosis: PBT’s remaining free parameter, the interval t_{ready} between evolution steps, is itself consequential and badly understood. Evolve too often and agents are judged before their new hyperparameters have had any effect; too rarely and the schedule cannot track the drift it exists to track.

Multiple-frequencies PBT runs sub-populations that evolve at different perturbation frequencies with an asymmetric migration rule between them, reporting that vanilla PBT can fall behind random search over long horizons, the same greediness disease FIRE-PBT treats (Doulazmi et al., 2025). A 2026 preprint takes the opposite tack for small populations, adjusting the interval through restarts that reuse weights and re-seed hyperparameters by time-varying BO (Chebykin et al., 2025).

On the off-policy side (algorithms that learn from a stored *replay buffer* of past experience rather than only from the freshest data), SEARL shares one replay buffer across the whole population, so the meta-search rides on data the population already paid for, a design worth remembering for our off-policy RLlib work (Franke et al., 2021). None of these ships in any framework we audited; Ray provides PBT, PBT replay, and a continuous-only PB2 (Chapter 10), which is exactly why this family is where an in-house implementation adds value.

6.6 What the product should take from this family

Three design consequences follow. First, schedules are a first-class output: our tuner must be able to return a hyperparameter trajectory, not only a point, and our run store must record the lineage that produced it; Figure 6.1’s bold path must be a queryable object, not a diagram. Second, the exploit and explore steps need checkpoint surgery (copy weights, swap optimizer state, resume), so the execution layer must treat checkpoints as cheap, frequent, and programmable, which is the strongest single argument for keeping Ray Tune underneath (its PBT scheduler already exercises this machinery, and RLlib checkpoints compose with it). Third, the population methods and the rung methods are complements, not rivals: the honest default is budget-dependent, and the product should encode that guidance rather than leaving it to each user’s optimism.

6.7 The verdict, argued from four angles

The chapter’s verdict, build the PB2-Mix-to-BG-PBT line as the flagship but confine the family behind the parallelism boundary, should survive scrutiny from four independent directions, and it does.

From mechanism: the family’s failure modes were derivable before any experiment. Truncation selection optimizes the current score, so it must cull slow starters (the crossing curves of Figure 5.1 guarantee victims exist); weight copying must collapse diversity in small populations, since after a few exploit events every agent descends from one early winner. Both pathologies follow from the mechanism as stated, both were then observed independently (FIRE-PBT’s ImageNet gap for the first, the random-search upsets in both experiments for the second), and each upgrade in the chapter is a targeted repair of one of them. A verdict whose supporting evidence was predicted by the mechanism is worth more than one discovered by accident.

From evidence: two experiments with no shared authors, designed under different incentives (one by the method’s inventors, one by outside evaluators with a budget-matched, seed-split protocol), agree wherever their regimes overlap and disagree exactly where the budget regimes diverge. Section 6.4 laid out both designs so this claim can be checked rather than taken.

From operations: the family’s cost shape, B machines for one wall-clock training run, is the shape of our actual constraint, a cluster with idle capacity and researchers with deadlines; and the one genuinely hard engineering requirement, live checkpoint surgery, is machinery Ray Tune and RLlib already productionize (Chapter 12). We are not building the risky part.

From decision theory: building the line is an option purchase. Our RL work is moving toward exactly the large-parallelism, schedule-shaped regime where the method’s advantage lives; the boundary rule caps the downside of misuse at small budgets, where the package will steer users to DEHB instead; and declining to build leaves schedules unreachable at any budget, which is a capability gap, not a savings.

And the falsification criterion, stated in advance: the boundary and the flagship are both re-tested in our validation suite, populations against DEHB and BO-plus-ASHA on our pinned Brax task at matched compute on both sides of the eight-worker line. If the population line cannot beat the rung methods even in its home regime, the flagship is demoted and this chapter gets rewritten, not defended.

Decisions from this chapter. One threshold governs everything here: can the study run eight or more full-length trainings in parallel? Read eight as a calibrated default rather than a law: it is our reading of where [Eimer et al. \(2023\)](#)’s budgets and BG-PBT’s population regime cross on workloads that are not ours, so the product should ship it as a configurable threshold that dispatches to the rung methods below the line, warns rather than refuses on override, and records the override in the study’s audit trail. V08 is what would replace the default with a measured crossover on our own tasks. Below the line the small-budget evidence favors DEHB and BO-plus-ASHA ([Eimer et al., 2023](#)). Above it:

- **PBT** (expose, do not build). Ray Tune already ships the reference distributed implementation; our package should surface it with the greediness warning attached (the FIRE-PBT ImageNet numbers, 71.7 versus 76.5 percent top-1, are the one-line version of that warning, [Dalibard and Jaderberg, 2021](#)).
- **PB2 extended to mixed spaces** (build; moderate cost). Ray’s PB2 covers continuous knobs only; the categorical extension (PB2-Mix’s bandit over categories, or directly the mixed-space kernels we build for the trust-region searcher) is the first genuine gap our package fills in this family.
- **BG-PBT** (build; the flagship; months). The strongest published design when a real population, a schedule-shaped answer, and architecture search are all wanted at once: best mean return on five of seven Brax tasks against tuned PPO, random search, PBT, and PB2, and still ahead of a sequential baseline given six times the compute ([Wan et al., 2022](#)). The costs to state plainly: it needs the trust-region mixed-space searcher as a prerequisite, its theory does not cover the generational transfer that makes it shine, and below the population threshold it loses to methods a tenth its complexity.
- **FIRE-PBT, MF-PBT, IPBT, SEARL** (watchlist, do not build now). Each fixes a real PBT pathology (greediness, the evolution interval, off-policy data reuse), none ships in a framework we audited, and none has the cross-paper evidence BG-PBT has. SEARL’s shared replay buffer is the one to revisit first when our off-policy work scales up.

6.8 What would overturn these verdicts

The flagship’s fate is tied to V08, whose demotion rule was fixed before any code existed: if BG-PBT cannot beat DEHB and BO+ASHA on both sides of the eight-worker boundary at matched compute, it is demoted to an option and the chapter rewritten with the population line’s status downgraded. The eight-worker boundary itself is empirical; if real studies consistently choose population methods below it or consistently avoid them above it, the boundary input is re-tuned and this section updated. PB2-Mix’s dependency on the trust-region searcher is structural; if that searcher is cut for other reasons, PB2-Mix either finds a different explore engine or is deferred. FIRE-PBT’s hold is lifted if a study on RL tasks shows it reliably outperforming the simpler PBT variants already included.

Part IV

**Several objectives, and knowledge we
already own**

Chapter 7

More than one objective

Several of our workloads do not have a single number to minimize, and the Hamiltonian network introduced in Section 2.1 is the cleanest case: a model that nails one-step prediction but leaks energy over long rollouts is not better than one that does the reverse; it is better at one thing and worse at another. When the exchange rate between objectives is known in advance, the problem collapses back to Section 2.2: fix the weights and minimize the combination. Usually the exchange rate is exactly what we do not know, and this chapter builds the machinery for optimizing honestly without inventing one. The build order mirrors the single-objective story: first the right notion of “better” (dominance), then the right notion of progress (hypervolume), then the acquisition functions and schedulers that optimize it.

7.1 Dominance, the Pareto front, and how to score progress

With objectives $f_1, \dots, f_m$ to minimize, configuration λ *dominates* λ' if it is no worse in every objective and strictly better in at least one: there is no argument for keeping λ' when λ is on the table. (Economists are on home ground in this whole chapter: this is Pareto dominance exactly as welfare theory uses it, and the scalarization results below are the optimization mirror of familiar aggregation problems.) Domination is only a partial order, and that is the entire difficulty of this chapter: most pairs of configurations are simply incomparable, one better at prediction, the other at energy conservation, and no amount of cleverness removes the incomparability without importing a preference.

The *Pareto front* is the image of the undominated configurations, the trade-offs that survive every argument (Figure 7.1). Surveys of multi-objective tuning sort methods by when preferences enter: before the search, which fixes a scalarization and returns one point, or after it, which approximates the front and lets the user choose with the trade-off in view (Karl et al., 2023; Morales-Hernández et al., 2022). Everything below is of the second kind unless said otherwise: the tuner returns an approximate front, and the exchange rate is made visible rather than assumed.

If the target is a set rather than a point, what does progress mean? The standard scalar score is the *hypervolume*: fix a reference point $\mathbf{r} \in \mathbb{R}^m$ that is worse than anything acceptable, and measure the volume of objective space that the current front dominates,

$$\text{HV}(P) = \text{vol}\left(\bigcup_{\mathbf{y} \in P} [\mathbf{y}, \mathbf{r}]\right), \quad (7.1)$$

the union of the axis-aligned boxes spanned between each front point $\mathbf{y}$ and the reference. The indicator goes back to Zitzler and Thiele (1999).

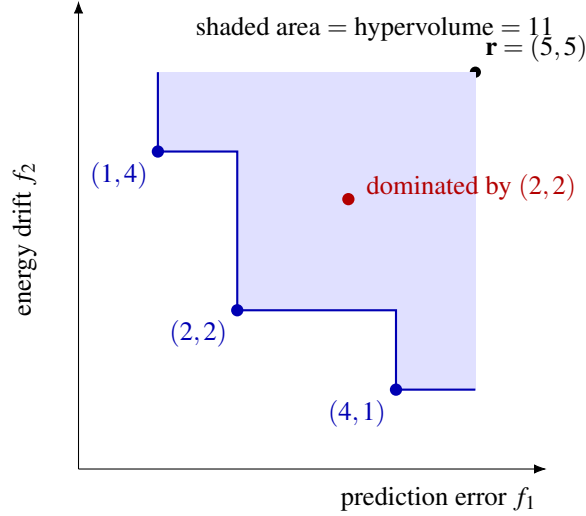


Figure 7.1: The Hamiltonian study’s objective plane (schematic, minimizing both axes). The three blue trials are mutually incomparable and form the current front; the red trial is dominated and argues for nothing. The shaded region is everything the front dominates up to the reference point, and its area, computed strip by strip in the text, is the hypervolume (7.1): the one number that grows only when the front genuinely advances or spreads.

A small computation makes it concrete. Take the Hamiltonian study with prediction error and energy drift both scaled to $[0, 5]$, reference point $\mathbf{r} = (5, 5)$, and three undominated trials at $(1, 4)$, $(2, 2)$, and $(4, 1)$. Slicing the dominated region into vertical strips: for first-objective values in $[1, 2)$ only the point $(1, 4)$ dominates, contributing a strip of height 1 and area 1; on $[2, 4)$ the point $(2, 2)$ takes over, height 3, area 6; on $[4, 5)$ the point $(4, 1)$ gives height 4, area 4. Total hypervolume: 11 of the 25 available. Adding a new trial at $(3, 1.5)$ would carve out fresh territory (a rectangle of width 1 and height 0.5 that no existing box covers), so the hypervolume rises; adding a dominated trial adds nothing. This is why the indicator earns its role as the standard yardstick: by (7.1) it rewards convergence toward the true front and spread along it at once, and any addition that dominates new territory strictly increases it.

Hypervolume’s cost is the catch. Computing and decomposing the dominated region has time complexity that grows super-polynomially in the number of objectives m (Daulton et al., 2021), which in practice confines hypervolume-driven methods to two to four objectives; beyond that, scalarization-based methods take over. All of our current use cases sit comfortably at $m \in \{2, 3\}$.

7.2 Scalarize many times, or evolve a population

The oldest trick converts the problem into many single-objective ones: draw a weight vector, optimize the scalarized objective, repeat with fresh weights, and collect the results. Whether this can even recover the whole front depends on the scalarization, and the failure of the obvious choice is geometric enough to draw.

Minimizing a weighted sum $\sum_i w_i f_i$ with fixed positive weights means sliding the line (in general, hyperplane) $\sum_i w_i f_i = c$ toward the origin until it last touches the attainable region: the touching point is the minimizer. A supporting line can only touch the region’s convex hull, so for *any* positive weights, the minimizer of a weighted sum sits on the convex hull of the attainable objective vectors, and Pareto points

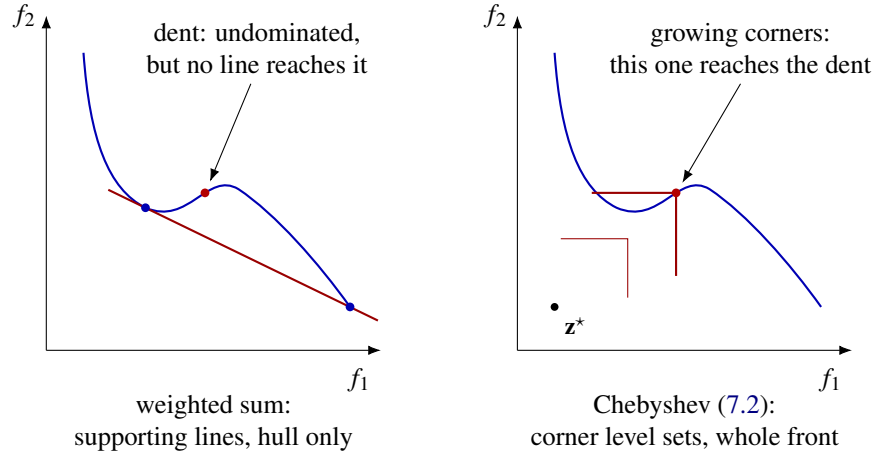


Figure 7.2: Why the obvious scalarization loses solutions (schematic). Left: minimizing a weighted sum slides a line toward the origin; it can only stop on the convex hull, so the concave stretch of the front is unreachable for every choice of weights. Right: the Chebyshev scalarization’s level sets are corners, which can touch any Pareto point, concave stretches included; sweeping its weights traces the whole front.

inside a concave stretch of the front are never selected, whatever the weights (Figure 7.2, left).

In the Hamiltonian study this is not a curiosity: if the best balanced models happen to sit in a dent of the front, every weighted-sum study we run will report only the two extremes, a model that predicts well and leaks, and a model that conserves and mispredicts, with the interesting compromises silently unreachable.

The scalarization of choice is therefore the Chebyshev form: with an ideal point z^* (each objective’s best value, in practice the best seen) and weights $w_i > 0$,

$$g_w(\lambda) = \max_{i=1,\dots,m} w_i (f_i(\lambda) - z_i^*), \quad (7.2)$$

the weighted *worst* gap to the ideal rather than the weighted sum of gaps. Its level sets are axis-aligned corners rather than lines, and a corner can nestle into a concave dent that no line can touch (Figure 7.2, right). The standard completeness statement (sweeping positive weights with an appropriate true ideal point recovers every Pareto point under convexity or boundary assumptions) is proven using the true utopian vector, not a best-seen point; the latter is usable in practice and provides a coverage advantage over weighted sums, but it is an operational approximation rather than a theoretical guarantee (Karl et al., 2023).

ParEGO runs exactly this inside BO: each iteration draws a random Chebyshev weight, scalarizes the observed data, and takes an EI step on the result (Knowles, 2006); random-scalarization bandits generalize the idea with guarantees (Paria et al., 2019). Scalarization scales gracefully in m and composes with every single-objective engine we build, which is why it remains the fallback when hypervolume machinery gets expensive.

Evolutionary methods attack the front directly with a population, using dominance itself as the fitness signal. NSGA-II (the name abbreviates non-dominated sorting genetic algorithm), the workhorse, sorts the population into non-domination ranks (rank 1: the undominated; rank 2: undominated once rank 1 is removed; and so on) and breaks ties within a rank by crowding distance, a dispersion measure that prefers points in sparsely covered stretches of the front, so selection pressure pushes toward the front while spreading along it (Deb et al., 2002). It needs generous evaluation budgets, which multi-fidelity variants

mitigate, and it is the default multi-objective sampler in Optuna, so we get a tested implementation for free when trials are cheap.

7.3 Bayesian optimization of the front, one repair at a time

When a trial costs hours, the population’s appetite disqualifies it, and we are back in Chapter 4’s situation: few, expensive, noisy evaluations, and a surrogate that has to make each one count. The machinery that results is best understood the way EI was: one natural idea, then a short chain of repairs, each forced by a failure that is easy to see once it is named.

7.3.1 EHVI: expected improvement, paid in area

The natural idea first. With one objective, EI asked how much a candidate would improve on the incumbent, averaged over the surrogate’s uncertainty. With several objectives there is no single incumbent; there is the staircase of Figure 7.1, and progress means adding area to what it dominates. *Expected hypervolume improvement* (EHVI) therefore scores a candidate by the expected increase of (7.1) if we were to evaluate it: the expected area of fresh territory.

One number makes the definition mechanical. Take the front of Figure 7.1 and a candidate configuration whose GP posterior, say, puts half its mass at the outcome (3, 1.5) and half at (3, 2.5). The first outcome lands in the dent and carves out the 1×0.5 rectangle we computed earlier, an area of 0.5; the second is dominated by (2, 2) and adds nothing. The candidate’s EHVI is $0.5 \cdot 0.5 + 0.5 \cdot 0 = 0.25$. Every candidate gets such a score, and the searcher proposes the maximizer, exactly EI’s decision logic with improvement measured in area. In practice the posterior is a continuous Gaussian rather than two points, so the average is estimated from posterior samples; nothing about the logic changes.

7.3.2 Making it computable in batches: qEHVI

Turning that definition into a working acquisition hits a geometric wall. Each posterior sample asks for the area a proposed outcome adds to a union of overlapping boxes, and we rarely propose one point at a time: Chapter 4’s batch argument applies unchanged, since idle workers do not care that the objectives are plural. The joint fresh area of q proposals is the area of a union over a union, and evaluating it naively by inclusion-exclusion sums over subsets of the batch, a count that grows exponentially in q .

Daulton et al. (2020) made the computation practical at batch sizes above one. The dominated region is decomposed once per front into disjoint boxes; each posterior sample’s contribution is then a sum over those pieces, exact rather than approximated, and the whole estimator is differentiable in the candidate coordinates, so the batch is optimized by gradient ascent like any other acquisition of Chapter 4.

7.3.3 Noise bends the front: qNEHVI

Now the failure our workloads force. Suppose one Hamiltonian trial, of genuinely middling quality, draws a lucky seed and reports (1.8, 1.8). That phantom point dominates the honest compromise at (2, 2), so the observed staircase bends around an outcome the true objectives never produced. Every genuinely improving candidate is now scored against territory the phantom already claims, EHVI near zero across the board, and the study stalls defending a front that does not exist. This is the winner’s curse of Section 4.2 again, with the lucky seed promoted from incumbent to front member.

qNEHVI applies the same repair noisy EI used there: stop letting raw observations define the front. The improvement is integrated over the surrogate posterior at the observed points, so the front itself is

treated as uncertain, and a phantom member holds exactly as much territory as the posterior believes it deserves (Daulton et al., 2021). The same paper removed the remaining cost wall, making joint batch proposals tractable where the naive computation grew exponentially in batch size.

7.3.4 Vanishing improvements again: qLogNEHVI

The last repair is inherited. Late in a good study, almost no candidate adds much area, and expected improvements shrink to the magnitudes where Section 4.2 showed plain EI underflowing to an exact floating-point zero; the underflow argument applies verbatim to hypervolume improvements. The fix is the same log-space computation, and the numerically stabilized variant qLogNEHVI is the recommended default in the BoTorch ecosystem (Ament et al., 2023). The lineage that looked like alphabet soup at the chapter’s door is thus three cumulative repairs to one idea, and the name now reads as a recipe: the q is the batch, the N is the noise treatment, the Log is the underflow repair, and EHVI is the idea they all serve.

7.3.5 Two extensions on file

Two further members solve problems we do not have yet, and one paragraph each is enough to know when to reach for them. For high-dimensional spaces, MORBO transplants the trust-region idea of Figure 4.4: several local regions each pursue hypervolume improvement with local models, coordinating through the shared front (Daulton et al., 2022a).

And when the objectives differ in evaluation cost, a profiler’s latency reading next to a full training run’s accuracy, the hypervolume knowledge gradient extends the machinery to decoupled evaluations: it decides not only where to evaluate next but which objective to measure there (Daulton et al., 2023).

Our product needs qLogNEHVI as the premium two-to-four-objective searcher and Chebyshev scalarization as the cheap many-objective fallback; both ride on the same GP infrastructure as Chapter 4.

7.4 Multi-objective meets multi-fidelity

The savings of Chapter 5 apply here too, with one twist: promotion decisions need a ranking, and “better” is now the partial order we started the chapter with. Promote-the-top-third is not defined when the top third is not defined.

MO-ASHA resolves this with the layering NSGA-II already taught us. Rung members are sorted into non-domination ranks, and rank 1 promotes before rank 2. Ties within a front are broken by a dispersion rule in objective space, either NSGA-II’s crowding distance or an epsilon-net that repeatedly picks the candidate farthest from those already chosen; the top fraction then promotes as usual. In its authors’ experiments, promotion rules aware of the whole front beat scalarized promotion in wall-clock terms (Schmucker et al., 2021), which is the geometry of Figure 7.2 surfacing inside the scheduler: a scalarized promotion inherits the scalarization’s blind spots, rung by rung.

The repair travels. MO-DEHB applies the same non-dominated-sorting-plus-crowding selection to DEHB’s evolutionary rungs (Awad et al., 2023), and the population family of Chapter 6 has its member too: MO-PBT swaps PBT’s scalar exploit ranking for non-dominated selection, evaluated on trade-offs such as precision against recall and accuracy against fairness (Dushatskiy et al., 2023).

These are exactly the schedulers we will pair with cheap objectives such as short-horizon energy drift in Hamiltonian networks. The drift at rung 1 is cheap to measure and, per Figure 5.1’s warning, worth a pilot check that its early rankings mean anything before we let it cull.

7.5 Constraints and preferences

Two pragmatic variants round out the toolkit. Hard constraints (memory ceilings, latency budgets, a minimum acceptable return) are handled by weighting acquisitions with the modeled probability of feasibility. A candidate whose acquisition score is 0.8 but whose modeled chance of fitting under the memory ceiling is 0.3 counts as $0.8 \times 0.3 = 0.24$: still on the table, discounted by exactly its chance of being usable. Noisy-observation treatments of the same idea exist (Letham et al., 2019), and trust-region versions scale it to many constraints at once (Eriksson and Poloczek, 2021).

Preferences short of hard constraints (“we care about drawdown roughly twice as much as turnover”) can be encoded by restricting the reference point or the weight distribution of the scalarization; this is cheap, transparent, and usually all a practitioner wants. We recommend exposing both, and resisting anything more elaborate until a real use case demands it.

7.6 The evidence, and the check it still lacks

Weighed the way Section 4.5 taught us to weigh, this chapter’s support has an unusual profile. A larger share than usual is mathematics that can be checked at a desk; the experiments that exist are the method authors’ own; and the tier that anchored the last two chapters, an independent budget-matched comparison by outsiders, has no counterpart here that our literature sweeps could find. Each of those three facts shapes a different part of the decision box.

Start with what the mathematics settles on its own. The weighted-sum blind spot is a one-page supporting-hyperplane argument, reproduced in full above; the Chebyshev form’s ability to reach every Pareto point is its geometric mirror (Karl et al., 2023); hypervolume strictly increases exactly when the front advances or spreads, the worked computation under Figure 7.1; and the cost of decomposing the dominated region grows super-polynomially in the number of objectives (Daulton et al., 2021), which draws the two-to-four boundary. These four facts alone decide the exclusion of weighted sums, the inclusion of Chebyshev as the fallback, the choice of progress score, and the boundary past which the premium searcher gives way. No experiment is being asked to carry those verdicts, which is fortunate, because experiments of the decisive kind are the scarce resource here.

The experiment behind the premium pick is the qNEHVI paper’s own evaluation, and its design earns a close look because of one choice: noise is a first-class experimental axis rather than a footnote. The comparison spans the modern acquisition families (information-based PESMO, MESMO, and PFES, the diversity-guided DGEMO, scalarizing TSEMO and MOEA/D-EGO, Thompson-sampling and plug-in variants of the hypervolume line) on synthetic and real-world problems, with observation noise scaled as a percentage of each objective’s range and pushed to twenty percent, a regime the authors note prior noisy work had probed only at one percent (Daulton et al., 2021).

Every method degrades as noise rises; qNEHVI degrades least and leads the noisy benchmarks in both sequential and batch settings, at competitive wall time thanks to the cached box decompositions. This is the authors’ own arena, so the selection-bias caution applies, but two things temper it. The winning move is the same integrate-over-the-posterior treatment whose single-objective version carried the noise evidence of Chapter 4, arriving here as transferred machinery rather than a fresh conjecture. And the axis the authors chose to stress, heavy observation noise, is not a flattering backdrop chosen for the method; it is the axis our workloads actually live on, seed-noisy RL returns and effective-sample-size measurements.

The experiment behind the multi-objective early-stopper is the MO-ASHA paper’s, three tasks chosen to be genuinely unlike one another: architecture search on a tabular benchmark whose exact Pareto front is knowable, fairness-accuracy trade-offs on a standard dataset, and Transformer language-model tuning,

all run as thirty-minute wall-clock races with objectives normalized by empirical quantiles so that no objective’s units dominate by accident (Schmucker et al., 2021). The comparison that matters inside it: promotion rules that see the whole front (non-dominated sorting with a dispersion tie-break) against three scalarized promotion rules, each given a hundred random weight vectors. The front-aware rules win in wall-clock hypervolume, which is the experimental echo of the geometry above: a scalarized promotion inherits the scalarization’s blindness, rung by rung.

What this family lacks is the third tier: no independent, budget-matched, seed-split study of multi-objective tuners on live training runs surfaced in our audit, nothing that does for this chapter what Eimer et al. (2023) does for Chapters 5 and 6. The authors’ arenas are also not ours; fairness datasets and tabular architecture-search benchmarks are a long way from Hamiltonian rollouts and sampler diagnostics. The consequence is not paralysis, because the geometry and the transferred noise machinery carry most of the weight; the consequence is that our own validation suite has to play the role the literature has not filled yet, and the falsification criterion below is written accordingly.

7.7 The verdicts, argued from four angles

The chapter recommends qLogNEHVI as the premium searcher inside the two-to-four-objective boundary, Chebyshev scalarization as the cheap universal fallback, NSGA-II by wrapping for bulk cheap trials, and MO-ASHA as the early-stopper. Four angles, one verdict.

From geometry: the load-bearing exclusions and boundaries follow from arguments a reader can verify without trusting a single benchmark: the hull argument disqualifies weighted sums, the corner level sets qualify Chebyshev, and the decomposition cost draws the objective-count boundary. Decisions anchored this way are immune to the selection-bias worries that haunt benchmark-anchored ones, and the experiments are needed only for the choices the geometry leaves open.

From economics: committing to weights before searching is assuming the exchange rate the study exists to discover; returning a front is handing the decision-maker the actual menu and letting preference be exercised when the trade-off is visible and the stakes are known. The Chebyshev-versus-weighted-sum contrast is the familiar one between corner-shaped and linear indifference curves, and the dent in Figure 7.2 is precisely where a linear indifference curve misreports what is attainable. A reader who would not let a colleague assume a client’s utility function before seeing the budget set should not let a tuner do it either.

From evidence: the experiments that exist, though author-run, stress the failure modes that matter to us rather than flattering ones, heavy observation noise for the acquisition and wall-clock promotion honesty for the scheduler, and their findings are the experimental shadows of theorems taught above rather than free-standing claims. Nothing in this family’s record contradicts any result we verified in the single-objective chapters, and the missing independent tier is named rather than papered over.

From operations: the whole chapter rides rails already bought. The premium searcher runs on Chapter 4’s GP stack and inherits its LogEI repair; the evolutionary default arrives by wrapping Optuna, whose maintenance we audited (Chapter 10); the early-stopper is a week of work on top of our ASHA; and the objective-count boundary is enforceable in software. The marginal cost of honest multi-objective tuning, given the rest of the product, is weeks, which changes the include-or-defer calculus in its favor.

And the falsification criterion, stated in advance: the validation suite adopts the Hamiltonian study, prediction error against energy drift, as a pinned two-objective task, run at matched budgets across seeds with final hypervolume and its uncertainty reported. The premium searcher must beat random-weight Chebyshev and the wrapped NSGA-II there; MO-ASHA’s cheap early signal, short-horizon drift, must pass the rank-correlation pilot of Chapter 5 before it is allowed to cull; and if the cheap fallback matches the premium searcher on our tasks, the premium tier is demoted to an option, the same rule the

single-objective chapters applied to themselves.

Decisions from this chapter. Pick by the number of objectives, the noise level, and the price of a trial.

- **qLogNEHVI on BoTorch** (include; the premium searcher; weeks). For two to four objectives with expensive, noisy trials, which is precisely the Hamiltonian and finance profile. It is the noise-aware, numerically repaired member of the hypervolume family, and the BoTorch ecosystem’s own recommended default (Daulton et al., 2021; Ament et al., 2023). Beyond four objectives its hypervolume arithmetic becomes the bottleneck, so the default switches to scalarization there; treat that as a calibrated cutoff on published complexity results rather than a prohibition, and make it a configurable threshold whose override is recorded in the study’s audit trail.
- **Chebyshev scalarization with random weights** (include; days). The cheap fallback that composes with every single-objective engine we own and has better coverage than weighted sums: it can reach concave stretches of the front that weighted sums cannot (Figure 7.2). Completeness under random weights requires the true ideal point and additional assumptions; the operational approximation using best-seen values gives better coverage in practice without the theoretical guarantee. This is also the answer for five or more objectives.
- **Plain weighted sums as a product default** (exclude). They silently discard the concave stretch of the front; if a user truly has fixed exchange rates, they have a single-objective problem and should declare it as one.
- **NSGA-II by wrapping Optuna** (include; free). For cheap trials in bulk, the evolutionary route is a tested default we get by wrapping, not building.
- **MO-ASHA with veto gates** (build; about a week over ASHA). The multi-objective early-stopper, and the veto machinery doubles as the correctness gate the sampler workload needs (Chapter 9).
- **MORBO and the hypervolume knowledge gradient** (defer). Each answers a situation we do not yet have: very high-dimensional Pareto search, and objectives measurable separately at different prices. Revisit on demand.

7.8 What would overturn these verdicts

qLogNEHVI’s tier-1 inclusion depends on V09: if it cannot beat random-weight Chebyshev scalarization and the wrapped NSGA-II on the Hamiltonian task by a margin that justifies its complexity, it is demoted to an option. MO-ASHA’s veto gates are tied to V10 and V13: if cheap fidelity signals (short-horizon drift) fail the rank-correlation pilot, or if a veto-failing configuration ever gets promoted in validation, the mechanism is disabled and full-fidelity MO defaults to the scalarized floor. Weighted-sum exclusion is reopened if the convexity assumption turns out to hold on our tasks, which V09’s results will show. MORBO’s deferral is reconsidered when trust-region BO is proven and the Pareto-aware TR extension becomes a marginal cost rather than a second research bet.

Chapter 8

Using what we already know: priors, transfer, and the price of a trial

A tuner that starts every study from ignorance wastes its first dozen trials rediscovering folklore: learning rates live near 3×10^{-4} , discount factors near 0.99, dropout rarely helps reinforcement learning. Every GP in Chapter 4 began with a zero-mean prior that believes nothing; every project lead we serve believes a great deal, and is usually right. We tune the same families of models over and over, so encoded experience is the cheapest compute we own. The literature offers three grades of it, in increasing order of machinery, plus a fourth option that replaces search entirely at large scale.

8.1 Expert priors over the optimum

The lightest grade lets a human say where the optimum probably lies. In π BO, the user supplies a prior density $\pi(\boldsymbol{\lambda})$ over the *location of the best configuration* (not over function values, which is the GP’s job), and the acquisition function is multiplied by that density raised to a power that decays with the number of observations,

$$\alpha_n^\pi(\boldsymbol{\lambda}) = \alpha_n(\boldsymbol{\lambda}) \cdot \pi(\boldsymbol{\lambda})^{\beta/n}, \quad (8.1)$$

where α_n is any standard acquisition (EI in the paper), n counts observations, and β sets how long the advice matters (Hvarfner et al., 2022b).

The design reads directly off the formula. Early on (n small) the exponent is large, the prior term dominates, and the search concentrates where the expert pointed; as evidence accumulates the exponent shrinks toward zero, $\pi(\boldsymbol{\lambda})^{\beta/n} \rightarrow 1$ for every $\boldsymbol{\lambda}$ in the prior’s support, and the acquisition converges back to its data-driven self on that support (Figure 8.1). Points assigned zero prior probability stay at zero. The decay is what makes the mechanism safe: the authors prove it preserves standard convergence behavior, and show large early gains when the prior is good and mild losses when it is wrong, since a wrong prior is overruled at the rate evidence arrives. A confidently wrong expert costs the study its opening moves, not its conclusion.

PriorBand carries the same idea into the multi-fidelity world of Chapter 5, where the object being steered is not one acquisition but a stream of configurations entering ASHA-style rungs. Each new configuration is drawn from a portfolio mixing three samplers: uniform exploration, the user prior, and local perturbations of the current incumbent, with the mixing weights depending on the rung. The prior is trusted more at higher rungs, where evaluations are expensive and the expert’s opinion matters most, and the incumbent component is switched on only once enough evidence has accumulated, which is what

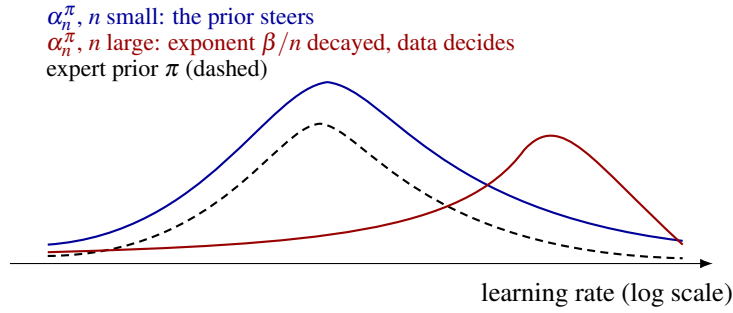


Figure 8.1: The π BO mechanism (8.1) over a study’s lifetime (schematic). The expert’s prior (dashed) multiplies the acquisition with an exponent that decays as observations arrive: early proposals concentrate where the expert pointed, late proposals go where the data says, and a wrong prior is forgotten rather than obeyed.

protects the method against a confidently wrong prior (Mallik et al., 2023). Priors of this kind should be a first-class field in our search-space schema from day one: every project lead we serve has them, and both mechanisms are simple to implement over components we are building anyway. The prior is also the natural home for the folklore this chapter opened with; writing $\pi(\lambda)$ down once beats re-teaching it to every study.

8.2 Transfer from our own past studies

The second grade learns from previous tuning studies instead of asking a human. The robust end of this spectrum is unglamorous and effective: initialize a new study with configurations that ranked well on similar past tasks (Feurer et al., 2015), or combine surrogates fit to past tasks into an ensemble whose weights reflect how well each past task’s surrogate predicts the current one, comparing on ranks so that tasks with different loss scales can vote together (Feurer et al., 2018). Because our product will keep every trial of every study in one store (Chapter 12), the warm start costs a database query: the run store is Figure 2.1’s bottom box earning its place.

The research frontier goes further and meta-learns the surrogate itself, and the vehicle deserves a proper introduction because it has appeared twice already (ifBO in Chapter 5) and will appear again. A *prior-fitted network* (PFN) is a transformer trained once, offline, on millions of synthetic datasets drawn from a chosen prior over functions. At use time it is shown the current study’s observed (configuration, score) pairs as context, and it outputs a predictive distribution for new configurations in a single forward pass.

Why that works is the part worth pausing on. Training on draws from the prior teaches the network to approximate the posterior predictive distribution under that prior, so the expensive part of Bayesian inference is paid once, by the pre-trainer, rather than at every step of every study. Where the GP of Section 4.1 refits (4.4) inside the loop, a PFN just reads its context. The model class changed; the economics changed with it.

PFNs4BO demonstrates the swap directly, pre-training PFNs that mimic GP and Bayesian-neural-network surrogates and slotting them into standard BO (Müller et al., 2023).

OptFormer showed the scale endgame: a transformer trained on hundreds of thousands of tuning trajectories from Google’s Vizier service imitates several tuners and improves on them when combined with expected improvement (Chen et al., 2022).

The same idea specialized to learning curves gives ifBO’s freeze-thaw surrogate (Chapter 5), and the line is still moving, with in-context amortization now reaching the acquisition function itself (Rakotoarison et al., 2026).

These pre-trained surrogates are the most credible path to tuners that get smarter as our run store grows, and one of them (ifBO) ships with usable code today. We rate them a second-wave investment: valuable, but only after the plain warm start is in place and earning.

8.3 Cost-aware search

The third grade treats the price of a trial as part of the decision. Trials rarely cost the same: in the running example, doubling network width or unroll length can quadruple wall-clock, so two candidates with equal expected improvement can differ fourfold in what they cost to evaluate, and an acquisition that ignores this will happily spend its budget in the expensive corner of the space. The classical repair prices the acquisition in the right currency: improvement per unit cost, $\alpha_{EI}(\lambda)/c(\lambda)$, with c a model of trial cost, already present in Snoek et al. (2012) as expected improvement per second. CARBO refines it with an explicit cheap-first initialization and a cost-cooling schedule, dividing by $c(\lambda)^\alpha$ with an exponent that anneals from one toward zero over the budget, so that early exploration favors cheap points and the endgame, where only the best candidates remain, ignores cost (Lee et al., 2020); the annealing acknowledges that a bargain matters when we are learning the space and stops mattering when we are choosing the winner.

CARBS pushes cost-awareness to its logical end for scaling studies. It maintains a Pareto front in the plane of cost versus performance, the machinery of Chapter 7 with cost promoted to a full objective, and searches locally around that front: separate Gaussian processes model performance, cost, and the front’s shape, candidates are Gaussian perturbations of current Pareto members, and the acquisition is expected improvement relative to the front weighted by expected cost, so the method tunes hyperparameters *while* the model grows and emits a scaling law for every hyperparameter as a byproduct. Its authors demonstrate it by reproducing the Chinchilla scaling result for language models and by solving the full ProcGen benchmark with a tuned but otherwise plain PPO (Fetterman et al., 2023). For our fixed-size studies, EI-per-cost with cost cooling is the sensible default; a CARBS-style cost frontier becomes interesting the day we tune models meant to be scaled up afterward.

8.4 When not to search at all: transfer across scale

For large networks there is a fourth option that is not search. The maximal update parametrization (μ P) rescales initialization and per-layer learning rates so that the size of each layer’s update stays controlled as network width grows; under that parametrization the optimal values of many hyperparameters become approximately invariant to width, so one tunes a small proxy model and transfers the result to the wide target (μ Transfer). Its authors validated the recipe by outperforming the published BERT-large and GPT-3 6.7B settings while spending a few percent of one pretraining run on tuning (Yang et al., 2022). The claim’s boundary is part of the result: the theory covers width; transfer across depth, batch size, and training horizon is empirical, with stated failure cases (regularization hyperparameters do not transfer, and depth transfer breaks for post-layer-norm transformers), and follow-up parametrizations have since extended the depth story (Dey et al., 2025).

More recent LLM-era work makes the third regime of Figure 8.2 explicit, fitting power laws for optimal hyperparameters as functions of scale and turning the largest tuning problems into curve lookups: compute-dependent learning-rate and batch-size laws appear in the DeepSeek LLM report (DeepSeek-AI,

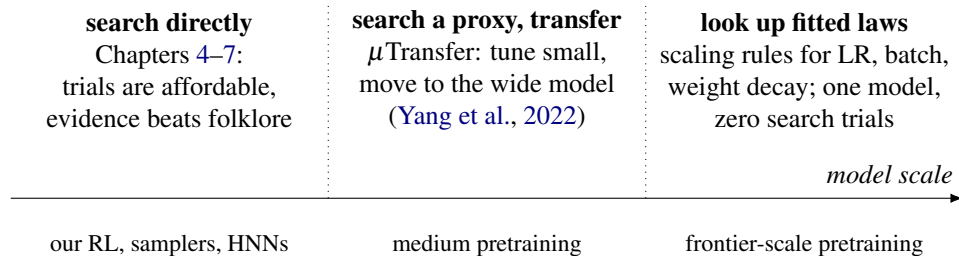


Figure 8.2: The three tuning regimes by model scale. Everything before this chapter lives in the left band. As trials become unaffordable, search moves onto small proxies whose optima provably or empirically transfer, and at the largest scales it gives way to fitted scaling laws. A good tuner should know which band a study is in and say so.

2024), learning-rate dependence on the token horizon has its own fitted law (Bjorck et al., 2025), weight decay and batch size get the same treatment (Bergsma et al., 2025), and a dedicated preprint fits joint optimum surfaces across model and data size (Li et al., 2025).

We track this line in Chapter 15 as guidance to encode rather than an algorithm to implement: the laws enter our product as priors and sanity checks, not as a searcher.

The strategic point for the product is the figure’s caption: at small scale we search, at medium scale we search on proxies and transfer, and at large scale search gives way to parametrization and extrapolation. A good tuner should know which regime it is in and say so.

8.5 The evidence, and how the failure mode was tested

Every claim in this chapter has the same shape: knowledge from outside the study, an expert’s hunch, a past study, a smaller model, someone else’s fitted law, can safely be spent inside it. For claims of that shape the success story is cheap to produce and nearly beside the point, because of course good advice helps. The evidence that matters is what happens when the imported knowledge is wrong, and the reason this chapter’s verdicts lean confident is that its sources, unusually, tested exactly that.

The π BO experiments are built around controllable advice quality. On a toy function and two surrogate tuning tasks, where the true optimum is known and priors of any desired quality can be manufactured, the authors ran the spectrum from accurate to “purposefully detrimental,” the wrong prior being a narrow density parked in the worst region of the space (Hvarfner et al., 2022b).

The findings match the mechanism read off formula (8.1): strong priors buy large early gains, and under sabotage the method climbs back to roughly the regret of its prior-free baseline, while the baseline that trusts the prior only at initialization keeps paying for it. The trade-off is stated rather than hidden: a larger β extracts more from good advice and forgives bad advice more slowly.

The upside case is demonstrated separately on a deep-learning pipeline, a 12.5-fold time-to-accuracy speedup over vanilla BO and a new best validation accuracy for that pipeline. Author-run experiments, as usual in this document’s sources; what tempers the caution here is that the safety half of the claim is also a theorem (the exponent β/n dies as evidence arrives, dragging the acquisition back to its data-driven self), so the experiment confirms an argument the reader can check rather than substituting for one.

PriorBand’s evaluation had to build its own arena first, a benchmark suite in which priors are part of the task definition rather than an afterthought: synthetic multi-fidelity surfaces plus image and language tasks derived from recorded large-model training runs, each equipped with good and deliberately bad priors, run at the small budgets that define deep-learning tuning (Mallik et al., 2023). The motivating

ablation is the honest part: naively biasing Hyperband’s sampler toward the prior wins when the advice is good and burns exactly proportional budget when it is bad, while a half-random hedge is robust but never competitive, and the rung-dependent trust schedule is the repair that keeps both properties. One design detail is worth copying into our own suite: every prior-using method in the comparison had the prior’s mode evaluated first, so nobody wins merely by being handed the expert’s best guess. Across tasks and prior qualities, PriorBand ranked most consistently at the top.

The transfer-across-scale evidence is differently shaped, and stronger for it. μ Transfer’s headline test pitted the recipe against external baselines its authors did not set: the published, expert-tuned settings of BERT-large and GPT-3 6.7B, beaten while spending a few percent of one pretraining run on proxy-model tuning (Yang et al., 2022), with the non-transfers (regularization strengths, depth under post-layer-norm) reported in the same paper. CARBS makes its case the same way, by reproducing a famous external result, the Chinchilla compute-optimal scaling law, and by solving the full ProcGen benchmark with an otherwise plain PPO (Fetterman et al., 2023). Replication of results the authors do not own is the closest thing this literature has to the adversarial arena of Section 4.5, and both transfer claims clear it.

The last tier is the one we deliberately did not audit. The scaling-law papers of the previous section report fitted power laws whose estimation we have not checked, and nothing in our product is allowed to depend on their exactness: they enter as priors and sanity checks, which is weight their evidentiary status can carry even if a particular fit is off. The demotion is a statement about our audit, not an accusation about their work.

8.6 The ordering, argued from four angles

The chapter’s verdict is an ordering: priors and warm starts first, cost-aware acquisition alongside, pre-trained surrogates adopted rather than trained, scaling knowledge encoded rather than implemented. Four angles converge on it.

From mechanism: the safety properties that make priors acceptable as a default are structural, not empirical. The decay exponent in (8.1) guarantees the expert’s influence is transient; PriorBand’s incumbent component stays off until evidence justifies it; in both, the worst case of bad advice is a study’s opening moves. Features whose failure modes are bounded by construction can be on by default; features whose failure modes are empirical questions cannot. That one sentence sorts most of the decision box.

From evidence: the sources tested the sabotage case, not only the success case, and the transfer methods validated against external baselines their authors did not control. Where our own audit is thinner, the scaling-law fits, the verdict steps down to encode-as-guidance in lockstep. Nothing here asks the reader to extend more trust than the checked evidence covers.

From economics: encoded experience is the cheapest compute we own. A written-down prior converts folklore into an asset that every study inherits for free; the run store turns past studies’ sunk compute into a warm start that costs a query; and EI-per-cost simply prices candidates in the currency the cluster actually charges, improvement per unit cost rather than improvement per trial. The whole chapter is arbitrage on knowledge the organization has already paid for once.

From operations: the entire first wave is days of work riding components that earlier chapters build anyway, a multiplier on the acquisition, a query over the run store, a divisor on the acquisition, and the single costly commitment, making the prior a first-class field of the search-space schema, costs nearly nothing on day one and a migration if retrofitted later. Adopting ifBO’s published surrogate rather than training our own keeps a research program off the product’s critical path.

And the falsification criterion, stated in advance: the validation suite reruns the pinned tasks under three conditions, no prior, the folklore prior, and a deliberately wrong prior in π BO’s sense, at our routine

budgets. The prior-weighted searcher must beat its prior-free self under the folklore prior and must recover to the prior-free final, within noise, under the wrong one; if bad advice costs more than the opening moves on our workloads, prior weighting is demoted from default to opt-in. The warm start faces its own test the day the run store holds enough studies: measured trial savings on a new study of a familiar family, or it stays a query we offer rather than a step we perform.

Decisions from this chapter. Ordered by payoff per week of engineering, which is the honest ranking here.

- **π BO’s prior-weighted acquisition** (include; days). Formula (8.1) is a few lines over the Tier 0 GP searcher, every project lead has a prior to give it, and its built-in decay means bad advice costs a study its opening moves rather than its conclusion, *provided the prior does not assign zero probability to the true region*. The search-space schema should therefore require a lower-bounded mixture or explicitly nonzero support everywhere, so that wrong advice can be overruled by evidence (Hvarfner et al., 2022b). Highest value density in the document under that guard.
- **PriorBand** (include; one to two weeks). The same idea where our savings actually live, the multi-fidelity rungs; the rung-dependent portfolio is what keeps a confidently wrong prior from doing damage (Mallik et al., 2023).
- **Warm starts from the run store** (include; days). A ranked query over past studies (Feurer et al., 2015). The prerequisite is the run store itself, which Tier 0 builds regardless.
- **RGPE-style surrogate ensembles** (second wave). Real gains, but only once the store holds enough completed studies to ensemble; building it before the data exists would be decoration.
- **Pre-trained surrogates (PFN family)** (adopt one, build none). We take ifBO’s published pre-trained model as our learning-curve surrogate (Chapter 5) and train nothing ourselves; OptFormer-scale training runs are a research program, not a package feature.
- **EI-per-cost with cost cooling** (include; days). One division and one annealing exponent on top of the GP searcher, *dividing by a predicted cost* rather than the realized cost the executor measures (Lee et al., 2020). A second GP fits log wall-clock time to configuration coordinates, refitting on the same cadence as the performance surrogate, with failed and killed trials entered as censored observations. The roadmap details the cold-start and calibration-logging protocol.
- **CARBS-style cost frontier** (hold). Becomes relevant the day a project tunes a model that is destined to be scaled up; none is on the books today (Fetterman et al., 2023).
- **μ P and the scaling-law literature** (encode, do not implement). These enter the package as priors, defaults, and sanity checks on the regime map of Figure 8.2, not as a searcher; at our current scales they advise rather than replace search.

8.7 What would overturn these verdicts

The π BO prior weight’s default-on status depends on V11: if gains under a good prior do not materialize, or if recovery from a wrong prior takes longer than the no-prior baseline, the feature is demoted to opt-in. PriorBand’s inclusion holds as long as its ASHA dependency does; if ASHA is demoted, PriorBand’s rung-sampling portfolio loses its substrate. Warm starts (V12) mature as the store fills; if trial savings remain undetectable after the store holds dozens of studies, the query surface is re-examined. EI-per-cost’s inclusion assumes executor cost metering is cheap; if metering overhead becomes visible in profiling, the feature becomes opt-in. RGPE’s second-wave hold is lifted when an ensemble that cheap proves itself on public suites, at which point the evidence section is updated and a new decision box opened.

Part V

From methods to our system

Chapter 9

Our workloads, concretely

The preceding chapters organized the literature by method. This one flips the table and asks, for each family of work in our repositories, what the evidence says we should actually do. Each section ends in a concrete default: the study we would configure today, with the chapters that justify each choice.

9.1 Reinforcement learning

The controlled evidence base here is unusually good, and we have already leaned on it: [Eimer et al. \(2023\)](#) tuned PPO, SAC, and DQN across eleven environments with matched budgets and seeds. Four of their findings should directly shape our defaults. First, hyperparameter importance is broad but concentrated: almost every swept knob mattered somewhere, importance analysis attributed most variance to one or two knobs per environment, and which knobs those were changed across environments, so the search space should include everything plausible rather than a hand-picked pair. This is the effective-dimensionality picture of Figure 3.1 with a twist: the important axes exist, but we cannot know in advance which they are, which is precisely the situation where random-style coverage of every axis, and GP lengthscales that learn to switch axes off (Section 4.1), earn their keep.

Second, the response surfaces themselves looked smooth and well behaved; the difficulty of tuning RL comes overwhelmingly from seed noise, not from a pathological objective. That is good news for every surrogate in Chapter 4, and it aims the engineering effort at noise handling (the noise-aware acquisitions of Section 4.2) rather than at exotic models.

Third, the seed noise is severe enough to corrupt conclusions: configurations tuned on one seed transferred poorly, incumbents evaluated on tuning seeds overstated test performance several-fold in the worst cases, and their protocol of tuning on a handful of seeds and reporting on disjoint test seeds is the minimum honest practice.

Fourth, at budgets of ten to sixty-four full training runs, DEHB was the most dependable tuner, with random search respectable and population methods behind; the population methods' natural habitat is the large-parallelism regime of Chapter 6, where BG-PBT's own experiments live.

Assembled into a default: an RL study below a few dozen full-run equivalents uses DEHB (or BO plus ASHA) over the full plausible space, evaluates each candidate on several tuning seeds, reports on disjoint test seeds, and holds the tuning budget fixed in any method comparison; a study with eight or more dedicated full-length workers and a schedule-shaped question upgrades to the population line, PB2-Mix now and BG-PBT when we build it.

The broader AutoRL literature, surveyed by [Parker-Holder et al. \(2022\)](#), adds the framing that hyperparameters, architecture, and even algorithm components form one design space, but the budget-

dependent boundary above is the piece with direct operational consequence for us.

Two later results sharpen the picture. The ARLBench benchmark, built from hyperparameter-landscape data across many environments, reports that RL tuning landscapes are less benign than the supervised-learning ones and selects small representative environment subsets that cut the cost of evaluating a tuner by roughly an order of magnitude, which is precisely what our validation suite needs (Becktepe et al., 2024). And a methodology paper on hyperparameter sensitivity warns that some reported algorithmic improvements in RL amount to increased reliance on tuning, one more reason to hold tuning budgets fixed when comparing methods (Adkins et al., 2024).

For meta-gradient methods that adapt hyperparameters inside a single run, such as STAC (Zahavy et al., 2020), our position is to treat them as properties of the RL algorithm, not of the tuner: worth enabling when RLLib exposes them, outside our product’s scope to implement.

One more RL-specific decision deserves sunlight: what number the tuner optimizes. Final evaluated return conflates stability with peak performance; area under the training curve rewards fast learners; return at a fixed step budget penalizes slow starters that would win given more steps. The three orderings genuinely disagree, and Figure 5.1 shows the disagreement in miniature: at the final step the blue configuration wins, on area under the curve the red one may, and a truncated budget flips the verdict again. The choice changes rankings, it interacts with early stopping (a curve that plateaus late looks like a curve worth killing), and it should be an explicit, recorded field of every study rather than a hidden default.

When return and stability both matter, nothing stops us from declaring them as two objectives and using the machinery of Chapter 7; the multi-objective RL literature makes the matching argument on the agent’s side, warning against collapsing genuinely conflicting objectives into a fixed linear combination (Hayes et al., 2022), which is Figure 7.2’s lesson wearing a different hat. We found no dedicated literature on multi-objective hyperparameter tuning for RL specifically; the generic machinery is the state of the art.

9.2 Neural-transport samplers

Our NeuTra-style work trains a normalizing flow to reshape a posterior, then runs Hamiltonian Monte Carlo in the warped space (Hoffman et al., 2019). Tuning has two loops, and the division of labor matters enough to draw (Figure 9.1).

The inner loop belongs to the samplers themselves: step size adapts by the dual-averaging scheme introduced with the No-U-Turn Sampler (NUTS), trajectory length by NUTS’s recursive doubling (Hoffman and Gelman, 2014) or, in the many-parallel-chains regime that accelerators favor, by gradient-based adaptation of the trajectory length against a mixing proxy, as in ChEES-HMC (Hoffman et al., 2021) and its successor that maximizes a bound on effective sample size per gradient along an estimated principal component (Sountsov and Hoffman, 2021).

Where good internal adaptation exists, the outer tuner should not fight it: a tuner that overrides the step size NUTS would have chosen is spending trials to re-derive dual averaging, badly. What remains for the outer loop is everything the samplers cannot adapt on their own: the flow’s architecture (depth, width, number of transformations), its optimizer and training length, the variational objective’s minibatch size, and the target acceptance rate handed to the inner adapters.

The outer objective already has a citable template, and it is the NeuTra paper’s own. Its authors selected step size and leapfrog steps by Bayesian optimization on an objective that rewards effective sample size (ESS, the number of independent draws a correlated chain is worth) per gradient evaluation while sharply penalizing configurations whose potential-scale-reduction diagnostic $\hat{R}$ departs from one (Hoffman et al., 2019); tuning samplers by BO goes back at least to Mahendran et al. (2012). Effective samples per gradient is the right currency because gradient evaluations dominate cost in both flow training

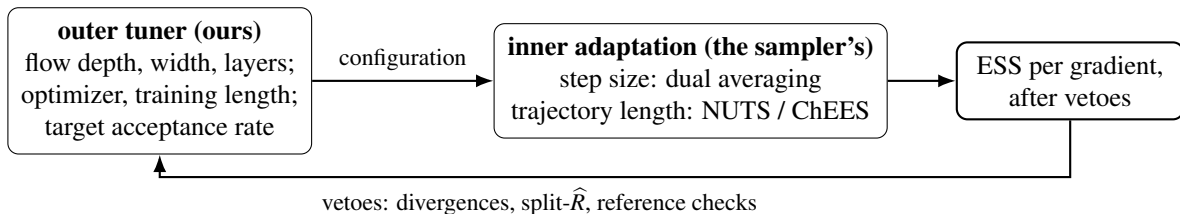


Figure 9.1: The two tuning loops of a neural-transport sampler. The inner adapters already solve step size and trajectory length; the outer tuner owns what they cannot reach, and its objective is efficiency (effective samples per gradient) gated by correctness vetoes that no efficiency score can buy off.

and leapfrog integration.

But ESS estimated from short chains is exactly the kind of proxy that Chapter 5 warned about, and sampler tuning has a sharper failure mode than slow starters: a chain that mixes quickly inside one mode of a multimodal posterior reports excellent short-run ESS while being wrong, the statistical equivalent of Figure 5.1’s sprinter, except that this sprinter never stalls visibly; it simply reports the wrong answer with growing confidence.

The remedy is to treat correctness diagnostics as vetoes rather than objectives: divergence counts, split $\hat{R}$ across independent chains, and, where a reference posterior is available on a pilot problem, agreement with it. A configuration that fails a veto is discarded no matter how efficient it looks; among configurations that pass, ESS per gradient is a legitimate objective. Multi-fidelity structure is still available (shorter chains, fewer chains), but rung promotion should require the vetoes to pass at every rung, which is exactly the kind of user-defined gate our scheduler interfaces must support (Chapter 12).

The concrete default: a single-objective study on ESS per gradient with divergence and split- $\hat{R}$ vetoes at every rung, ASHA over chain length as the scheduler, the GP searcher of Chapter 4 over the flow and training knobs, and the inner adapters left switched on with only their target acceptance rate exposed to the search.

9.3 Hamiltonian and physics-structured networks

Hamiltonian neural networks learn a scalar function whose symplectic gradient supplies the dynamics; conservation comes from the architecture rather than from a penalty term, and the original paper reports models that conserve an energy-like quantity where an unconstrained baseline drifts (Greydanus et al., 2019). Notably for us, its authors chose their hyperparameters by a coarse grid over learning rate, layer width, activation, and batch size, which is precisely the procedure Chapter 3 retired; the field’s own tuning practice trails its modeling practice, and that gap is our opportunity.

Tuning these models well is multi-objective in the evaluation even when the training loss is a single term: short-horizon prediction error, long-rollout state error, and energy drift move separately, the training loss sees only the first, and the scientific value lives in the second and third.

The axes of Figure 7.1 were chosen with this workload in mind, and the study template is the one that figure illustrates: a two-or-three-objective search (prediction error, long-rollout error or energy drift, optionally training cost) solved with qLogNEHVI or MO-ASHA from Chapter 7, over a space that includes integrator choice and step size, width and depth, activation (smoothness matters when the learned function is differentiated), learning rate, and loss weights where present. Where variants do carry multi-term losses (the pixel-observation task in the original paper combines the dynamics loss with autoencoder terms), the weights joining them are exactly the knobs an outer tuner should own.

Trials are cheap, minutes rather than hours, so this is also the workload where the tuner’s own overhead (Section 2.3) is most visible and the local execution backend of Chapter 12 earns its keep.

A neighboring literature balances competing loss terms inside training rather than outside it: gradient-statistics reweighting for physics-informed networks (Wang et al., 2020a), weights set from neural-tangent-kernel eigenvalues (Wang et al., 2020b), and relative loss balancing schemes compared systematically by Bischof and Kraus (2025). The division-of-labor rule from the sampler subsection transfers unchanged: where internal balancing works, let the trainer balance and let the tuner search what the trainer cannot, and when in doubt, the internal balancer is a competitor our outer loop must beat at matched cost, not a rival to ignore.

9.4 Architecture search, in moderation

Our NAS folder is full of ambitious methods, and the sober reading of the field is that a small team should adopt its cheap findings and skip its expensive machinery. The thousand-paper survey of White et al. (2023) recommends random search as the mandatory baseline for any NAS claim (Chapter 3’s discipline, enforced field-wide) and records that simple quantities such as parameter count and FLOPs (floating-point operation counts) are surprisingly competitive with the leading zero-cost proxies, whose reliability degrades on larger search spaces. Systematic bench-marking of those proxies found their value to lie in complementing a learned performance predictor rather than in standalone selection (Krishnakumar et al., 2022).

On the one-shot side, the DARTS family is documented to produce degenerate architectures across a range of search spaces without careful regularization (Zela et al., 2020), and controlled evaluations found much of the reported progress attributable to training tricks rather than to the search itself, with hand-designed macro-structure mattering more than the searched cells (Yang et al., 2020).

Meanwhile the benchmark built for joint architecture-and-hyperparameter search motivates itself with the observation that good hyperparameters can matter more than the best architecture (Bansal et al., 2022).

For us, the conclusion is comfortable: architecture search should mean width, depth, and a few structural flags included as ordinal and categorical axes in the ordinary search space (as BG-PBT does for policy and value networks, with the ordinal-aware kernels of Section 4.4 doing the modeling), multi-fidelity doing the heavy lifting, parameter count reported as an objective when size matters, and no dedicated weight-sharing subsystem. That covers every architecture question our current projects actually ask, at a small fraction of the complexity.

Because “in moderation” is the kind of phrase that drifts, the boundary is worth stating as a scope rather than as a mood. Inside it: width and depth as ordinal axes, a small number of structural flags (normalization choice, residual connections, activation family) as categorical axes, parameter count and measured FLOPs available as objectives for the multi-objective machinery of Chapter 7 rather than as proxies substituting for training, and architecture coordinates mutating through the same explore step every other coordinate uses, which is what BG-PBT’s architecture handling amounts to once the weight-sharing is stripped out (Chapter 6). Outside it, deliberately: topology and cell search, one-shot or weight-sharing supernet, a zero-cost proxy subsystem, grammar-based or hierarchical architecture spaces (Schrodi et al., 2023), and differentiable search of any flavor. The exclusions are not a claim that those methods fail; they are the claim this chapter’s evidence actually supports, which is that they cost a subsystem each and buy nothing our workloads have asked for. A project that needs them needs a different governing document, and the honest move then is to write one rather than to widen this sentence.

One consequence is worth flagging where it lands. Changing width or depth mid-run invalidates the weights a population method wants to inherit, so a population arm that mutates an architecture coordinate

pays for the change: either it restarts the trial from initialization, or it transfers what it can by distillation from the parent, which is the route BG-PBT takes (Wan et al., 2022) and which Chapter 6 treats as part of the population line’s price rather than as free.

9.5 The compute bill, and the shortcuts we are asked about

Tuning reinforcement learning is expensive enough that the fair question is not “which method” but “can we afford any of this,” so the bill deserves to be written out. A study’s cost factors cleanly into three multiplied terms: the number of candidate configurations examined, times the seeds per candidate that Chapter 9’s protocol demands, times the training steps per trial. In the running example the last factor alone is up to 150 million environment steps, hours on several accelerators, so a naive 64-candidate, 3-seed, full-length study is roughly two hundred full training runs before anything has been learned about tuning at all.

Every lever this document recommends attacks one factor: multi-fidelity scheduling cuts steps per trial by an order of magnitude (Chapter 5, and it is the biggest single lever); model-based search and warm starts cut the number of candidates (Chapters 4 and 8); the seed protocol stops compute being spent optimizing luck; population methods make the trials *be* the training run rather than a rehearsal for it (Chapter 6); and ARLBench-style representative benchmark subsets cut the cost of evaluating tuner changes by roughly an order of magnitude (Becktepe et al., 2024), which is what makes our own validation suite affordable.

Against that backdrop, three architecture-search shortcuts come up whenever the bill is discussed, and each deserves a plain description and a verdict.

One-shot search trains a single “supernet” that contains every candidate architecture as a sub-network with shared weights; candidates are then scored by inheriting weights from the supernet instead of being trained separately, so one big training run amortizes thousands of evaluations. The economics are attractive and the evidence is not. The DARTS family of one-shot methods is documented to produce degenerate architectures across search spaces without careful regularization (Zela et al., 2020); controlled re-evaluations attribute much of the reported progress to training tricks rather than the search, with hand-designed macro-structure mattering more than the searched cells (Yang et al., 2020); and the field’s own thousand-paper survey responds by making random search the mandatory baseline (White et al., 2023).

Two further points are specific to us rather than to the literature, and we label them as our inference: supernet training assumes a stable supervised training signal, while RL’s data distribution shifts under the policy as it learns; and our architecture questions concern small policy and value networks, where there is not enough training to amortize for the supernet’s economy to pay. Verdict: no one-shot subsystem, for reliability reasons first and fit reasons second.

Zero-shot search (zero-cost proxies) scores architectures at initialization, without any training, from quantities like gradient statistics at the first batch; the promise is architecture selection for the price of a forward-backward pass. The systematic benchmark of these proxies found their value to lie in complementing a learned performance predictor rather than in standalone selection, with reliability degrading on larger spaces (Krishnakumar et al., 2022), and the survey notes that plain parameter count and FLOPs are surprisingly competitive with the sophisticated proxies (White et al., 2023). That last fact is the actionable one: params and FLOPs are free, we already record them in the run store, and they can serve as objectives or tie-breakers today. Verdict: report the free quantities; build no proxy subsystem.

Grammar-based spaces attack a different problem than cost. A context-free grammar generates architectures the way a grammar generates sentences: production rules expand high-level structure into

blocks and blocks into operations, which compactly defines hierarchical search spaces vastly larger than the usual cell catalogs, searched with Bayesian optimization over hierarchical kernels (Schrodi et al., 2023); NePS ships support for such spaces (Chapter 10). The honest reading for a manager: this is a solution to *expressiveness*, letting one search whole macro-structures rather than a fixed menu, and a larger space raises rather than lowers the number of evaluations needed. It becomes relevant the day a project genuinely needs to discover topology; every architecture question our projects ask today is width, depth, and flags, where the benchmark evidence says hyperparameters matter at least as much as the architecture anyway (Bansal et al., 2022). Verdict: not in the package now; NePS is the reference implementation if a topology-search project ever lands.

The summary sentence for the section: the compute problem in RL tuning is real, and its solution in our package is the stack of levers in the first paragraph, not an architecture-search shortcut; the shortcuts either lack reliability evidence (one-shot), reduce to free quantities we already record (zero-shot), or solve a problem we do not have (grammar spaces).

Decisions from this chapter. Per workload, the study we configure today; and the architecture-search verdicts.

- **RL, routine budgets** (a few dozen full runs): DEHB or BO-plus-ASHA over the full plausible space, several tuning seeds, disjoint test seeds, objective recorded explicitly. **RL, large parallelism** (8+ workers): the population line, PB2-Mix now, BG-PBT when built.
- **Neural-transport samplers**: GP searcher over flow and training knobs, ASHA over chain length, ESS-per-gradient objective, divergence and split- $\hat{R}$ vetoes at every rung; leave the samplers' internal adapters on.
- **Hamiltonian networks**: two-to-three-objective study (prediction error, energy drift or rollout error, optionally cost) with qLogNEHVI or MO-ASHA, on the local execution backend.
- **Architecture search**: width, depth, and flags as ordinary search axes; parameter count and FLOPs reported as free objectives. **One-shot supernets**: exclude (reliability evidence against, RL mismatch). **Zero-cost proxy subsystem**: exclude (free quantities capture most of the value). **Grammar-based spaces**: exclude for now (solves expressiveness, not cost; no current project needs topology discovery); NePS is the reference if that changes.

9.6 What would overturn these verdicts

The routine-budget RL verdict (DEHB or BO+ASHA) is decided by V07; whichever loses is excluded and this section rewritten. The sampler study's veto gates are tied to V13: if vetoes prove unreliable (failing configurations promoted, or survivors ranking poorly against the reference posterior), correctness checking is pulled back into training code and the tuner's veto hook is disabled. The Hamiltonian study's reliance on the premium MO searcher follows V09's outcome; if qLogNEHVI is demoted, the study defaults to scalarization. The workload-template abstraction itself (four matched studies, not a general tuner) holds unless colleagues request a fifth workload family with evidence that it differs structurally from the four, at which point a new template is argued and added or the request is accommodated by parameterizing an existing one.

Chapter 10

The open-source libraries, one by one

The method chapters told us what is worth running; this one asks who actually ships it, and in what condition. Every version number, maintenance signal, and feature claim in this chapter was checked against the project’s own repository, package index, release notes, or documentation on August 22, 2026, and re-checked on August 23; the summary appears in Table 10.1. Maintenance status is a fact with a date on it, not a permanent verdict, and it matters here as much as algorithmic content: several famous names below are no longer safe foundations. A useful way to read the section is to sort the field by what each project fundamentally is: a few are *execution frameworks* that run and schedule trials, most are *search engines* that propose configurations and expect someone else to run them, and two are *services* wrapping a search engine behind a network API. Our product needs exactly one good member of the first kind and a curated set of the second.

10.1 Execution frameworks

Ray Tune. Part of Ray (release 2.57.0, August 11, 2026, Apache 2.0), Tune runs distributed trials natively on a Ray cluster, with fault tolerance and a stable Tuner interface. Its scheduler side is genuinely strong: ASHA, Hyperband, median stopping, a BOHB scheduler, and the only mainstream native distributed implementations of population-based training, PBT with replay, and PB2, with the documented requirement that the trainable support checkpoint save and restore (Liaw et al., 2018; Moritz et al., 2018). Two findings from the release history temper the picture. First, across the five Ray releases from December 2025 to August 2026, every Tune-specific change was a documentation restructure, example, or bug fix; no new algorithm or capability landed in that window. Second, the roster of bundled search integrations has been deliberately pruned: Dragonfly, SigOpt, scikit-optimize, and FLAML’s searchers are gone from the current documentation, leaving wrappers for Ax, bayesian-optimization, BOHB (whose upstream is unmaintained, see below), HEBO, Hyperopt, Nevergrad, Optuna, and ZOOpt. Core Tune has no native multi-objective support: the tuner takes one metric and one mode, and Pareto workloads pass through the Optuna searcher, which accepts lists of metrics. The fair summary: a first-class execution and scheduling substrate whose own search intelligence is coasting, which is precisely the shape of dependency we want underneath a search layer of our own.

Syne Tune. Originally from AWS, now in its own community organization (release 0.16.0, August 3, 2026, Apache 2.0, about 420 GitHub stars), Syne Tune is the closest existing thing to the product this document proposes: a broad method zoo (multi-fidelity, multi-objective, transfer, and population-based training all claimed in its README) behind a backend abstraction that can execute trials locally, on

SageMaker, or inside a simulator that replays tabular and surrogate benchmarks faster than real time (Salinas et al., 2022). That simulator is unique in the field and exactly the right idea for cheap algorithm regression tests. The concerns are social rather than technical: a small community and a post-corporate maintenance arrangement, with no Ray backend, so adopting it as our substrate would mean owning a second execution stack beside the one RLlib already requires. We take its backend-abstraction design and its simulator idea as blueprints rather than its code as foundation.

Katib. Kubeflow’s Kubernetes-native tuner (v0.19.0, October 2025, Apache 2.0) runs trials as Kubernetes workloads defined by custom resources, with early stopping and a set of suggestion services that mostly wrap Optuna, Goptuna, and Hyperopt. It is the right shape for an organization whose platform is Kubeflow; on a Ray shop it duplicates the execution layer we already have while adding no search capability we could not get from the wrapped engines directly.

Determined and NNI. Two cautionary entries. Determined, the training platform whose adaptive ASHA implementation was well regarded, has had no repository activity since March 2025 and no release since then either, without any formal notice; do not build on it. NNI, Microsoft’s once-popular AutoML toolkit, is formally archived (last release September 2023), and its fourteen thousand GitHub stars are a trap for anyone ranking libraries by popularity.

10.2 Search engines

Optuna. Optuna (4.9.0 in June 2026, 5.0 release candidate in August 2026, MIT) is where sampler development is most active in 2026 (Akiba et al., 2019). The 5.0 line promotes a multivariate TPE with a constant-liar strategy for parallel workers to the default sampler; recent 4.x releases grew the GP sampler into constrained and multi-objective territory (a log-EHVI acquisition, plus batch-proposal strategies), an automatic sampler-selection module ships through the OptunaHub plugin registry, and a Rust core accelerates the standard samplers. Its define-by-run interface remains the best available way to express conditional search spaces, and NSGA-II is the default for multi-objective studies, with the multi-objective TPE variant (Ozaki et al., 2020) now folded into the standard sampler rather than shipped separately. The limitation is the mirror image of Ray Tune’s: Optuna executes nothing. Its documented multi-node story is a relational database that workers poll, with an explicit warning about file-based storage over NFS, and it has no population-based training. Optuna behind Ray Tune’s OptunaSearch is the pairing both projects document, and it is the pairing new tools reach for: DataRobot’s syftr, an agent-workflow optimizer released in 2025, states in its README that it builds on Ray for distributed execution and Optuna for define-by-run search and multi-objective optimization.

Ax and BoTorch. Meta’s pair covers the model-based frontier: BoTorch (0.18.1, June 2026, MIT) supplies the Gaussian-process machinery and the modern acquisition functions, including the qLogNEHVI family that Chapter 7 recommends, and its own documentation warns users off the older non-log variants; Ax (1.3.1, June 2026, MIT) wraps it in a clean ask-tell client aimed at practitioners, defaulting to noisy expected hypervolume improvement for multi-objective studies (Balandat et al., 2020). BoTorch’s README is candid that end users not doing BO research should use Ax instead. Neither executes trials; both are exactly the acquisition engine our GP searchers should be built on rather than reimplemented.

SMAC3. The AutoML group’s random-forest Bayesian optimizer (2.4.0, April 2026, BSD) remains actively developed, with multi-fidelity and multi-objective support across its facades and a 2.4.0 change of

default surrogate to scikit-learn’s random forest ([Lindauer et al., 2022](#)). It is the canonical engine for large conditional spaces and the tuner used as the sequential baseline in the BG-PBT paper. Its ergonomics are single-machine and academic; we would wrap it for awkward spaces rather than adopt its frame.

NePS. Neural Pipeline Search from the same group (0.16.0, March 2026, Apache 2.0, under a hundred stars but actively developed) is the reference home of the prior-and-multi-fidelity methods we care about: PriorBand, ifBO, and a multi-objective variant with expert priors, plus grammar-based hierarchical architecture spaces. Its parallelism model is workers coordinating through a shared results directory, simple and DDP-friendly, though the same design that Optuna’s documentation warns can race over NFS. For us it is a source of reference implementations and a correctness oracle for our own PriorBand and ifBO ports, not a substrate.

DEHB and HpBandSter. The two AutoML-group bandit engines are both frozen, in different ways. DEHB (0.1.2, mid 2024) announces in its README that it is maintained for stability but no longer actively developed; the algorithm remains the evidence-backed recommendation for small-budget RL tuning (Chapter 9), so the pragmatic path is to implement DEHB against our own interfaces with the frozen package as a reference. HpBandSter, the reference BOHB implementation, opens its README with “not maintained anymore”; notably, Ray Tune’s BOHB searcher still depends on it.

Hyperopt, Nevergrad, FLAML. Hyperopt (0.3.0, July 2026) received its first release since 2021, a compatibility revival with no new methods; it matters today mainly as the legacy TPE many codebases still import. Nevergrad (1.0.12, April 2025, sixteen months without a release at our check date) remains a broad evolutionary toolbox whose NGOpt wizard picks among CMA-ES, differential evolution, and dozens of other solvers by problem features; a good non-BO baseline for bake-offs. FLAML (2.6.0, April 2026, active) is the cost-frugal specialist: its CFO and BlendSearch searchers start cheap and spend more only when justified, with explicit fields for the cost-related hyperparameters ([Wang et al., 2021](#)); its Ray integration, however, has disappeared from Ray’s documentation, so using it means wiring it ourselves.

OpenBox. An academic black-box optimization service (0.9.0, March 2026, MIT) with an unusually broad claim sheet: multi-objective, constraints, transfer, early stopping, and a REST service mode ([Li et al., 2021](#)). Activity is modest and burst-like, and the service architecture duplicates infrastructure a Ray shop already runs. Worth watching, not depending on.

10.3 Services and partial offerings

Open-source Vizier. Google’s Vizier (client-server, Apache 2.0) publishes the most complete description of an industrial default anywhere: a Matérn-5/2 GP with MAP-fit hyperparameters, an output-warping stack that tames outliers, a deliberately explorative UCB coefficient, trust regions, and an evolutionary acquisition maximizer, benchmarked in its own report against algorithms accessed through Ray Tune ([Song et al., 2022, 2024](#)). The repository is active but the package on PyPI had not been released for eighteen months at our check date. Its value to us is the design disclosure more than the dependency.

Weights and Biases sweeps. The open pieces are the client and a small suggestion library (random, grid, a Bayesian sampler, Hyperband stopping) whose standalone package last shipped in 2022; orchestration and state live in the proprietary service. Fine as a convenience for W&B customers; not an engine to build on.

Table 10.1: Open-source tuning libraries as checked on August 22, 2026. “Maintained” reflects release and commit activity at that date, not a prediction. Stars are approximate GitHub counts.

Library	Release (date)	Status	One-line reading
Ray Tune	Ray 2.57.0 (2026-08)	active; Tune coasting	best executor; search layer aging
Optuna	4.9.0; 5.0rc1 (2026-08)	very active	best samplers; no executor
Ax	1.3.1 (2026-06)	active	polished BO client over BoTorch
BoTorch	0.18.1 (2026-06)	active	the acquisition engine to reuse
Syne Tune	0.16.0 (2026-08)	active, small team	broadest methods; design blueprint
SMAC3	2.4.0 (2026-04)	active	conditional-space BO plus racing
NePS	0.16.0 (2026-03)	active, small team	PriorBand and ifBO reference code
DEHB	0.1.2 (2024-07)	stability-only	algorithm good; package frozen
HpBandSter	0.7.4 (2018-11)	unmaintained	BOHB reference; do not depend
Hyperopt	0.3.0 (2026-07)	compatibility only	legacy TPE
Nevergrad	1.0.12 (2025-04)	slowing	evolutionary baseline (NGOpt)
FLAML	2.6.0 (2026-04)	active	cost-frugal searchers
OpenBox	0.9.0 (2026-03)	sporadic	broad claims, small team
OSS Vizier	0.1.24 (2025-02)	repo active, releases stalled	reference design disclosure
Katib	0.19.0 (2025-10)	active	Kubernetes-native; redundant on Ray
Determined	0.38.1 (2025-03)	dormant	no activity since early 2025
NNI	3.0 (2023-09)	archived	formally retired
Orion	0.2.7 (2023-03)	dormant	no release in over three years
W&B sweeps	lib 0.2.0 (2022-08)	minimal open core	orchestration is proprietary

New arrivals worth a footnote. The visible new energy of 2025 and 2026 sits in agentic tooling rather than classical search: an official Optuna server exposing studies to language-model agents, and a fast-growing project that has coding agents iterate on training configurations directly. None of these publish controlled evidence against the methods of Chapters 4–8 at matched budgets, so they change nothing in our plan, but they are worth rechecking in a year.

Part VI

The product proposal

Chapter 11

The product, concretely

Everything before this page weighed evidence; this part spends it. The reader changes here too: the chapters so far served whoever must judge our verdicts, and the chapters from here serve whoever must build from them, starting the week the plan is approved. The two jobs meet in one requirement, so we state it before any design.

Concrete has a checkable meaning in this part, and it is not “more detail.” A verdict from the decision boxes counts as delivered only when four anchors exist for it: a contract (which interface, schema, or component carries it), a test (which entry of the validation suite would catch its failure), a schedule slot (which tier and week builds it), and a price (whose weeks pay for it). The master table of Chapter 15 already lists every verdict with its tier and cost; this part’s job is to supply the two missing anchors, contracts and tests, and to close the loop so that no verdict lives only in a box.

The part runs in build order. This chapter says what the product is and walks one study through it on day one. Chapter 12 holds the architectural design, two interfaces, two adapters, and a database, unchanged from the verdict it argued. The chapters after it pin the contracts and the validation suite, and Chapter 15 closes the part with the schedule and the bill.

11.1 What we ship, and to whom

The product is an internal Python package: a tuning layer that our four workload families call, running on Ray Tune as its execution layer, exactly as Chapter 12 argues. It is not a service, not a platform, and not a research program; every research bet it contains (the learning-curve scheduler, the population flagship) is scheduled behind a working default, not in front of it.

The customers are the projects of Chapter 9, and the package meets each one with a configured study rather than a menu: the routine-budget RL study, the large-parallelism RL study, the sampler study with its correctness vetoes, and the two-objective Hamiltonian study. A researcher who knows only their own workload should be able to run the matching study without reading this part of the document; the chapters exist so that the person *extending* the package can reconstruct why each default is what it is.

What ships, in one sentence per layer: a search-space schema that knows the three knob kinds, conditional structure, and an optional prior; a searcher interface and a scheduler interface with wrapped implementations behind them (Optuna’s TPE and the GP searcher on one side, ASHA and the median rule on the other); a run store that records configurations, seeds, curves, costs, and lineage; the seed protocol as the default study shape rather than an option; and the validation suite, which is a deliverable of the same rank as the tuner, because every falsification promise made in Chapters 3 through 8 is an entry in it.

11.2 A study on day one

The fastest way to make the product concrete is to run one study through it in words, at tier 0, using only components that exist and are maintained today plus our thin glue. Nothing in this walk waits on research; that is what “wire up what exists” meant in Chapter 15, and the walk doubles as the acceptance test for tier 0.

A researcher opens a study for the running example: PPO on the Brax task, routine budget, the situation of Chapter 9. They declare the fifteen-knob space of Chapter 2 in the schema, unchanged and with no knob added: learning rate log-uniform between 10^{-5} and 10^{-2} , the ordinal widths, depths, and batch sizes as ordered integers, and the two spectral-normalization flags as categories. The BG-PBT space carries no conditional knob, so this study declares a flat space; the schema’s conditional support is exercised by the annealing switch of Chapter 2 and by the off-policy and sampler workloads of Chapter 9, and the capability test for it lives there rather than here. The schema also accepts the folklore prior (learning rate near 3×10^{-4} , discount near 0.99) as a first-class field. On day one nothing consumes it; recording it from day one is deliberate, because Chapter 8 showed that priors are cheap to store, expensive to retrofit, and eventually worth real trials.

They declare the objective explicitly, mean evaluated return aggregated over the declared tuning seeds, because Chapter 9 showed that final return, area under the curve, and truncated return genuinely disagree, and a hidden default here silently changes every ranking downstream. This tuning-seed mean is what the searcher observes and optimizes. After selection is frozen, the winner (or Pareto front) is evaluated once on disjoint test seeds that the tuner never saw, and that held-out performance is what the final report quotes.

This seed protocol is the study’s default shape: search driven by tuning-seed feedback, final evaluation on test seeds. This is not an option the researcher enables; it is the study shape, and opting *out* is the act that requires writing something down. The evidence for that inversion is the seed-noise record of Chapters 2 and 9, and the protocol is what makes the test set a genuine held-out evaluation rather than part of the search feedback loop.

The searcher on day one is the wrapped TPE; the scheduler is ASHA with the median rule beside it, granting first rungs at one tenth of full fidelity. While the study runs, every trial streams its curve into the run store with its configuration, seed, rung history, and bill, and the store computes one diagnostic the researcher did not ask for: the rank correlation between rung scores and final scores, logged per study. That number is Chapter 5’s crossing-curves hazard turned into a measured quantity of our own workloads, and the validation suite consumes it later.

What the researcher gets back is the incumbent configuration with its test-seed mean and spread, the full record of trials in the store, and the study’s own audit trail: which trials were stopped at which rungs, and what the early-versus-final correlation was. What they do not get on day one, stated as plainly: no schedules (the population line is tier 2), no premium multi-objective searcher (the Hamiltonian study runs on the wrapped evolutionary default until the qLogNEHVI searcher lands), and no prior-weighted acquisition yet. Each absence has a tier and a week in Chapter 15; none of them blocks the day-one study from being useful, which is the point of the tier ordering.

11.3 The register, and how this part uses it

Chapter 15’s Table 15.1 is the spine of this part: every verdict, its tier, its cost, and the chapter holding its evidence. The two chapters that follow extend each row with its remaining anchors. The contracts chapter names the interface or schema field that carries the verdict, so that a reader can point from “priors are

first-class” to the line of the schema that makes it true. The validation chapter names the suite entry that would catch the verdict failing, and it inherits its content from promises already made: every “argued from four angles” section in Chapters 3 through 8 ended with a falsification criterion, and the suite is those criteria collected, priced, and scheduled rather than restated.

Two bookkeeping rules keep the register honest while this part is written. A verdict with no plausible suite entry is marked untestable in place, with a sentence saying why, rather than receiving a decorative test. And any design sentence in this part that cannot cite a chapter of evidence is labeled an engineering judgment, so the reader always knows which statements the survey backs and which ones the build team owns.

11.4 What would overturn this chapter

The day-one walk is this chapter’s load-bearing claim: tier 0 is composition, not construction. It fails, and this part gets rewritten, if the walk cannot be reproduced in software within the tier 0 budget of Chapter 15, if any step turns out to require a component the audit of Chapter 10 did not verify as maintained, or if the seed-protocol default proves too expensive to be the default on our routine studies. The first validation-suite run is scheduled to test exactly this walk, and the suite, not this chapter, has the last word.

Chapter 12

Is Ray Tune the right base, and what we should build on it

The question that started this document can now be answered with the evidence laid out. Yes, Ray Tune is the right base, provided we are clear about which layer of the problem it solves. Tune should run our trials; it should not choose them. The search intelligence, which is where the field has moved fastest and where Tune has moved least, belongs in a layer we own. This chapter turns that sentence into a design, and the design is small: the whole proposal is two interfaces, two adapters, and a database.

12.1 The case for Tune underneath

Three facts carry the decision. First, we already operate Ray: RLlib training runs on the same cluster, speaks the same actor model, and produces the checkpoints that population-based methods need to copy and resume. A second execution stack would be pure overhead. Second, Tune’s execution machinery is the strongest available in open source for our shape of work: multi-node trial scheduling, fault tolerance, checkpoint-aware schedulers, and the only mainstream native distributed PBT and PB2 (Chapter 10). The population family of Chapter 6 is the part of our roadmap with the most demanding executor requirements, exploit steps are checkpoint surgery performed live on a running population, and Tune is the one framework that has already productionized them. Third, the composition we propose is the one the ecosystem itself converges on: Optuna and Ray document each other as the intended pairing, a 2025 industrial optimizer built its stack exactly this way, and Google’s published account of its own tuning service benchmarks against algorithms accessed through Ray Tune (Song et al., 2024).

The risks deserve equal daylight, and Chapter 10 documented them with dates. Tune’s own search layer has been coasting: across the five Ray releases from late 2025 through August 2026, every Tune change was a fix or a documentation edit, several of its remaining searcher integrations wrap upstreams that are frozen or formally unmaintained, and core Tune still has no native concept of a Pareto front, which after Chapter 7 we regard as a first-class requirement rather than a nicety. None of this argues against Tune as an executor; all of it argues against depending on Tune for algorithms. There is also a real overhead concern at the small end: for the cheap trials of our Hamiltonian-network work (Section 2.3), cluster machinery can cost more than the training it schedules.

12.2 The design

The consequence is a three-layer design, drawn in Figure 12.1. At the top, a search layer that is the actual product: searchers (ask-and-tell proposers: GP with qLogEI and qLogNEHVI on BoTorch, TPE via Optuna, CMA-ES, the mixed-space trust-region searcher that BG-PBT needs) and schedulers (ASHA and the median rule, their multi-objective variants with veto gates, the population controllers), all written against two small interfaces of our own. The split between the two interfaces is exactly the split drawn in Figure 2.1 at the start of the document: searchers answer “which configuration next,” schedulers answer “how much budget, and does this trial live”; every method surveyed decomposes into the two roles, and keeping the roles separate is what lets ASHA schedule trials proposed by any searcher.

In the middle, thin adapters: one maps our interfaces onto Ray Tune’s searcher and trial-scheduler protocols and is the default; one runs trials in local processes for laptop-scale studies where Ray does not earn its overhead; the design leaves room for a queue-based backend later, following the backend abstraction that Syne Tune demonstrated (Salinas et al., 2022). At the bottom, alongside the executors, a run store that records every trial of every study: configuration, seeds, fidelity-indexed curves, cost, diagnostics, and lineage (which weights were copied from whom, for population methods). The store is what turns the transfer methods of Chapter 8 from papers into features, warm starts are a query against it, and π BO priors and fitted cost models are trained from it, and it is also where a study’s evidence lives when we later ask whether the tuner is earning its keep.

To make the layering concrete, walk one trial of the running example through it. The GP searcher (top layer) proposes a configuration by maximizing qLogEI over the mixed space; the MO-ASHA scheduler assigns it a rung-1 budget of two million steps. The Ray adapter translates both decisions into a Tune trial with a checkpoint interval; RLLib trains it on the cluster and streams evaluation returns back. At the rung boundary the scheduler ranks the rung (non-dominated sorting if the study is multi-objective), checks the trial’s veto fields, and either promotes it, at which point the adapter resumes it from its checkpoint, or stops it. Every intermediate return, the cost, the seed, and the promotion decision land in the run store. Nothing in the top layer knows Ray exists; nothing below the adapters knows what an acquisition function is. That ignorance is the design.

Owning the two interfaces, rather than coding directly against Tune’s, is what keeps the Tune risks bounded. If Tune stagnates further or Ray’s APIs shift again, the adapters change and the product does not. It is also what lets the same searcher serve both executors, so a Hamiltonian study on a workstation and a PPO population on the cluster run the same code paths above the adapter line.

Three features must be first-class in those interfaces because retrofitting them is painful, and each traces to a specific finding earlier in the document. Multiple objectives and constraint vetoes: a scheduler must be able to rank partial results by non-dominated sorting and to refuse promotion to a trial that failed a correctness diagnostic, whatever its score; this is the promotion machinery of Chapter 7 and the veto gates that Figure 9.1 showed the sampler workload demanding. Seeds: a trial is a configuration times a seed, aggregation across seeds is the tuner’s job, and tuning seeds are disjoint from test seeds by construction, the protocol whose absence corrupted the RL results surveyed in Chapter 9 (Eimer et al., 2023). Schedules: the outcome of a population study is a trajectory with lineage, storable and replayable, the bold path of Figure 6.1 as a first-class object (Tune’s own PBT-replay scheduler shows the shape).

12.3 Alternatives we considered

Adopting Syne Tune outright would buy the broadest method zoo in one package, and its simulator backend is a genuinely good idea we intend to copy; against that stand a small maintainer base, no Ray

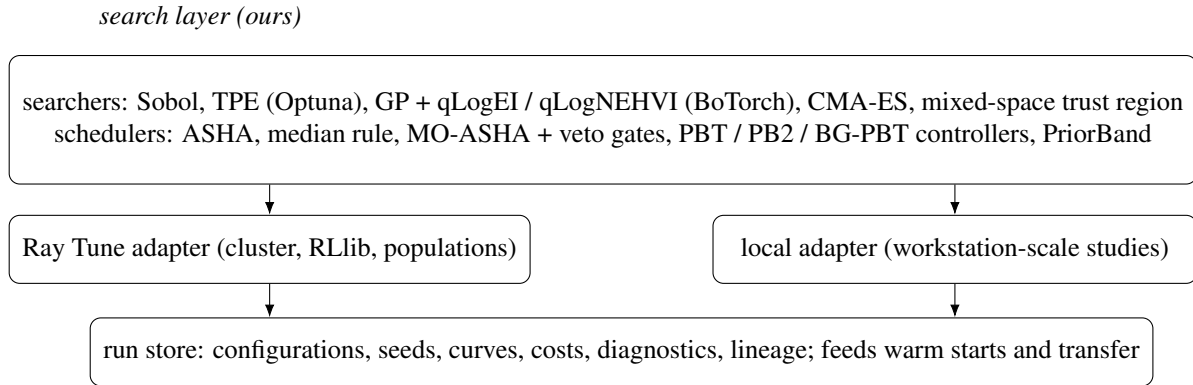


Figure 12.1: The proposed layering. Algorithms live above the adapter line and survive any change of executor; Ray Tune supplies distributed execution and checkpoint surgery; a shared run store accumulates the evidence that priors, warm starts, and transfer methods consume. The two interfaces at the adapter line are the searcher/scheduler split of Figure 2.1, made contractual.

backend, and a method set that still lacks the BG-PBT-class population line we care about.

Building on Optuna alone would give us the best samplers and nothing to run them with: its documented multi-node story is a shared relational database, and there is no population machinery. NePS has the freshest prior-and-fidelity algorithms and we will use it as a reference implementation and correctness oracle, but its filesystem coordination and small community rule it out as a substrate.

Katib solves execution for Kubernetes shops; we are a Ray shop. Writing our own executor from scratch is the option nobody should pick while Tune exists and is maintained. The remaining risk in our choice is Ray itself drifting in a direction that breaks Tune; the adapter boundary is the insurance, and the annual cost of that insurance is two small interfaces.

Chapter 13

Interfaces and data contracts

Chapter 12 argued that the whole proposal is two interfaces, two adapters, and a database. This chapter makes those five things exact enough to code against. The discipline throughout: every field and every method either names the verdict it carries, with the chapter holding the evidence, or is labeled an engineering judgment that the build team owns. A reader should be able to point from any line of the eventual code review back to a line of this survey, or know that no such line exists.

One vocabulary note before the details. A *study* is the unit a researcher opens: one search space, one objective declaration, one budget, many trials. The contracts below are organized the way a study meets them: first what the researcher declares, then the two decision-makers (searcher and scheduler), then the store that remembers everything, then the executors that pay the bills.

13.1 The study declaration

A study begins with a declaration, and the declaration is where several chapters' verdicts become unavoidable rather than optional.

The *objective declaration* names the quantity being optimized, its direction, and its evaluation protocol. For RL studies that means choosing among final evaluated return, area under the training curve, and return at a fixed budget, explicitly, because Chapter 9 showed the three orderings genuinely disagree; the field is mandatory and recorded, never defaulted silently. Multi-objective studies declare a vector of such objectives, and the product treats the front as the study's output from that moment (Chapter 7).

The *seed protocol* is part of the declaration, with defaults rather than blanks: a number of tuning seeds per candidate and a disjoint set of test seeds for the final report, the protocol of Eimer et al. (2023) that Chapters 2 and 9 adopted. Opting out is possible and is itself recorded, so a study that skipped the protocol says so in its own audit trail.

The *budget declaration* states the study's size in full-run equivalents and the number of concurrently available full-length workers. The second number exists because of Chapter 6's boundary: at eight or more dedicated workers with a schedule-shaped question the population line becomes eligible, and below it the product steers to the rung methods. The declaration is what lets the product enforce that boundary in software rather than trusting each user's optimism.

13.2 The search-space schema

The schema is a tree of knob declarations, one per hyperparameter, and its fields track exactly the structure Chapter 2 taught.

kind

One of continuous, ordinal, categorical. Three values, not the four kinds of Chapter 2, because the fourth is structural rather than typed: a conditional knob is a continuous, ordinal, or categorical knob that also carries a predicate, so conditionality lives in **condition** below. Integer-valued knobs are ordinals, the ordinal kind being defined as an integer with a natural order, and there is no separate integer kind to choose. The distinction that remains is load-bearing downstream: ordinals are modeled as ordered, never as unordered labels, because respecting the order produced measurable gains in the BG-PBT ablations (Chapter 4, Section 4.4), and categorical axes route to overlap kernels or per-category machinery rather than being embedded and hoped about.

range and transform

Bounds for continuous and ordinal knobs, the value list for categorical ones, and an optional transform, with log-uniform as the expected choice for scale-type knobs; the 90 percent of a uniform draw wasted in the top decade (Chapter 3) is the argument, and the same flag drives input warping in the GP searcher.

condition

An optional parent-knob-and-value predicate. A conditional knob exists only when its predicate holds, which makes the space a tree rather than a box; TPE consumes this natively, and searchers that cannot must say so through their capability flags below rather than silently sampling meaningless coordinates (Chapters 2 and 4).

prior

An optional distribution over the knob, the expert’s belief about where the optimum lies. First-class from day one because retrofitting it is expensive and every project lead has one (Chapter 8); consumed by the π BO weight and the PriorBand portfolio once tier 1 lands, recorded but inert before that.

note

Free text for units and meaning. An engineering judgment with a purpose: the schema doubles as the study’s documentation, and Chapter 2’s knob tour is the register this field is meant to echo.

13.3 The searcher interface

A searcher proposes configurations and learns from results. The interface is deliberately small.

propose(n, state)

Return n configurations to evaluate next. Batch proposals are the normal case, not the extension, because idle workers were Chapter 4’s standing argument; a GP searcher implements $n > 1$ by joint batch acquisition and handles still-running trials by fantasies, exactly the machinery of Section 4.2.

observe(record)

Ingest one trial record from the store, including partial-fidelity observations; a searcher that only understands final scores declares that limitation.

capabilities

A static declaration: which knob kinds, whether conditionals are supported, whether multiple objectives are, whether a prior is consumed, maximum sensible dimension. The product refuses a mismatched study loudly at declaration time. The flag exists because the audit of Chapter 10 found exactly such mismatches shipping silently (a continuous-only PB2 is the standing example), and refusing early is cheaper than diagnosing a silently degenerate study.

snapshot and restore

Serialize internal state, so long studies survive restarts and population methods can fork searcher state along a lineage. Engineering judgment on the mechanism; the requirement itself follows from the population line’s design (Chapter 6).

Behind this interface, tier 0 ships wrapped implementations (Optuna TPE, the BoTorch GP with qLogEI and dimension-scaled priors, CMA-ES, Sobol and log-uniform random), tier 1 adds qLogNEHVI, Chebyshev scalarization, the π BO weight, and EI-per-cost as decorators on the GP searcher, and the trust-region mixed-space searcher arrives as the shared core of high-dimensional single-objective search and the population line’s explore step (Chapters 4 and 6).

13.4 The scheduler interface

A scheduler allocates budget across running trials. Its contract has one more verb than the classical statement, because the population line needs it.

report(trial, fidelity, value)

Called as metrics stream in; returns continue, stop, or pause. ASHA and the median rule live entirely behind this verb, which is why they ask nothing of training code beyond streaming a metric (Chapter 5).

promote()

Return paused or new trials to run when a worker frees, with rung bookkeeping (η , initial budget r) internal to the scheduler. Asynchronous promotion against results-so-far is the ASHA rule; a PASHA flag stops growing the top rung once rankings stabilize.

gate(predicate)

Register a veto evaluated at every rung. A trial failing its gate is discarded regardless of its score; this is the sampler workload’s divergence and split- $\hat{R}$ machinery generalized into a hook (Chapter 9), and MO-ASHA’s correctness gates use the same mechanism (Chapter 7).

exploit(loser, winner) and explore(config)

The population verbs: copy weights and hyperparameters from winner to loser through the executor’s checkpoint surgery, then perturb or re-propose via a searcher. Keeping these on the scheduler interface, with the searcher consulted for explore, is precisely the entanglement Chapter 6 described, made explicit instead of implicit.

Every scheduler decision lands in the store with its reason, because the audit trail is a product feature, not a log file: the researcher of Chapter 11’s walk reads exactly this record.

13.5 The run store

The store is one database with three kinds of rows, and several chapters’ promises are really promises about these rows.

Trial rows carry the configuration, seed, rung history with per-rung scores and costs, the full learning curve, final objective values as a vector, veto outcomes, and a parent-trial pointer. The parent pointer is the population line’s lineage: with it, a hyperparameter *schedule* is a walk up the tree, queryable after the study, which is what “schedules are a first-class output” meant concretely in Chapter 6.

Study rows carry the declaration in full (space, objective, seeds, budget, boundary inputs), the incumbent or front over time, and the study-level diagnostics, of which one is computed always: the rank correlation between rung scores and final scores. Chapter 5 promised that the crossing-curves hazard would become a measured property of our workloads; this row is where the measurement lives, and the validation suite consumes it.

Query surfaces are part of the contract because two verdicts depend on them: warm starting a new study from configurations that ranked well on similar past studies (Chapter 8) is a ranked query over trial rows, and hypervolume-over-budget reporting (Chapter 15) is a scan over study rows. Storage technology is an engineering judgment; the schema above is not.

13.6 The executor adapters

Two adapters, per the architecture chapter. The Ray Tune adapter maps trials to Tune trials and must expose checkpoint surgery as first-class operations: save, copy weights and optimizer state across trials, resume. That is the machinery PBT exercises inside Tune today and the single strongest reason Chapter 12 kept Tune underneath; our adapter’s job is to keep it reachable from the scheduler’s exploit verb. The local adapter runs minutes-scale trials in-process for the Hamiltonian regime, where Chapter 9 showed the distributed stack’s overhead is the visible cost. Both adapters meter per-trial cost themselves, because EI-per-cost consumes measured cost, not the user’s folklore (Chapter 8).

13.7 The architecture factory

One contract exists only because Chapter 9 decided that architecture search means width, depth, and a few structural flags rather than topology. Those coordinates sit in the ordinary search space, so the searcher and scheduler interfaces above need no new verbs for them; what they do need is a way to turn a configuration into a model, and a way to say when two configurations’ models can share weights. Both belong to the user’s workload, not to the tuner, which is why they are a contract rather than an implementation.

build(config)

Return an initialized model from the architecture coordinates of a configuration. The workload supplies this; the product never constructs architectures itself, because the space of things a user might mean by “depth” is not ours to enumerate. The factory is also what makes the excluded methods excluded by construction: a contract that maps coordinates to a model has no place to put a supernet.

measure(model)

Return the size objectives, parameter count and FLOPs per forward pass, for a built model. These are measured on the constructed model rather than predicted from the coordinates, which is the whole reason Chapter 9 was willing to treat size as an objective and unwilling to treat proxies as a substitute for training. Multi-objective studies consume these through the ordinary objective vector (Chapter 7).

compatible(config_a, config_b)

Declare whether weights from one configuration’s model can be loaded into another’s. Pure hyperparameter changes are compatible; a width or depth change is not. The scheduler’s exploit verb consults this before checkpoint surgery, and an incompatible answer routes the trial to the transfer policy below instead of copying weights into a shape that cannot hold them. A workload that gets this predicate wrong produces silent garbage, so the conformance suite tests it directly: every declared compatible pair must survive a load, and every incompatible pair must refuse one.

transfer(parent, child)

The policy for an incompatible inheritance: restart the child from initialization, or warm-start it by distillation from the parent, which is BG-PBT's route (Chapter 6). The choice is declared per study, not inferred, and its cost is charged to the trial that caused it rather than hidden in the population's overhead. Distillation arrives with the population line in tier 2; before then the only legal policy is restart, and the factory says so through its capability flags.

The searcher's **capabilities** declaration gains one flag to match: whether the searcher models architecture coordinates as ordinals and categoricals with the rest of the space, or requires them held fixed. A searcher answering the latter is not eligible for a study whose space includes architecture axes, and the product refuses the combination at declaration time for the same reason it refuses a continuous-only searcher on a categorical space.

13.8 What would overturn this chapter

Contracts are falsified by construction work, and the suite is arranged to do it early: if the tier 0 wraps cannot be expressed behind these interfaces without holes, if the reference-parity tests of Chapter 14 need interface changes to pass, or if the population verbs prove insufficient for a faithful PBT run on the Ray adapter, the contracts change and this chapter is rewritten before the code doubles down. The interfaces are the product's most expensive thing to change later; that is why they get a chapter, a register mapping, and the first weeks of the calendar rather than an afternoon.

Chapter 14

The validation suite

A tuning product can fool its builders the same way a configuration fools a tuner: by being evaluated on the problems it was tuned on. This document has met the trap twice, as seed overfitting in Chapter 9 and as the proxy bias of Figure 5.1, and the defense is the same both times: decide the evaluation before the thing being evaluated exists. This chapter is that decision. It collects every falsification promise made in Chapters 3 through 11 into numbered tests with pass rules, prices the expensive ones, and ties each build tier’s completion to named entries rather than to the feeling of being done.

14.1 Three layers

The suite runs at three costs, and a test’s layer is part of its identity.

Layer one is cheap and broad: surrogate and tabular benchmarks (YAHPO-Gym, HPOBench, and the RL-specific ARLBench), where a full tuner comparison costs CPU-minutes because the training runs were paid for once by others (Pfisterer et al., 2022; Eggenberger et al., 2021; Becktepe et al., 2024), plus recorded-curve replays in the style of Syne Tune’s simulator (Chapter 10), where a scheduler’s decisions are re-executed against stored learning curves in seconds. A scheduler bug found in a CPU-minute replay is a bug not found in a GPU-week study; layer one exists to make that sentence routine. The multi-fidelity-with-priors suite from the PriorBand line covers the prior-handling code specifically (Mallik et al., 2023).

Layer two is realistic and ours: one pinned task per workload family. A PPO task on a Brax environment; a flow-plus-HMC sampler task scored by ESS per gradient under divergence and split- $\hat{R}$ vetoes, with a reference posterior computed once at great length; a Hamiltonian network task scored on prediction error and energy drift. Layer two is where verdicts about *our* workloads are settled, at GPU prices, under the statistical discipline of Section 14.3.

Layer three is standing: diagnostics that run inside every real study the product ever executes. The rank correlation between rung scores and final scores is logged always (Chapter 5); the seed protocol’s presence or explicitly recorded absence is part of every study row (Chapter 13). Layer three is how the suite keeps learning after release, and it costs nothing because the store records it anyway.

14.2 The register of tests

Table 14.1 is the register. Each entry names the claim under test, the protocol in one line, the rule that passes or vetoes, the chapter whose promise it discharges, and the tier gate it belongs to. The subsections after it expand the entries whose protocols need more than a line.

Table 14.1: The validation register. Layer 1 runs on tabular benchmarks and replays (CPU); layer 2 on the pinned tasks (GPU); layer 3 inside every production study. “Gate” names the tier that cannot be declared done until the entry is green.

ID	Claim under test	Pass rule or veto	Evid.	Gate
V01	each wrapped searcher/scheduler matches its reference implementation on tabular benchmarks	final and anytime scores within tolerance of reference	10	T0
V02	scheduler logic is correct under asynchrony	replayed decisions on recorded curves match specification; stragglers and drops simulated	5	T0
V03	the suite itself catches wiring bugs	seeded defects (off-by-one rung, dropped seed, swapped objective sign) are caught by V01–V02	11	T0
V04	every shipped searcher earns its complexity	beats the log+Sobol floor at matched budget on pinned tasks, each searcher tested at the gate of the tier that ships it: the tier-0 GP with qLogEI and the wrapped TPE at the tier-0 gate, the tier-1 and later searchers at theirs	3, 4	T0/T1
V05	the transform matters as claimed	GP with and without log-warping on a pinned task; warped must not lose	3	T0
V06	early stopping is nearly free	searcher+ASHA matches searcher-at-full-fidelity final quality within noise at a fraction of compute; rung correlations logged	5	T1
V07	the small-budget RL verdict holds here	DEHB vs BO+ASHA at matched 16-run budget on the pinned RL task	5, 6	T3
V08	the population line does not lose in its home regime	population line vs DEHB and BO+ASHA on both sides of the eight-worker line at matched compute; 3 reps/arm/side (declared exception, Sec. 14.4); consistent home loss demotes the flagship, disagreement reports unresolved	6	T2
V09	the premium MO searcher earns its tier	qLogNEHVI vs random-weight Chebyshev and wrapped NSGA-II on the Hamiltonian task; hypervolume with uncertainty; demote to option on a tie	7	T1
V10	cheap MO fidelity signals may cull	short-horizon drift passes the rank-correlation pilot before MO-ASHA may kill on it	7	T1
V11	priors are safe to default-on	no-prior vs folklore-prior vs wrong-prior runs; gains under good, recovery within noise under wrong, else demote to opt-in	8	T1
V12	warm starts pay	measured trial savings on a new study of a familiar family once the store holds enough studies	8	T1+
V13	sampler correctness cannot be bought by speed	veto-failing configurations never promoted; survivors ranked by ESS/gradient; pilot agreement with the reference posterior	9	T1
V14	tier 0 is composition, not construction	the day-one walk of Chapter 11 reproduced end to end within the tier-0 budget	11	T0
V15	the curve-model bet is live before we pay for it	ifBO matches or beats ASHA on the curve-informative public suites before any integration work	5	T3

14.3 The statistical protocol

Every layer-two comparison follows one protocol, copied from the discipline of Eimer et al. (2023) and from this document’s own warnings. Fixed tuning seeds and disjoint test seeds. Budgets matched on measured cost, with the tuner’s own overhead included in the bill. Results reported as performance-over-budget curves with uncertainty bands, never as single numbers. And a repetition count that is derived rather than declared: a pilot campaign estimates the between-run variance, and the sample size follows from that variance and the entry’s decision margin at 80 percent power. The ten repetitions this document has quoted since Chapter 9 are a planning heuristic for the calendar and the budget, not a substitute for that calculation. Where the pilot shows low variance against a wide margin, fewer repetitions suffice and the protocol records why; where it shows high variance against a narrow one, the budget rises or the claim softens, and the entry says which.

Because the budget is finite, every entry also carries a maximum sample size, and reaching it without crossing the decision threshold is a third outcome alongside pass and fail: *inconclusive at budgeted precision*. That is a real verdict with a real consequence, which is that the capability does not become a default on evidence that never arrived. Pilot variance estimates are recorded before the full campaign begins and are not revised during it, since a variance estimate updated mid-campaign is a decision threshold updated mid-campaign wearing a disguise. V08 remains the one entry whose arms are too expensive for any of this at full strength; it runs three repetitions per arm per side under the restricted question, asymmetric demotion rule, and release semantics of Section 14.4, and any future exception is written into this section before its campaign runs, never after its results are seen.

Verdicts then come in three strengths, and the suite’s reports use the distinction explicitly. Hard vetoes (a diverged run promoted, a veto-failing configuration surviving, a protected seed leaking into tuning) fail a test regardless of scores. Descriptive differences (one curve above another within overlapping bands) are recorded and never rank anything. A ranking is claimed only when the repetitions support it statistically; when a result shows one method ahead within noise, the honest summary is that they tied, and a tie against a simpler method is a loss for the complex one wherever the register says so (V04, V09). The suite exists to catch regressions and to rank where gaps are real, not to manufacture leaderboard drama.

14.3.1 Register shorthand resolved

The table above uses unquantified phrases to fit on one page; this section pins each one to a concrete rule before any entry runs.

“Within tolerance” (V01): Anytime and final scores differ by no more than 0.5% of the reference method’s final value on each benchmark, and rank orderings match exactly. The tolerance is loose by design—V01 gates implementation correctness, not optimizer superiority.

“Within noise” (V06, V11): The 95% bootstrap confidence interval of the difference in means lies entirely inside the preregistered non-inferiority margin $[-\delta, +\delta]$: for V06, δ is 5 percent of the full-fidelity baseline’s final quality; for V11, 10 percent of the no-prior baseline’s. Bootstrap by resampling tuning runs with replacement, 10,000 iterations, empirical percentiles. The distinction from an interval that merely overlaps zero is the whole point. Overlapping zero says a difference was not detected, which a small enough sample says about everything; an interval contained in the margin says any difference that survives is too small to care about. An interval that overlaps zero but spills past $\pm\delta$ is recorded as inconclusive, not as a tie, and a tie is claimed only in the contained case. Both margins are set here, before the campaigns, as the smallest difference that would change what the product recommends.

“A fraction of compute” (V06): Early stopping reaches the same final quality, within the non-inferiority margin above, at no more than one-third the total compute of full-fidelity search. Compute

is the accelerator-seconds and wall-clock the executor actually recorded, which puts startup, training, evaluation, checkpoint save and load, queue wait, and the optimizer’s own overhead on the bill; the savings claim is about resources consumed, not about an idealization of them. Full-run equivalents survive as a normalized secondary measure for readability, computed by dividing total cost by the mean cost of one full-fidelity trial in the baseline arm, and they are never the number the gate reads. The threshold applies to accelerator-seconds. Where wall-clock diverges materially from it through scheduling or parallelism, both are reported and the verdict follows the accelerator-seconds, since that is the resource the budget buys. V06 passes only if ASHA delivers the factor-of-three saving on measured cost.

“Measured trial savings” (V12): The warm-started study reaches the same final objective value in at least 20% fewer trials than a cold start on the same task. Both arms use the same searcher; only the initialization differs. Below 20%, warm starting is not worth defaulting on.

“Hypervolume with uncertainty” (V09): Report hypervolume over budget with 95% confidence bands from bootstrap resampling of tuning runs. Rank methods by their final hypervolume; demote the premium method (qLogNEHVI) to an option if the interval overlaps the simpler method’s (Chebyshev) or if it loses outright.

“Enough studies” (V12): The run store must contain at least ten completed studies from the same model family (same architecture class, same task domain) before V12 runs. Below that, the warm-start query has no data to learn from and the test is meaningless.

All numerical gates (0.5%, one-third, 20%, 0.6 for V10, 1.01 for V13) are judgments recorded before the data arrives, not fit to results. The V10 and V13 protocols later in this chapter set their thresholds explicitly; the phrases above inherit the same discipline.

14.3.2 Sampling levels and paired designs

A tuning comparison has more than one source of randomness, and conflating them is the fastest way to a confident wrong answer. Five levels are in play: the optimizer seed (which tuning run), the workload training seed (which initialization or environment instance), the evaluation seed (which test episodes), the task (which benchmark), and the configuration draw (which candidates the searcher happened to try). Treating the leaves as independent samples when the branches are shared is pseudoreplication: it shrinks the reported interval without adding information, and every gate downstream inherits the false precision.

The first defense is pairing. Where a comparison admits it, the methods share the optimizer seed set, the task and environment instances, and any baseline configurations drawn once and evaluated by all arms. Pairing removes the between-instance variance from the contrast rather than averaging over it, which buys power at no extra compute, and it applies wherever the methods being compared consume seeds the same way. Where they do not, because a population method spends seeds on a schedule a sequential method has no analogue for, the entry says so and falls back to unpaired sampling with the larger sample size the power calculation then demands.

Uncertainty is computed by hierarchical bootstrap rather than by resampling individual trials. The resampling unit is the complete tuning run; when runs span several tasks, tasks are resampled and then runs within the sampled tasks, preserving the nesting. The statistic of interest is recomputed on each of 10,000 bootstrap samples and reported at empirical percentiles. This propagates every level’s variance without assuming a distributional form for any of them, and it is the procedure behind every interval quoted in this chapter.

14.3.3 Development and acceptance tasks

The trap this chapter opened with applies to the suite itself. A pinned task that implementers run daily while debugging stops being a held-out measurement of anything; it becomes the thing the implementation was tuned on, and the gate it anchors measures fit to the validation set. So each workload family carries two disjoint task sets. Development tasks are named in week one of tier 0, recorded in the validation manifest, and free to run as often as anyone likes. Acceptance tasks are chosen by an engineer not implementing the capability under test, revealed at gate time, and recorded in the gate manifest with their data checksums.

Both numbers are reported, and the gap between them is itself a diagnostic: a capability strong on development tasks and weak on acceptance tasks has overfit the suite, which is a finding rather than a mystery. Promotion decisions weigh the acceptance results, because those are the only ones the implementation has not seen.

Task sets rotate on a version, not on a feeling. When a tier's gates are recorded, new tasks enter development and some older development tasks may migrate to acceptance for the next tier, with version numbers and dates in the task registry. The registry is what makes task-selection bias auditable after the fact, which matters precisely because the person selecting is usually the person hoping.

14.3.4 Multiple comparisons and endpoints

Several entries ask more than one question. V04 compares searchers across tasks and budget points; V09 crosses objectives with comparators; V13 inspects many posterior marginals at once. Twenty independent comparisons at the conventional threshold produce one spurious winner on average, which is enough to promote a searcher that does nothing.

Each entry therefore predeclares one primary endpoint, the single comparison the tier's completion depends on, and labels the rest. The primary endpoint is judged at $\alpha = 0.05$ uncorrected, because there is one of it. Secondary comparisons that feed a ranking claim are controlled family-wise at $\alpha = 0.05$ by the Holm–Bonferroni stepdown procedure (Holm, 1979), which is uniformly more powerful than plain Bonferroni at the same guarantee and needs no assumption about dependence among the tests. Comparisons that rank nothing and decide nothing, the descriptive kind this chapter has already licensed, need no correction, and saying so is what keeps multiplicity control from becoming a tax on honest reporting.

V04's primary endpoint, as the worked instance: whether the tier-0 GP with qLogEI beats the log-plus-Sobol floor on the acceptance task at matched budget. Comparisons among GP variants, or across development tasks, are secondary and corrected. Every entry's primary endpoint is written into its protocol before its campaign begins, for the same reason its thresholds are.

14.3.5 Immutable artifacts and preregistration

None of the above binds unless the plan provably predates the data, so every gate campaign emits a signed manifest before it runs. The manifest is JSON and carries the protocol version and entry identifier (V04-T0-v1.2 and the like), SHA-256 checksums of task code, data files, and environment configuration, the exact tuning and test seed sets per method and repetition, the analysis script with its thresholds and tests committed to version control while the results do not yet exist, a full dependency lock and hardware description, the commit hash of the code under test, and a creation timestamp with a signature.

Results are written as versioned JSON with the manifest hash embedded, under `validation/artifacts/`, a path deliberately not ignored by `.gitignore`. Reanalysis is permitted and expected; what it may not do

is quietly replace the first analysis. A changed threshold, a changed script, or a changed seed set after results are visible produces a new manifest at an incremented version with the reason written down, and both versions survive.

The point is not ceremony. Researcher degrees of freedom (Simmons et al., 2011) are invisible in a report and obvious in a diff, and this is the cheapest available way to put them in a diff: a reviewer can check that the analysis plan predates the data, that the tested code matches the history, and that the tasks and seeds were not chosen after the fact to flatter a result the team had already grown attached to.

14.4 The two expensive campaigns

Two entries dominate the compute bill and deserve their design stated in advance.

V07 races DEHB against BO+ASHA at a matched sixteen-full-run budget on the pinned RL task, ten repetitions each. The cost driver is the pinned task’s full-run price, so the task is sized in week one with ARLBench’s guidance on representative subsets, which cut tuner evaluation cost by roughly an order of magnitude in their study (Becktepe et al., 2024). As a sizing illustration, labeled an estimate: if one full run costs two accelerator-hours on the chosen environment, the campaign is $2 \times 16 \times 10 \times 2 = 640$ accelerator-hours, under a month on a single dedicated node and a weekend on eight. If the chosen environment cannot fit that envelope, the environment is wrong, not the protocol.

V08 is the flagship’s trial, run late, when the population line exists: the population method against DEHB and BO+ASHA on both sides of the eight-worker boundary at matched total compute. It is the one entry where the ten-repetition rule collides with the price of the test, because a single population arm already costs eight full-length trainings, and ten repetitions on both sides of the boundary is a quarter-million accelerator-hour campaign we will not buy. The exception is therefore stated as a protocol rather than waved through. The repetition count drops to three per arm per side, the smallest number that still yields a spread rather than a point, and the reduction is paid for by narrowing the question: V08 does not ask which method is best, only whether the boundary sits where Chapter 6 says it does. Three arms, two sides, three repetitions, and the ordering of arms within a side is what the entry reports, with the observed spread shown always.

The stopping rule is written in advance, because a cheap test with a flexible endpoint is worse than no test. If all three repetitions on a side agree in ordering, the side is called and no further repetitions are bought. If they disagree, the entry reports *unresolved at this budget* and the flagship keeps whatever status it had going in, which is a hold rather than a promotion; the one outcome V08 may never produce is a promotion on evidence that did not separate. The demotion direction is asymmetric and deliberately so: a consistent loss in the flagship’s home regime demotes it on three repetitions, because the claim being falsified is a strong one and a strong claim can be broken by a weak test, whereas confirming the boundary would need the full protocol we declined to buy. That asymmetry is the honest content of a three-repetition campaign, and it is why V08 is written here as a falsifier of Chapter 6’s threshold rather than as its validation.

The result states and their release consequences are named too, since a falsifier with no defined pass state cannot gate a tier. V08 opens at *hold: insufficient evidence* and can reach three places. *Demote*: all three repetitions on a side agree that the population line loses in its home regime, in which case the flagship becomes an experimental option and the roadmap’s defaults revert to whichever of DEHB or BO+ASHA won V07. *Hold*: the repetitions disagree, or the home-regime side is inconclusive, in which case the entry stays at *unresolved at this budget* and the flagship keeps its tier-2 experimental status without becoming a default recommendation. *Pass*: reachable only through the full sequential campaign this entry declined to buy, at up to ten repetitions per arm per side with preregistered margins and decision boundaries. The

three-repetition run is a tripwire, not promotion evidence, and no accumulation of tripwires becomes one.

Tier 2 completion therefore requires that V08 be neither demoted nor blocked by an implementation failure; a hold is acceptable completion. That is the honest reading of a capability whose audience is users at eight or more dedicated workers: the population line ships as an option with Chapter 6’s boundary documented as guidance rather than enforced as a default, and the campaign that would promote it waits for tier 3 or a post-release cycle, if demand and budget ever arrive together.

14.5 Two entries whose protocol needs pinning

V10 and V13 both gate a capability on a threshold, so the threshold has to be a number chosen before the data arrives rather than after.

V10 asks whether a short-horizon energy-drift measurement may cull a Hamiltonian trial. The pilot runs a fixed set of one hundred configurations drawn uniformly from the search space, each at a single fixed seed, to full fidelity (the study’s declared endpoint), recording drift at the candidate rung (proposed as one third of the endpoint) and at the end. The gate metric is Spearman rank correlation between the two drift vectors (one value per configuration, not per run, since single-seed eliminates within-config variance). Compute the bootstrap 95% confidence interval by resampling configurations with replacement 10,000 times and taking empirical percentiles of the resampled correlations. MO-ASHA may kill on that signal only if the interval’s lower bound clears 0.6. Below it, the scheduler still runs but the signal is advisory, promotion decisions falling back to the objective that does correlate. The number 0.6 is a judgment, not a result from the literature, and it is recorded here so that a later argument about it is an argument about a written-down choice. The hundred-trial campaign costs about 133 full runs on the Hamiltonian task, budgeted under layer-two validation.

V13 asks whether the sampler study’s correctness vetoes hold. Three components, each with its own rule. The vetoes themselves are absolute: any configuration with a divergent transition or split- $\hat{R}$ above 1.01 is never promoted, and a single promotion of a veto-failing configuration fails the entry outright. The sampling protocol that feeds the vetoes: four independent chains per configuration, each from a dispersed random initialization, 2,000 warmup draws discarded plus 2,000 retained draws per chain, for 8,000 posterior samples total per configuration. Split- $\hat{R}$ is computed by splitting each chain in half (eight half-chains) and comparing within-half-chain variance to between-half-chain variance per the rank-normalized folded formula of [Vehtari et al. \(2021\)](#); the veto fires if any marginal exceeds 1.01 or if effective sample size falls below 400. Divergent transitions are reported by the sampler itself (a numerical-integration failure during the Hamiltonian trajectory). A bimodal or multimodal target is detected by visual inspection of marginal trace plots during the pilot; if found, chains are re-initialized from dispersed modes rather than randomly, per Bayesian workflow recommendations.

Survivors are ranked by effective sample size per gradient evaluation, the cost measure the executor records rather than one the user supplies. And agreement with the reference posterior is checked on the pilot: the reference is computed once at great length using the same four-chain protocol but with 20,000 retained draws per chain, and each candidate configuration’s marginal means and variances are compared to the reference marginals. Agreement is defined as: the candidate’s posterior mean falls within one standard error of the reference posterior mean, where the standard error of each mean is $\sigma/\sqrt{\text{ESS}}$, using each sampler’s own reported effective sample size and posterior standard deviation. An entry that ranks a fast, subtly wrong sampler above a slow, correct one is the failure this test exists to catch, which is why speed enters only after the vetoes and the agreement check have passed.

14.6 Sabotage entries

Two entries test the testers. V03 seeds defects into the wiring, an off-by-one rung, a dropped seed, a flipped objective sign, and requires the parity and replay layers to catch them; a suite that cannot catch a planted bug is decoration, and this entry is the reason we can trust green elsewhere. V11 seeds bad advice, the purposefully detrimental prior in π BO’s own sense, and requires the mechanism’s safety half, recovery to the prior-free baseline, to hold on our tasks and budgets, not only in the paper (Chapter 8).

14.7 What the suite costs, and when it runs

Layer one is effectively free and runs on every change: minutes of CPU per full pass, which is why it can gate merges. Layer three is free by construction. Layer two is a budget line: V04, V05, V06, V09, V10, V11, and V13 together are dominated by the RL entries and the ten-repetition rule; on the sizing illustration above, a full tier-1 gate pass is of the order of a few hundred accelerator-hours, scheduled as a campaign with a manifest, not run ad hoc. V07 and V08 are separate, priced campaigns. All layer-two numbers are estimates until the week-one task sizing; the envelope, one GPU-week class of compute per tier gate, is the planning figure the calendar of Chapter 15 carries.

14.8 Gates, and what failure does

A tier is done when its register entries are green, and each entry’s failure has a named consequence rather than a shrug. Wiring failures (V01–V03, V14) block the tier and are repaired. Evidence failures execute the demotion already promised in the verdict’s chapter: V04 removes the losing searcher from defaults, V09 demotes the premium searcher to an option, V11 demotes priors to opt-in, V08 demotes the flagship. The register of `product_register.json` records, for every verdict, which entry anchors it; a verdict whose entry fails does not quietly survive in the code while dying in the appendix.

14.9 What would overturn this chapter

The suite is itself under test. If V03’s planted defects pass, the parity layer is rebuilt before anything else proceeds. If the layer-two envelope exceeds its planning figure by more than a small factor after week-one sizing, the pinned tasks are resized and this chapter’s estimates corrected in place. And if layer three’s standing diagnostics show our real studies violating the assumptions the pinned tasks embody, fidelities decorrelating where the suite assumed they correlate, the pinned tasks follow the evidence, because a suite that drifts from the workloads it guards inherits exactly the proxy bias it was built to catch.

Chapter 15

What to build, in what order

The survey supports a fairly confident ordering of implementation work. The principle behind the ordering: ship the methods with the strongest evidence-per-line-of-code first, and let each tier’s infrastructure carry the next tier. Every item below names the chapter that justifies it, so that when the evidence changes, and Chapter 10 is a reminder of how fast this field’s ground shifts, we can trace which decisions the change touches. Table 15.1 collects every verdict in one place; the decision boxes at the ends of Chapters 3 through 9 carry the one-paragraph rationales, and the listed chapter holds the full teaching and evidence for anyone who wants to challenge a line.

15.1 Tier 0: wire up what exists (weeks)

The execution substrate plus existing searchers already covers the sturdy defaults from Chapters 3 to 5: Sobol and log-uniform random sampling (the two free upgrades of Chapter 3), ASHA and the median stopping rule, TPE via Optuna’s sampler for conditional spaces, CMA-ES for cheap continuous problems, and a single-objective GP searcher with qLogEI (either BoTorch directly or Optuna’s GP sampler), with the dimension-scaled lengthscale priors of Section 4.4 switched on from the start, since they cost one line. None of this is novel; all of it is table stakes, and the deliverable of this tier is really our own searcher and scheduler interfaces (Chapter 12) proven against known-good implementations, plus the run store that records every trial with its configuration, seed, fidelity, curve, and cost. A tier is done when the validation suite below says our wiring matches the reference implementations, not when the code merely runs.

15.2 Tier 1: the differentiators (the core product)

Four capabilities justify building something of our own, because no single existing library ships them together on our execution substrate.

Multi-objective, properly. A qLogNEHVI searcher for two to four objectives, Chebyshev scalarization as the many-objective and cheap fallback (the division Figure 7.2 argued), MO-ASHA as the multi-fidelity scheduler, hypervolume-over-budget reporting so that a study’s progress is one honest curve, and Pareto fronts as first-class study outputs (Chapter 7). This is the piece our Hamiltonian and finance workloads ask for most directly, and the piece the audit of Chapter 10 found no executor shipping natively.

Priors and warm starts. The π BO acquisition weight (8.1) and the PriorBand sampling portfolio over ASHA rungs, both consuming a prior declared in the search-space schema; plus quantile-based warm starting from the run store (Chapter 8). Cheap to build, immediately felt by every user, and the mechanism’s built-in decay means a wrong prior costs a study its opening moves rather than its conclusion

Table 15.1: Every build decision in one place. “Cost” is engineering effort to a first useful version; tiers are this chapter’s build order; the last column is the chapter that teaches the method and holds the evidence for the verdict. Read a row, then challenge it against its chapter’s decision box.

Capability	What it buys us	When it applies	Cost	Verdict	Ch.
Sobol + log sampling	honest coverage of every axis	always; also the GP’s opening phase	days	T0	3
Random-search baseline	the bar every searcher must clear	validation suite	days	T0	3
ASHA + median rule	kills bad trials early; the biggest single saving	any workload with learning curves	days	T0	5
TPE (wrap Optuna)	conditional and categorical spaces from few trials	mixed spaces, cheaper trials	days	T0	4
GP + qLogEI (BoTorch)	the most information per expensive trial	costly, noisy trials	weeks	T0	4
Dimension-scaled priors	reliability past ten dimensions	always on	hours	T0	4
qLogNEHVI	noisy Pareto fronts, two to four objectives	Hamiltonian, finance	weeks	T1	7
Chebyshev scalarization	fallback at any objective count; random weights approximate the front when best-seen values proxy the ideal point	five-plus objectives, or cheap engine	days	T1	7
MO-ASHA + veto gates	early stopping for fronts; correctness gates	Hamiltonian, samplers	week	T1	7
π BO + PriorBand	expert head start that forgets bad advice	every study	1–2 wks	T1	8
Warm start from run store	recurring studies start smart	repeated model families	days	T1	8
EI-per-cost + cooling	budget-aware proposals	unevenly priced trials	days	T1	8
PB2-Mix, then BG-PBT	schedules and architecture at scale	RL with 8+ parallel workers	months	T2	6
DEHB	most consistent small-budget RL tuner	10–64 full runs	weeks	T3	5
ifBO scheduler	curve-shape-aware budget allocation	once pause/resume is solid	weeks	T3	5
HEBO-style robustness	warping and ensembles on the GP	messy real objectives	weeks	T3	4
CARBS cost frontier	tuning while a model scales	scaling-bound projects	later	hold	8
One-shot / zero-cost / grammar NAS	see Section 9.5’s verdicts	no current project	—	skip	9

provided the product schema requires a lower-bounded mixture or explicitly nonzero support everywhere, so that wrong priors remain wrong rather than becoming unfalsifiable (Chapter 8’s decision box carries this guard).

Cost-aware acquisition. EI-per-cost with cost cooling (Lee et al., 2020) on top of the Tier 0 GP searcher. This needs a *predictive* cost model, because acquisition prices a candidate before it runs and realized cost cannot do that; what the executor supplies is the training data for the model, not the price itself. The specification is deliberately modest: a second GP on $\log(\text{wall-clock seconds})$, fit to the executor’s measured per-trial costs from the run store, with the same kernel family as the performance surrogate; log space because cost is positive and varies multiplicatively with width, depth, and unroll length, the three coordinates that dominate it. Cold start uses the user’s declared cost hint if present and a uniform prior otherwise, with the cheap-first initialization of CARBO carrying the first rounds until the model has data. Failed and killed trials enter as censored observations at their elapsed time (a lower bound on cost, not the cost), because dropping them would teach the model that the expensive corner is cheap. The model refits with the performance surrogate on the same cadence, and its calibration is logged as a standing diagnostic beside the rung correlation, so a badly-fit cost model is visible rather than silently distorting proposals.

15.3 Tier 2: the population line for large-parallelism RL

Ray already ships PBT and a continuous-only PB2, so the work starts one rung up: extend the explore step to mixed spaces (PB2-Mix’s bandit over categories, or directly the Casmopolitan-style kernels), then build BG-PBT proper: the mixed-space trust-region searcher with ordinal-aware kernels, generational restarts, and distillation hooks left to the training code (Chapter 6). The official BG-PBT code is available to crib from, and RLlib checkpointing supplies the surgery PBT needs. This is the flagship for the large-parallelism regime, and the product should encode the budget boundary from Eimer et al. (2023): below a few dozen full-run equivalents, steer users to DEHB or BO plus ASHA instead. Encoding the boundary is not a footnote; it is the difference between a product that embodies the evidence of Chapter 6 and one that merely implements its algorithms.

15.4 Tier 3: the frontier, selectively

Three bets with good expected value once Tier 1 is earning. A learning-curve scheduler in the ifBO style, using the published pre-trained freeze-thaw surrogate, exploiting our pause-and-resume plumbing (Chapter 5); the decision problem it solves is Figure 5.3, and the pre-trained surrogate (Chapter 8) is why the bet is affordable. DEHB, which the RL evidence recommends and which also gives us a non-GP engine for awkward spaces; the upstream package is frozen (Chapter 10), so this is an implementation against our interfaces with the original as a reference. And HEBO-style robustness (input and output warping, noise-aware acquisition ensembles) folded into our GP searcher (Cowen-Rivers et al., 2022), which Section 4.1.1 argued is where the challenge-winning gains actually came from. A CARBS-style cost-performance frontier is the natural fourth once a scaling-bound project appears (Chapter 8).

15.5 What we deliberately skip

Differentiable NAS and a bespoke zero-cost-proxy subsystem (Chapter 9); gradient-based HPO through the training loop; a meta-learned transformer surrogate trained by us (we adopt pre-trained ones where they ship); and any LLM-driven configuration proposer, which as of this writing has no controlled evidence of

beating the methods above at matched cost on our kinds of workloads. Each skip is revisitable when the evidence changes; the skips are recorded here precisely so that revisiting them is a decision rather than a drift.

15.6 The calendar, and its dependencies

The tiers above say what and why; a build needs when and after what. The calendar below assumes one dedicated engineer with part-time research support for the campaign runs, an assumption the manager owns, not the survey; with a second engineer the two independent tracks (the multi-objective stack and the population line) run in parallel and the total compresses by roughly a third. Weeks are planning estimates in the same spirit as the cost column of Table 15.1.

Weeks 1–2: freeze. The interfaces and store schema of Chapter 13 are frozen against a written review, the pinned tasks of Chapter 14 are sized to their compute envelope, and the layer-one harness (benchmark parity and curve replay) is stood up first, because every later gate runs through it.

Weeks 3–6: tier 0. The wrapped searchers and schedulers go behind the interfaces; the run store lands with lineage and the standing diagnostics; the Ray and local adapters pass checkpoint surgery through. Gate: V01–V05 and V14 green. V04 tests the tier-0 GP and TPE at this gate; tier-1 searchers face it at the tier-1 gate. The tier-0 gate is deliberately mostly wiring tests plus the one method floor check, because tier 0 makes no novelty claims, only composition claims.

Weeks 7–12: tier 1. The multi-objective stack first (qLogNEHVI, Chebyshev fallback, MO-ASHA with veto gates, weeks 7–9), then priors, warm start, and cost-aware acquisition (weeks 10–11), then the tier-1 campaign week: V04, V06, V09, V10, V11, V13, with V12 maturing later as the store fills. Dependencies bind the order: MO-ASHA extends the tier-0 ASHA; PriorBand consumes its rungs; warm start queries the store that tier 0 built; EI-per-cost decorates the tier-0 GP searcher.

Weeks 13–20: the population line’s prerequisite. The trust-region mixed-space searcher is built once and serves twice, as the high-dimensional single-objective searcher of Chapter 4 and as the explore engine of the population line; PB2-Mix lands on it in weeks 17–20 using Ray’s existing PBT machinery through the exploit and explore verbs of the scheduler interface.

Weeks 21–30: the flagship. BG-PBT proper (generational restarts, architecture coordinates, distillation hooks left to training code), closing with the V08 boundary campaign around week 30, whose demotion rule was fixed in Chapter 6 long before this code exists.

Weeks 31 onward: tier 3, each item behind its own gate. DEHB against our interfaces, raced in V07; the ifBO scheduler only after V15 shows the bet live on public suites and only once pause-and-resume is proven by the population work; HEBO-style robustness folded into the GP searcher last, because it is an upgrade to a working thing rather than a missing thing.

15.7 The bill

Engineering: roughly six weeks to the tier-0 gate, six more to the tier-1 gate, eighteen for the population line through its campaign, and about twelve of selective tier-3 work; call it forty-two engineer-weeks, ten months of one engineer or six of two, before tier-2 electives. These are the table’s per-item costs rolled up, not new estimates.

Compute: layer one is CPU-minutes and free in practice; each layer-two tier gate is budgeted at the one-GPU-week class, and the two priced campaigns (V07, and V08 with its both-sides-of-the- boundary design) are of the same order each, per the sizing arithmetic of Chapter 14. The planning envelope for the

whole program is therefore a few GPU-weeks, spent in named campaigns with recorded manifests rather than ad hoc runs.

Maintenance: the standing cost of the architecture is the one Chapter 12 priced, two small interfaces kept current against Ray, plus pinned dependency versions reviewed on their release cadence.

15.8 Risks, and the tripwires that catch them

Ray or Tune drifts under us. Tripwire: the layer-one replay and parity tests run against every Ray upgrade before it is adopted; the adapter boundary is the insurance, and its annual premium was priced in Chapter 12.

Upstream churn in the searcher libraries. The audit of Chapter 10 already caught BoTorch deprecating acquisition names and Optuna changing default samplers at 5.0; mitigation is pinned versions, thin wrappers, and the store recording the wrapped library's version in every study row, so a result is never orphaned by an upgrade. Tripwire: V01 parity on any dependency bump.

The population evidence fails to transfer to our cluster. That is not a risk to manage but a measurement to take: V08 exists to take it, and its failure path (demote the flagship, keep the rung methods) was committed to in writing before the build began.

The seed protocol proves too expensive as a default. Layer three records every opt-out; a high opt-out rate on real studies is the tripwire, and the response is a cheaper default protocol argued in an update to Chapter 9, not a silent loosening.

The pinned tasks are mis-sized. Week 1–2 sizing exists for this, and Chapter 14's own overturn clause governs the repair.

15.9 How we will know the tuner works

The full answer is now Chapter 14: the register of tests, the statistical protocol, the priced campaigns, and the rule that a tier is done when its gate entries are green. What this chapter adds is only the schedule on which those gates fire, weeks 6, 12, and 30 on the one-engineer calendar, and the standing rule that no tier is declared done by the feeling of momentum.

Glossary

Plain-language definitions of every term of art the document leans on, for reading without the original papers at hand. Terms are grouped by the section that introduces them.

The problem (Chapter 2)

trial One training run with one fixed configuration and one recorded random seed.

incumbent The best configuration found so far in a study.

fidelity The resource level of a partial evaluation: epochs trained, environment steps consumed, chain length run. Low fidelity is cheap and only loosely predictive of full results.

searcher The half of a tuner that decides which configuration to try next.

scheduler The half of a tuner that decides how long trials run, which are stopped early, and which paused ones resume.

run store The database recording every trial's configuration, seed, curves, cost, and diagnostics; the raw material for warm starts and transfer.

seed protocol Evaluating candidates on several training seeds and reporting results on separate test seeds the tuner never saw, so that a study optimizes configurations rather than luck.

Model-free search (Chapter 3)

log-uniform sampling Drawing a scale-type knob (like a learning rate) uniformly in its exponent, so each factor-of-ten range gets equal attention.

Sobol points A deterministic sequence that spreads samples more evenly than independent random draws, avoiding accidental clusters and gaps.

effective dimensionality The number of knobs that actually move the objective; usually far smaller than the number of knobs, but which ones matter varies by problem.

CMA-ES An evolutionary optimizer that keeps a Gaussian sampling distribution and reshapes it toward directions where recent samples scored well; needs hundreds of cheap evaluations to shine.

differential evolution An evolutionary rule that proposes new points by adding scaled differences of existing population members; the engine inside DEHB.

Bayesian optimization (Chapter 4)

surrogate A cheap statistical model of the expensive objective, fit to the trials seen so far and consulted thousands of times between real trials.

Gaussian process (GP) The default surrogate: a Bayesian regression over functions whose predictions come with calibrated uncertainty; the same mathematics as kriging in spatial statistics.

kernel The GP's assumption about how similar the objective is at two configurations, as a function of the distance between them.

lengthscale The distance along one axis over which the objective decorrelates under the kernel; a long fitted lengthscale means that axis barely matters.

marginal likelihood The score maximized to fit kernel parameters; it trades off fitting the data against model complexity automatically.

input/output warping Transforming the search axes and the observed losses so they look well behaved (stationary, roughly Gaussian) to the surrogate; where HEBO's challenge-winning gains came from.

acquisition function The rule that converts the surrogate's beliefs into the next trial, trading off promising means against informative uncertainty.

expected improvement (EI) The classic acquisition: the expected amount by which a candidate beats the incumbent, an option-value computation with the incumbent as strike.

LogEI The numerically repaired EI family that optimizes the logarithm through stable formulas; identical optima, usable gradients.

UCB Optimism under uncertainty: score each point by the best value its posterior plausibly allows, then pick the best score.

fantasies Posterior samples standing in for the unknown outcomes of still-running trials, so new proposals can be made without waiting.

TPE A sampler that models where good and bad trials sit in configuration space and proposes where the good-to-bad density ratio peaks; native to conditional spaces, blind to interactions.

trust region A box around the incumbent inside which the surrogate is trusted; grows after successes, shrinks after failures, restarts on collapse.

ask-and-tell The interface shape where the tuner is asked for a configuration and later told the result, with no control inversion.

Multi-fidelity (Chapter 5)

rung One fidelity level in a race: all trials at a rung are compared at the same budget.

successive halving Race many configurations at a small budget, keep the best fraction, multiply their budget, repeat.

Hyperband A portfolio of successive-halving races from aggressive to conservative, hedging the choice of starting budget.

ASHA Asynchronous successive halving: promote the moment a trial is in the top fraction of results seen so far, never waiting for a rung to fill; the standard distributed scheduler.

median stopping rule Kill any trial running below the median of completed trials at the same step; the cheapest insurance.

BOHB / DEHB Racing schedulers whose new entrants come from a model (TPE) or from differential evolution instead of random sampling; DEHB is the best-evidenced small-budget RL tuner.

freeze-thaw / learning-curve scheduling Keeping a basket of paused trials and allocating the next budget slice by extrapolating each trial's curve with uncertainty.

ifBO The learning-curve scheduler whose extrapolator is a transformer pre-trained once on synthetic curves, so no model is refit inside the loop.

Population methods (Chapter 6)

PBT Training a population whose members periodically copy the weights and hyperparameters of better members (exploit) and perturb them (explore); the output is a schedule, not a point.

truncation selection The exploit rule: bottom performers restart from copies of top performers.

lineage / schedule The chain of copies and perturbations behind the final best agent; the hyperparameter trajectory it traces is the method's real product.

PB2 PBT with the explore step replaced by a time-varying GP bandit maximizing a UCB rule over performance increments.

BG-PBT The flagship population method: trust-region mixed-space BO for the explore step, plus generational architecture search with distillation to warm-start new networks.

distillation Training a new network to imitate a trained one before switching to the real objective; what makes changing architecture mid-run affordable.

Multiple objectives (Chapter 7)

dominance / Pareto front One configuration dominates another if it is at least as good everywhere and better somewhere; the front is the set of undominated trade-offs, exactly as in welfare economics.

hypervolume The area (volume) of objective space dominated by the current front, up to a fixed reference point; the standard scalar progress measure for front-seeking methods.

scalarization Collapsing several objectives into one number; the Chebyshev form (weighted worst gap to the ideal) can reach every Pareto point, the weighted sum cannot.

EHVI / qNEHVI / qLogNEHVI Expected hypervolume improvement and its batched, noise-aware, numerically repaired successors; the premium acquisition for two to four objectives.

non-dominated sorting Layering a population into ranks (undominated; undominated once rank one is removed; and so on), used for selection and for rung promotion.

crowding distance A tie-breaker preferring points in sparsely covered stretches of the front, so selection spreads along it.

veto gate A correctness check (divergences, $\hat{R}$) that discards a trial regardless of its score; distinct from an objective.

Priors, transfer, and scale (Chapter 8)

π BO Multiplying the acquisition by an expert’s prior over where the optimum lies, raised to a power that decays as evidence arrives; steers early, forgets bad advice.

PriorBand The same idea inside multi-fidelity racing: new configurations drawn from a portfolio of uniform, prior, and around-the-incumbent samplers, weighted by rung.

warm start Seeding a new study with configurations that ranked well on similar past studies, straight from the run store.

prior-fitted network (PFN) A transformer pre-trained on synthetic datasets from a chosen prior so that, shown a study’s observations as context, it outputs posterior predictions in one forward pass; approximate Bayesian inference paid for once, offline.

EI-per-cost / cost cooling Dividing the acquisition by predicted trial cost, with an exponent annealed toward zero so bargains matter early and stop mattering at the end.

μ P / μ Transfer A network parametrization under which good hyperparameters become approximately width-invariant, so one tunes a small proxy and transfers to the large model.

scaling laws (for tuning) Fitted formulas giving optimal learning rate, batch size, or weight decay as functions of model and data scale; at the largest scales they replace search outright.

Architecture search (Chapter 9)

one-shot NAS / supernet Training one network containing every candidate architecture as a weight-sharing sub-network, then scoring candidates without retraining; economical in principle, unreliable in the documented evaluations.

zero-cost proxy A score computed at initialization, without training, meant to predict a trained architecture’s quality; best evidence says useful only alongside a learned predictor, with plain parameter count and FLOPs surprisingly competitive.

grammar-based search space Architectures generated by production rules like sentences from a grammar, compactly defining huge hierarchical spaces; solves expressiveness, not compute.

Sampler diagnostics (Chapter 9)

ESS per gradient Effective sample size earned per gradient evaluation; the efficiency currency for tuning MCMC, since gradients dominate cost.

$\hat{R}$ (potential scale reduction) A convergence diagnostic comparing chains; values away from one veto a configuration no matter how efficient it looks.

divergence (HMC) A numerical failure of the integrator flagged by the sampler; counted and used as a veto.

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